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[Intervention Review]

Interventions for preventing and treating kidney disease in IgA vasculitis

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ABSTRACT

Background

IgA vasculitis (IgAV), previously known as Henoch-Schönlein purpura, is the most common vasculitis of childhood but may also occur in adults. This small vessel vasculitis is characterised by palpable purpura, abdominal pain, arthritis or arthralgia and kidney involvement. This is an update of a review first published in 2009 and updated in 2015.

Objectives

To evaluate the benefits and harms of different agents (used singularly or in combination) compared with placebo, no treatment or any other agent for (1) the prevention of severe kidney disease in people with IgAV with or without kidney involvement at onset, (2) the treatment of established severe kidney disease (macroscopic haematuria, proteinuria, nephritic syndrome, nephrotic syndrome with or without acute kidney failure) in IgAV, and (3) the prevention of recurrent episodes of IgAV-associated kidney disease.

Search methods

We searched the Cochrane Kidney and Transplant Register of Studies up to 2 February 2023 through contact with the Information Specialist using search terms relevant to this review. Studies in the Register are identified through searches of CENTRAL, MEDLINE, and EMBASE, conference proceedings, the International Clinical Trials Register (ICTRP) Search Portal and ClinicalTrials.gov.

Selection criteria

Randomised controlled trials (RCTs) comparing interventions used to prevent or treat kidney disease in IgAV compared with placebo, no treatment or other agents were included.

Data collection and analysis

Two authors independently determined study eligibility, assessed the risk of bias and extracted data from each study. Statistical analyses were performed using the random-effects model, and the results were expressed as risk ratio (RR) for dichotomous outcomes and mean difference (MD) for continuous outcomes with 95% confidence intervals (CI). Confidence in the evidence was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

Main results

Twenty studies (1963 enrolled participants) were identified; one three-arm study has been assessed as two studies. Nine studies were at low risk of bias for sequence generation (selection bias), and nine studies were at low risk of bias for allocation concealment (selection bias). Blinding of participants and personnel (performance bias) and outcome assessment (detection bias) was at low risk of bias in four and seven studies, respectively. Nine studies reported complete outcome data (attrition bias), while 10 studies reported expected outcomes, so were at low risk of reporting bias. Five studies were at low risk of other bias.

Interventions for preventing and treating kidney disease in IgA vasculitis (Review)

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Eleven studies evaluated therapy to prevent persistent kidney disease in IgAV with or without kidney involvement at presentation. There was probably no difference in the risk of persistent kidney disease any time after treatment (5 studies, 746 children: RR 0.74, 95% CI 0.42 to 1.32) or at one, three, six and 12 months in children given prednisone for 14 to 28 days at presentation of IgAV compared with placebo or supportive treatment (moderate certainty evidence). There may be no differences in the risk of any persistent kidney disease with antiplatelet therapy (three studies) or heparin (two studies) in children with or without any kidney disease at study entry, although heparin may reduce the risk of proteinuria by three months compared with placebo or no specific treatment (2 studies, 317 children: RR 0.47, 95% CI 0.31 to 0.73). One study comparing montelukast with placebo found no differences in outcomes as assessed by severity scale scores.

Nine studies examined the treatment of severe IgAV-associated kidney disease. In two studies (one involving 56 children and the other involving 54 adults), there may be no differences in efficacy outcomes or adverse effects with cyclophosphamide compared with placebo or supportive treatment. In two studies, there may be no differences in the numbers achieving remission of proteinuria with intravenous (IV) cyclophosphamide compared with mycophenolate mofetil (MMF) (65 children evaluated) or tacrolimus (142 children evaluated). In three small studies comparing cyclosporin with methylprednisolone (15 children), MMF with azathioprine (26 children), or MMF with leflunomide (19 children), it is unclear whether the treatment had any effect on the numbers in remission or the degree of proteinuria between treatment groups because of small numbers of included participants. In one study comparing plasmapheresis, cyclophosphamide and methylprednisolone with cyclophosphamide and methylprednisolone, there may be no difference in the numbers achieving remission. One study compared fosinopril with no specific therapy and reported fosinopril reduced the number of participants with proteinuria. No studies were identified that evaluated the efficacy of therapy on kidney disease in participants with recurrent episodes of IgAV.

Authors' conclusions

There are no substantial changes in conclusions from this update compared with the initial review or the previous update despite the addition of five studies. From generally low to moderate certainty evidence, we found that there may be little or no benefit in the use of corticosteroids or antiplatelet agents to prevent persistent kidney disease in children with IgAV in participants with no or minimal kidney involvement at presentation. We did not find any studies which evaluated corticosteroids in children presenting with IgAV and nephritic and/or nephrotic syndrome, although corticosteroids are recommended in such children in guidelines. Though heparin may be effective in reducing proteinuria, this potentially dangerous therapy is not justified to prevent serious kidney disease when few children with IgAV develop severe kidney disease. There may be no benefit of cyclophosphamide compared with no specific treatment or corticosteroids. While there may be no benefit in the efficacy of MMF or tacrolimus compared with IV cyclophosphamide in children or adults with IgAV and severe kidney disease, adverse effects, particularly infections, may be lower in MMF or tacrolimus-treated children. Because of small patient numbers and events leading to imprecision in results, it remains unclear whether cyclosporin, MMF or leflunomide have any role in the treatment of children with IgAV and severe kidney disease. We did not identify any studies which evaluated corticosteroids

PLAIN LANGUAGE SUMMARY

Interventions for preventing and treating kidney disease in IgA vasculitis (Henoch-Schönlein Purpura)

What is the issue?

IgA vasculitis (IgAV), previously known as Henoch-Schönlein Purpura, causes inflammation of small blood vessels in children and rarely in adults. Symptoms and signs include a skin rash of small red spots and larger bruises, particularly on the bottom and legs, tummy pain, pain and swelling of joints, and occasionally bleeding from the gut. About a third of children have kidney involvement with blood and protein found in the urine on testing. In most children, kidney involvement is mild (small amounts of blood and protein in the urine only), and it resolves completely, but a few children have persistent kidney disease that may progress to kidney failure.

What did we do?

We looked at information from 20 randomised controlled trials (RCT), which included 1963 participants. Eleven studies included children with IgAV with mild or no kidney involvement. Five studies compared prednisone tablets given for 14 to 28 days with placebo tablets or no treatment, five studies compared medications that reduce blood clotting, and one study compared montelukast (a medication usually used in children with asthma) with a placebo. Nine studies included children with moderate or severe kidney involvement. Five studies compared different medications which suppress the immune system (cyclophosphamide, mycophenolate mofetil, tacrolimus, cyclosporin, leflunomide, azathioprine). One study compared plasma exchange (where the patient's plasma is removed and replaced with normal plasma) and cyclophosphamide and methylprednisolone with cyclophosphamide and methylprednisolone alone. The last study compared fosinopril, which reduces the amount of protein in the urine, with no treatment.

What did we find?

We wanted to see whether the tested treatments prevented or treated persistent kidney disease at six to 12 months after the onset of IgAV. We found no definite benefits of prednisone or other treatments in preventing more serious kidney involvement in children with none or mild kidney involvement at study entry. We did not find any studies which evaluated prednisone in children presenting with IgAV and severe kidney involvement, although it is recommended for such children in treatment guidelines. In children with severe kidney involvement, we found no benefit of any medication that suppresses the immune system or of plasma exchange in treating kidney involvement in IgAV. As in other kidney diseases, we found that the ACE inhibitor, fosinopril, reduced the number of children with protein in the urine.

Conclusions

There are few data from RCTs examining interventions to prevent or treat kidney disease in people with IgAV. We found no evidence that giving prednisone at the onset of IgAV reduces the risk of serious kidney disease subsequently. We found no evidence that some agents are more effective than others in treating kidney involvement when it occurs. However, the numbers of people studied were too small to exclude a benefit of treatment, so further studies are required. No serious side effects were reported.

SUMMARY OF FINDINGS

Summary of findings 1. Prednisone versus placebo or supportive treatment for preventing persistent kidney disease in patients with IgAV (Henoch-Schönlein Purpura)

Prednisone versus placebo or supportive treatment for preventing persistent kidney disease in patients with IgAV (Henoch-Schönlein Purpura)

Patient or population: patients with IgAV

Settings: all settings

Intervention: prednisone

Comparison: placebo or supportive treatment

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Certainty of the evidence (GRADE)
	Assumed risk	Corresponding risk			
	Placebo or supportive treatment	Prednisone			
Persistent kidney disease at any time after treatment	143 per 1000	106 per 1000 (60 to 189)	RR 0.74 (0.42 to 1.32)	746 (5)	⊕⊕⊕⊖ Moderate ¹
Children with any continuing kidney disease at 3 months	199 per 1000	165 per 1000 (92 to 303)	RR 0.83 (0.46 to 1.52)	655 (4)	⊕⊕⊕⊖ Moderate ²
Children with any continuing kidney disease at 6 months	100 per 1000	51 per 1000 (24 to 111)	RR 0.51 (0.24 to 1.11)	379 (3)	⊕⊕⊕⊖ Moderate ²
Children with any continuing kidney disease at 12 months	84 per 1000	89 per 1000 (32 to 244)	RR 1.06 (0.38 to 2.91)	455 (3)	⊕⊕⊕⊖ Low ^{2,3}
Any continuing kidney disease at 3 months (study with high risk of bias excluded)	243 per 1000	238 per 1000 (170 to 330)	RR 0.98 (0.70 to 1.36)	487 (3)	⊕⊕⊕⊕ High
Any continuing kidney disease at 12 months (study with high risk of bias excluded)	105 per 1000	146 per 1000 (79 to 272)	RR 1.39 (0.75 to 2.59)	287 (2)	⊕⊕⊕⊖ Moderate ^{3,4}
Number developing severe kidney disease	14 per 1000	22 per 1000 (6 to 85)	RR 1.58 (0.42 to 6)	418 (2)	⊕⊕⊕⊖ Low ^{3,5}

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in the footnotes. The **corresponding risk** (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: Further research is very unlikely to change our confidence in the estimate of effect

Moderate certainty: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate

Low certainty: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate

Very low certainty: We are very uncertain about the estimate

- ¹ Two studies had unclear or biased allocation concealment & were not blinded
- ² One study had inadequate allocation concealment & no blinding & one study had large loss to follow-up
- ³ 30% loss to follow-up in largest included study
- ⁴ Small numbers of patients and events
- ⁵ Small numbers of events

BACKGROUND

Description of the condition

IgA vasculitis (previously known as Henoch-Schönlein Purpura) is a primary non-thrombocytopenic small vessel, non-granulomatous vasculitis that typically presents acutely. It is the most common systemic vasculitis in children occurring between the ages of three and 15 years, with an incidence of 3 to 27 cases per 100,000 child population (Piram 2017; Ruperto 2010). Clinically the disease is characterised by a tetrad of features, including palpable purpura, arthritis or arthralgia, abdominal pain and kidney disease (Saulsbury 1999). IgA vasculitis (IgAV) is classified as vasculitis with IgA₁-dominant immune deposits affecting small vessels, predominantly capillaries, venules or arterioles (Jennette 2013). Kidney involvement, which occurs in 40% to 50% of children with IgAV, is the most important complication of IgAV since it is the only complication associated with long-term morbidity in children and adults. Kidney involvement is clinically manifested by microscopic or macroscopic haematuria, proteinuria, nephrotic syndrome and reduced kidney function. In a systematic review of studies of unselected patients with IgAV (Narchi 2005), kidney involvement occurred in 34% of children; 80% had isolated haematuria, proteinuria or both, while 20% had acute nephritic syndrome or nephrotic syndrome. Kidney disease, if it did occur, developed early - by four weeks in 85% and by six months in nearly all children. Persistent kidney disease (hypertension, reduced function, nephrotic or nephritic syndrome) occurred in 1.8% of children overall, but the incidence varied with the severity of the kidney disease at presentation. In general, the prognosis for long-term kidney function in IgAV is excellent in children with microscopic or macroscopic haematuria alone. However, patients with nephrotic syndrome and reduced kidney function frequently show a progressive course to end-stage kidney disease (ESKD). In a study of 78 children with IgAV and kidney involvement presenting to two paediatric nephrology services, 44% of children presenting with acute nephritic syndrome, nephrotic syndrome or both, compared with 13% presenting with haematuria, proteinuria, or both, had hypertension or impaired kidney function at a mean follow-up of 23.4 years (Goldstein 1992).

Description of the intervention

Corticosteroid therapy may be used in the acute phase of IgAV largely to manage severe abdominal pain. Controversy has existed as to whether corticosteroids can prevent the development of kidney involvement, reduce its severity, or both in IgAV. An earlier systematic review concluded that early corticosteroid therapy might reduce the risk of developing persistent kidney disease (Weiss 2007), but two other reviews concluded that the benefit of corticosteroids in preventing persistent kidney disease remained unproven (Wyatt 2001; Zaffanello 2007). There is also considerable uncertainty about the efficacy of therapies to prevent progression to chronic or ESKD in children with IgAV-associated acute nephritis or nephrotic syndrome. Corticosteroid therapy, azathioprine, mycophenolate mofetil (MMF), cyclophosphamide, calcineurin inhibitors (cyclosporin, tacrolimus), antiplatelet therapy, anticoagulants, and plasmapheresis have been used in such patients (Bergstein 1998; Du 2012; Flynn 2001; Foster 2000; Iijima 1998; Niaudet 1998; Ronkainen 2003; Shenoy 2007) with varying results. However, the data came largely from observational studies rather than from randomised controlled trials (RCTs). Recently European consensus guidelines have been developed

for the management of IgAV, including the management of IgAV nephritis (Ozen 2019). These guidelines separate kidney involvement in IgAV into mild, moderate and severe degrees of IgAV nephritis. Corticosteroids are recommended as the primary treatment for IgAV nephritis for mild or moderate nephritis, with corticosteroids with intravenous (IV) cyclophosphamide recommended for severe nephritis.

How the intervention might work

IgAV nephritis is due to a systemic vasculitis with deposition of immune deposits of IgA₁ in the mesangium, activation of the alternative complement pathway and inflammation. Therefore, it has been argued that medications which treat other immune diseases, including kidney diseases, would have a role in preventing or treating IgAV nephritis. In particular, it was postulated that corticosteroids could prevent the development of significant IgAV nephritis in children presenting with IgAV (Ozen 2019). The use of other immunosuppressive agents is based on their efficacy in treating other immune complex diseases such as systemic lupus erythematosus. Urokinase, dipyridamole and warfarin have been used because of their roles in inhibiting the mediators of glomerular damage (Kawasaki 2004). Angiotensin-converting enzyme inhibitors (ACEi) and angiotensin-receptor blockers (ARB) would be expected to reduce proteinuria via effects on intraglomerular haemodynamics (Ozen 2019).

Why it is important to do this review

Although multiple treatment modalities have been used to prevent or treat IgAV nephritis, there is no consensus on the efficacy of various therapies. The aims of this systematic review were to determine the benefits and harms of different interventions used to prevent or treat persistent kidney disease in IgAV in children and adults. The scope was deliberately broad because RCTs in IgAV are few, and variability in the spectrum of kidney disease included in the relevant studies was very likely. This update of this systematic review, originally published in 2009 (Chartapisak 2009) and updated in 2015 (Hahn 2015), aimed to incorporate any further data from RCTs to provide additional evidence for or against the use of corticosteroids or other therapies to prevent IgAV nephritis, for or against the use of immunosuppressive agents to treat established IgAV nephritis and to determine the efficacy of ACEi or ARB in reducing proteinuria in IgAV nephritis.

OBJECTIVES

To evaluate the benefits and harms of different agents (used singularly or in combination) compared with placebo, no treatment or any other agent for:

- The prevention of severe kidney disease in patients with IgAV without kidney disease at presentation.
- The prevention of severe kidney disease in patients with IgAV and minor kidney disease (microscopic haematuria, mild proteinuria) at presentation.
- The treatment of established severe kidney disease (macroscopic haematuria, proteinuria, nephritic syndrome, nephrotic syndrome with or without acute kidney failure) in IgAV.
- The prevention of recurrent episodes of IgAV-associated kidney disease.

METHODS

Criteria for considering studies for this review

Types of studies

All RCTs and quasi-RCTs (RCTs in which allocation to treatment was obtained by alternation, use of alternate medical records, date of birth or other predictable methods) looking at the benefits and harms of different therapeutic modalities for the prevention or treatment of kidney disease in IgAV. If cross-over studies were identified, the first period of randomised cross-over studies was to be included.

Types of participants

Inclusion criteria

Patients of any age with IgAV with or without kidney disease manifestations (microscopic haematuria, macroscopic haematuria, proteinuria, nephrotic syndrome, acute nephritic syndrome, reduced function, acute kidney failure).

Exclusion criteria

Patients with other forms of primary or secondary glomerulonephritis (GN), such as IgA nephropathy, mesangiocapillary GN, membranous GN, systemic lupus erythematosus, rapidly progressive GN not associated with IgAV, other systemic vasculitides.

Types of interventions

Inclusion criteria

- Immunosuppressive agents, including corticosteroids, alkylating agents (cyclophosphamide, chlorambucil), azathioprine, MMF, cyclosporin, tacrolimus and rituximab
- Anticoagulants and antiplatelet agents, including warfarin, dipyridamole, aspirin, heparin
- ACEi and ARB
- Fish oil
- Immunoglobulin G, plasma exchange, antibody therapy

The above agents used individually or in combination were compared with placebo or no specific therapy or compared with other agents

- Different durations, frequencies or modes of delivery of the same interventions.

Exclusion criteria

Studies of therapies with Traditional Chinese Medicines and non-pharmacological interventions were excluded.

Types of outcome measures

1. ESKD (including dialysis and transplantation)
2. Significant increase in serum creatinine (SCr) as defined by the investigators
3. Significant reduction in glomerular filtration rate (GFR) as defined by the investigators
4. Hypertension due to IgAV-associated kidney disease
5. Development, persistence or worsening of proteinuria as defined by the investigators

6. The development or persistence of nephrotic syndrome, nephritic syndrome, acute kidney insufficiency
7. Death
8. Biopsy results including per cent of crescent formation, chronicity index, sclerosis, and fibrosis
9. Quality of life (QoL)
10. Complications of therapy e.g. infection, bleeding, neutropenia, hypertension.

Primary outcomes

1. Reduction in kidney function including ESKD, acute kidney insufficiency or significant increase in SCr, significant reduction in GFR or both as defined by the investigators
2. Development, persistence or worsening of proteinuria, development of nephrotic syndrome or acute nephritic syndrome as defined by the investigators
3. Complications of therapy e.g. infection, bleeding, leucopenia, hypertension.

Secondary outcomes

1. Biopsy results, including per cent of crescent formation, chronicity index, sclerosis, and fibrosis
2. QoL
3. Hypertension due to IgAV-associated kidney disease
4. Death

Search methods for identification of studies

Electronic searches

We searched the [Cochrane Kidney and Transplant Register of Studies](#) up to 2 February 2023 through contact with the Information Specialist using search terms relevant to this review. The Register contains studies identified from the following sources:

1. Monthly searches of the Cochrane Central Register of Controlled Trials (CENTRAL)
2. Weekly searches of MEDLINE OVID SP
3. Handsearching of kidney-related journals and the proceedings of major kidney conferences
4. Searching the current year of EMBASE OVID SP
5. Weekly current awareness alerts for selected kidney and transplant journals
6. Searches of the International Clinical Trials Register (ICTRP) Search Portal and ClinicalTrials.gov.

Studies contained in the Register are identified through searches of CENTRAL, MEDLINE, and EMBASE based on the scope of Cochrane Kidney and Transplant. Details of search strategies, as well as a list of handsearched journals, conference proceedings and current awareness alerts, are available in the Specialised Register section of information about [Cochrane Kidney and Transplant](#).

See [Appendix 1](#) for search terms used in strategies for this review.

Searching other resources

1. Reference lists of review articles, relevant studies and clinical practice guidelines.
2. Letters seeking information about unpublished or incomplete trials to investigators known to be involved in previous studies.

Data collection and analysis

Selection of studies

The search strategy described was used to obtain titles and abstracts of studies that might be relevant to the review. The titles and abstracts were screened independently by two authors, who discarded studies that were not applicable. However, studies and reviews that might have included relevant data or information on studies were retained initially. Three authors independently assessed retrieved abstracts and, if necessary, the full text of these studies to determine which studies satisfied the inclusion criteria.

Data extraction and management

Data extraction was carried out independently by three authors using standard data extraction forms. Studies reported in non-English language journals were translated before assessment. When more than one publication of one study was identified, reports were grouped together, and the publication with the most complete data was used in the analyses. Where relevant outcomes were only published in earlier versions, these data were used. Any discrepancy between published versions was highlighted. Where necessary, authors were contacted for additional information about their studies. Disagreements were resolved by discussion.

Assessment of risk of bias in included studies

The following items were assessed independently by two authors using the risk of bias assessment tool (Higgins 2022) (Appendix 2).

- Was there adequate sequence generation (selection bias)?
- Was allocation adequately concealed (selection bias)?
- Was knowledge of the allocated interventions adequately prevented during the study?
 - Participants and personnel (performance bias)
 - Outcome assessors (detection bias)
- Were incomplete outcome data adequately addressed (attrition bias)?
- Are reports of the study free of suggestion of selective outcome reporting (reporting bias)?
- Was the study apparently free of other problems that could put it at risk of bias?

Measures of treatment effect

For dichotomous outcomes (number with any kidney disease), results were expressed as risk ratio (RR) with 95% confidence intervals (CI). For continuous outcomes (severity or duration of haematuria or proteinuria, SCr, GFR), the mean difference (MD) with 95% CI were calculated.

Unit of analysis issues

We planned to include data from the first part of any cross-over study if the data could be separated. However no cross-over studies were identified.

Dealing with missing data

Any further information required from the original author was requested by written correspondence, and any relevant information obtained was included in the review. We aimed to analyse available data in meta-analyses using the intention-to-treat (ITT) data. However, where ITT data were only available

graphically or were not provided, and additional information could not be obtained from the authors, available data were used in analyses. Attrition rates (e.g. drop-outs), losses to follow-up and withdrawals were assessed.

Assessment of heterogeneity

We first assessed the heterogeneity by visual inspection of the forest plot. We then quantified statistical heterogeneity using the I^2 statistic, which describes the percentage of total variation across studies that is due to heterogeneity rather than sampling error (Higgins 2003). A guide to the interpretation of I^2 values was as follows:

- 0% to 40%: might not be important
- 30% to 60%: may represent moderate heterogeneity
- 50% to 90%: may represent substantial heterogeneity
- 75% to 100%: considerable heterogeneity.

The importance of the observed value of I^2 depends on the magnitude and direction of treatment effects and the strength of evidence for heterogeneity (e.g. P-value from the χ^2 test or a CI for I^2) (Higgins 2022).

Assessment of reporting biases

We planned to assess for reporting bias using funnel plots. However we did not identify sufficient studies on any intervention to allow this assessment.

Data synthesis

We pooled data using the random effects model but we also analysed the fixed effect model to ensure robustness of the model chosen.

Subgroup analysis and investigation of heterogeneity

We planned subgroup analyses to explore possible sources of heterogeneity among participants (severity of kidney disease, kidney pathology, age), interventions (agent, dose and duration of treatment) or associated risk of bias that might explain any observed heterogeneity of treatment effects. Examination of these possible between-study differences by subgroup analysis was not possible because of insufficient study data.

Sensitivity analysis

Sensitivity analysis was undertaken where significant heterogeneity among studies existed, and single studies appeared to be responsible for this heterogeneity. Where required, results were reported with and without the inclusion of such single studies.

Summary of findings and assessment of the certainty of the evidence

We presented the main results of the review in a Summary of findings table. This table presents key information concerning the certainty of the evidence, the magnitude of the effects of the interventions examined, and the sum of the available data for the main outcomes (Schunemann 2022a). The Summary of findings table also includes an overall grading of the evidence related to each of the main outcomes using the GRADE (Grades of Recommendation, Assessment, Development and Evaluation) approach (GRADE 2008; GRADE 2011). The GRADE approach defines the certainty of a body of evidence as the extent to which one can

be confident that an estimate of effect or association is close to the true quantity of specific interest. The certainty of a body of evidence involves consideration of within-trial risk of bias (methodological quality), directness of evidence, heterogeneity, precision of effect estimates and risk of publication bias (Schunemann 2022b). We presented the following outcomes in the Summary of findings table.

- Number with any persistent kidney involvement at different time points
- Number with severe kidney involvement

RESULTS

Description of studies

Results of the search

In the 2009 review, 909 reports were identified. After screening titles and abstracts, 13 reports underwent full-text review. Ten studies (11 reports) were included (Dudley 2013; He 2002; Huber 2004; Islek 1999; Jauhola 2010; Mollica 1992; Peratoner 1990; Ronkainen 2006a; Tarshish 2004); and one three-armed study was listed as two studies for analysis purposes (Yoshimoto 1987a; Yoshimoto 1987b). Two studies were excluded (Hui-Lan 2001; Jin 2003).

In the 2015 update, 37 new reports were identified. There were six reports of three new studies (CESAR 2010; Fuentes 2010; Xu

2009) and 11 reports of five already included studies (Dudley 2013; He 2002; Jauhola 2010; Ronkainen 2006a; Tarshish 2004). Sixteen new studies (16 reports) were excluded. Four studies (4 reports) were identified prior to publication, but no data were available and were listed as awaiting classification. Additional information was obtained for four studies (Dudley 2013; Fuentes 2010; Jauhola 2010; Ronkainen 2006a).

For this 2023 update, we searched the Cochrane Kidney and Transplant Register of Studies up to 2 February 2023 and identified 30 new reports. There were five reports of five new studies (Du 2016; Geng 2021; Liu 2019e; Tian 2015; Zhang 2021a) and three reports of two already included studies (CESAR 2010; Fuentes 2010). Three reports of three new ongoing studies were identified (NCT02532777; NCT02532790; NCT02540720), and 19 new studies were excluded (19 reports).

We reassessed and reclassified three studies awaiting classification: one study was included (Wu 2014c), one study was excluded (Ding 2014), and one study was deleted (not randomised). NCT00301613 was completed in 2006; however, no publication of the results has been identified. One previously excluded study was deleted as it was not randomised.

A total of 19 studies (34 reports), 1963 randomised participants (Figure 1) were included, 37 studies were excluded, one study is awaiting classification, and there are three ongoing studies.

Figure 1. Flow diagram of study selection

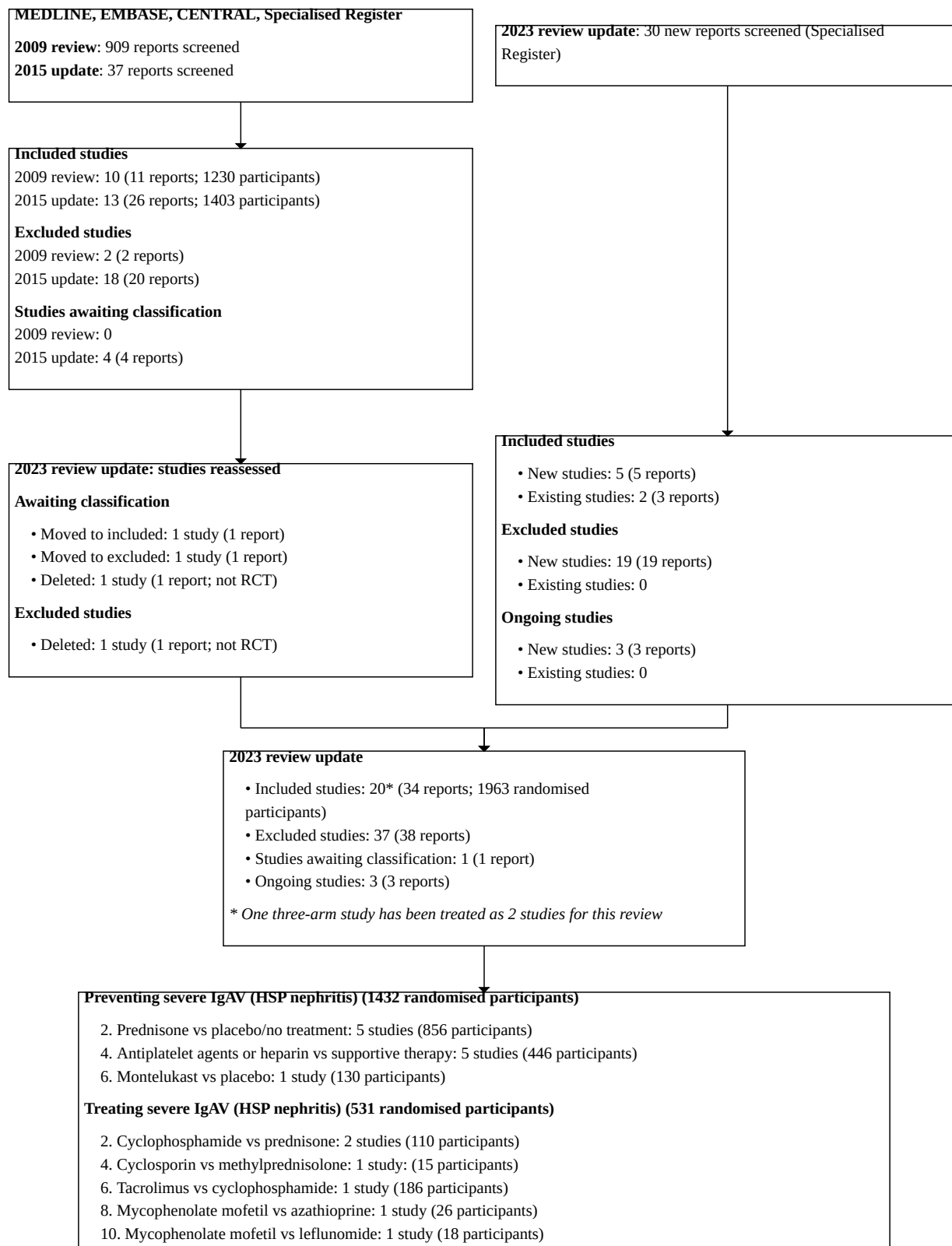


Figure 1. (Continued)

8. mycophenolate mofetil vs azathioprine: 1 study (26 participants)
10. Mycophenolate mofetil vs leflunomide: 1 study (18 participants)
12. Mycophenolate mofetil vs cyclophosphamide: 1 study (68 participants)
14. Fosinopril + standard therapy vs standard therapy: 1 study (48 participants)
16. Plasmapheresis + cyclophosphamide + methylprednisolone vs cyclophosphamide + methylprednisolone: 1 study (60 participants)

Included studies

In this 2023 update, a total of 20 studies (36 reports) enrolling 1963 participants were included (CESAR 2010; Du 2016; Dudley 2013; Fuentes 2010; Geng 2021; He 2002; Huber 2004; Islek 1999; Jauhola 2010; Liu 2019e; Mollica 1992; Peratoner 1990; Ronkainen 2006a; Tarshish 2004; Tian 2015; Wu 2014c; Xu 2009; Yoshimoto 1987a; Yoshimoto 1987b; Zhang 2021a). One study (Yoshimoto 1987a; Yoshimoto 1987b) compared two different interventions with a single control intervention and was treated as two studies for this review. CESAR 2010 included 54 adult patients, but the other studies only included children. Five studies were available in abstract form only (Du 2016; Fuentes 2010; Islek 1999; He 2002; Yoshimoto 1987a; Yoshimoto 1987b). Eighteen studies were published in English; two studies published in Chinese were translated before assessment.

Eleven studies (1432 participants) examined the prevention of progressive kidney disease in participants with IgAV with or without kidney involvement at presentation. Five studies (856 enrolled participants) examined the effects of short-duration corticosteroids (14 to 28 days) on preventing persistent IgAV-associated kidney disease at six to 12 months after presentation in comparison with placebo (Dudley 2013; Huber 2004; Ronkainen 2006a) or supportive treatment (Islek 1999; Mollica 1992). Three studies included children with kidney disease at randomisation (Dudley 2013; Huber 2004; Ronkainen 2006a). Participants considered to have established IgAV-associated kidney disease (proteinuria > 300 mg/L or haematuria > 10 red blood cells/high power field) were excluded from Ronkainen 2006a, while Dudley 2013 and Huber 2004 included children with any degree of kidney disease at randomisation. Islek 1999 and Mollica 1992 only included children who had no haematuria or proteinuria at presentation.

Three studies (Peratoner 1990; Yoshimoto 1987a; Yoshimoto 1987b) with 129 participants (all children) compared antiplatelet agents (dipyridamole, cyproheptadine and salicylates) with supportive treatment, and two studies (He 2002; Tian 2015) (317 children) compared heparin with placebo or no specific treatment. Peratoner 1990 provided outcome data separately for children with and without kidney disease at presentation. Tian 2015 included participants with kidney involvement but did not specify the severity, while the other studies only included children with no kidney disease at randomisation.

Dudley 2013 used urinary protein-creatinine ratio (UPCR) as the primary endpoint, while in nine other studies, the primary endpoint of kidney disease was defined by a composite of haematuria and proteinuria. The primary outcome in Dudley 2013 at 12 months was only available in 247 of the 296 children who had a 12-month follow-up visit. Data on the number of children with haematuria or

proteinuria were used in analyses at one and three months. At each time point, the number of children with available data was less than the number undergoing follow-up.

Wu 2014c compared montelukast sodium with placebo in 84 children with IgAV without kidney involvement and in 46 children with IgAV with haematuria and proteinuria and reported results according to a symptom severity score.

Nine studies (531 participants) examined the treatment of severe IgAV-associated kidney disease (nephrotic range proteinuria, ISKDC grade II-IV changes on biopsy).

- Tarshish 2004 (56 children entered/evaluated) compared cyclophosphamide with no specific treatment.
- Jauhola 2010 (15 children entered/evaluated) compared cyclosporin with methylprednisolone.
- CESAR 2010 (54 adults) compared cyclophosphamide and prednisone with prednisone alone in adults with severe biopsy-proven IgAV kidney disease.
- Fuentes 2010 (26 children entered/evaluated) compared MMF with azathioprine; both treatment groups received prednisone.
- Du 2016 (18 children entered/evaluated) compared MMF with leflunomide.
- Liu 2019e (60 children entered/evaluated) compared double filtration plasmapheresis together with cyclophosphamide and corticosteroids with cyclophosphamide and corticosteroids alone.
- Geng 2021 (65 children entered/evaluated) compared MMF with IV cyclophosphamide, with both groups receiving corticosteroids.
- Zhang 2021a (186 children entered/170 evaluated) compared tacrolimus with IV cyclophosphamide with all three groups receiving corticosteroids; participants in the third arm of Zhang 2021a were excluded from analyses as they received tacrolimus with tripterygium glycosides (Chinese medicines were excluded from this review).
- Xu 2009 (48 children) compared fosinopril with supportive treatment, with the primary endpoint being the number with remission of proteinuria.

In the nine studies evaluating interventions for severe IgAV-associated kidney disease, the primary endpoint was defined by a composite of proteinuria and reduced kidney function in three studies (Jauhola 2010; Liu 2019e; Tarshish 2004), while CESAR 2010 used a Birmingham Vascular Activity Score (BVAS) of zero at six months as indicating complete disease remission. In the remaining five studies (Du 2016; Fuentes 2010; Geng 2021; Liu 2019e; Zhang 2021a), the primary endpoint was remission of proteinuria,

usually measured by 24-hour urinary protein excretion. [Jauhola 2010](#) reported data on included randomised and non-randomised patients. Using information obtained from the authors, only randomised patients were included in the study analyses.

Outcomes were assessed at six to at least 12 months in nine studies ([CESAR 2010](#); [Dudley 2013](#); [Fuentes 2010](#); [Huber 2004](#); [Jauhola 2010](#); [Mollica 1992](#); [Peratoner 1990](#); [Ronkainen 2006a](#); [Wu 2013b](#)), at 15 months in one study ([Geng 2021](#)) and at two years in two studies ([Jauhola 2010](#); [Zhang 2021a](#)). [Tarshish 2004](#) reported the outcomes at the end of the study without providing detailed information on the duration of the study. [Ronkainen 2006a](#) provided a further long-term outcome at eight years. The remaining seven studies did not specify the timing of the outcome assessment ([Du 2016](#); [He 2002](#); [Islek 1999](#); [Liu 2019e](#); [Tian 2015](#); [Xu 2009](#); [Yoshimoto 1987a](#); [Yoshimoto 1987b](#)).

No studies examining warfarin, ARB alone, fish oil, immunoglobulin G, rituximab or dapsons were identified.

Excluded studies

Thirty-seven studies were excluded. Reasons for exclusion included:

- Wrong or unclear study design ([Fukui 1989](#); [Jiang 2003](#))
- Wrong population ([Gao 2018a](#); [Liu 2008](#); [NCT02939573](#); [NCT03647852](#); [NCT04008316](#); [Xia 2016](#); [Xiong 2019](#))
- Interventions not relevant to this review ([Chen 2010](#); [ChiCTR2100050353](#); [ChiCTR-INR-17013850](#); [Cui 2020](#); [Ding 1995](#); [Ding 2014](#); [Ding 2019](#); [Ding 2021](#); [Hui-Lan 2001](#); [Jia 2001](#); [Jin 2003](#); [Jin 2021](#); [Kim 1987](#); [Li 2022](#); [Wang 2011](#); [Wang 2021a](#); [Wu 2013b](#); [Xie 2009](#); [Yang 2010](#); [Yi 2007](#); [Zhang 1984](#); [Zhang 2019b](#); [Zhao 2009](#); [Zhu 2016](#)).

Risk of bias in included studies

[Figure 2](#) and [Figure 3](#) describe the graphical representation of the risk of bias assessment for all studies.

Figure 2. Risk of bias: Review authors' judgements about each methodological quality item presented as percentages across all included studies.

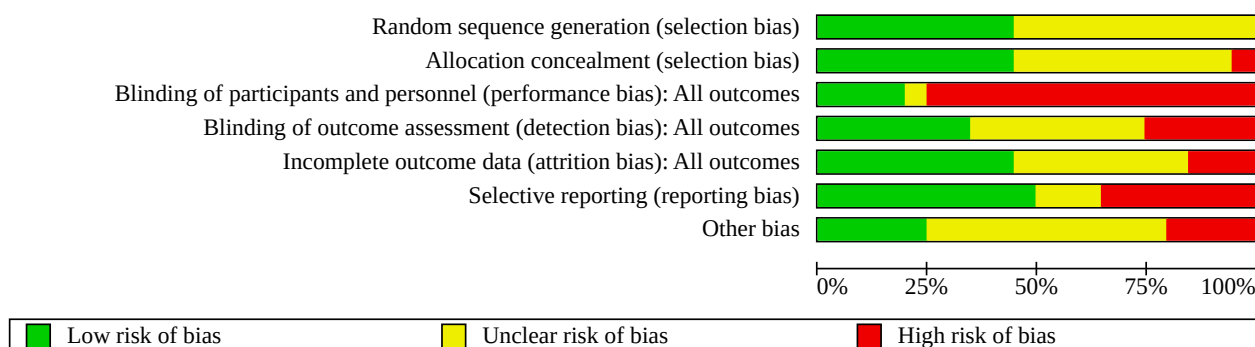


Figure 3. Risk of bias: Review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
CESAR 2010	+	+	-	-	+	+	+
Du 2016	?	?	-	+	?	-	-
Dudley 2013	+	+	+	+	-	+	+
Fuentes 2010	+	+	-	-	+	+	-
Geng 2021	+	?	-	+	+	+	?
He 2002	?	?	?	?	?	+	?
Huber 2004	+	+	+	+	+	+	+
Islek 1999	?	?	-	?	?	?	-
Jauhola 2010	+	+	-	-	-	+	?
Liu 2019e	?	?	-	+	+	+	?
Mollica 1992	?	-	-	?	-	-	?
Peratoner 1990	?	?	-	?	?	-	?
Ronkainen 2006a	+	+	+	+	+	-	+
Tarshish 2004	?	+	-	?	?	+	?
Tian 2015	?	?	-	-	+	-	?
Wu 2014c	+	+	+	?	?	-	+
Xu 2009	?	?	-	-	+	+	?

Figure 3. (Continued)

Xu 2009	?	?	-	-	+	+	?
Yoshimoto 1987a	?	?	-	?	?	?	-
Yoshimoto 1987b	?	?	-	?	?	?	?
Zhang 2021a	+	+	-	+	+	-	?

Allocation

Nine studies (CESAR 2010; Dudley 2013; Fuentes 2010; Geng 2021; Huber 2004; Jauhola 2010; Ronkainen 2006a; Wu 2014c; Zhang 2021a) were determined to be at low risk of bias for random sequence generation; the risk of bias was unclear in the remaining 11 studies.

Nine studies (CESAR 2010; Dudley 2013; Fuentes 2010; Huber 2004; Jauhola 2010; Ronkainen 2006a; Tarshish 2004; Wu 2014c; Zhang 2021a) were determined to be a low risk of bias for allocation concealment, one study was at high risk of bias (Mollica 1992), and the remaining 11 studies had unclear risk of bias.

Blinding

Performance bias was at low risk in four studies (Dudley 2013; Huber 2004; Ronkainen 2006a; Wu 2014c), at unclear risk in one study (He 2002), and at high risk of bias in the remaining 12 studies.

Detection bias was at low risk in seven studies (Du 2016; Dudley 2013; Geng 2021; Huber 2004; Liu 2019e; Ronkainen 2006a; Zhang 2021a), at high risk in five studies (CESAR 2010; Fuentes 2010; Jauhola 2010; Tian 2015; Xu 2009) and at unclear risk in the remaining seven studies.

Incomplete outcome data

Nine studies were considered to be at low risk of attrition bias (CESAR 2010; Fuentes 2010; Geng 2021; Huber 2004; Liu 2019e; Ronkainen 2006a; Tian 2015; Xu 2009; Zhang 2021a). Three studies were at high risk of attrition bias (Dudley 2013; Jauhola 2010; Mollica 1992) because of loss to follow-up or exclusion of data from analyses. In the eight remaining studies, there was insufficient information provided to determine whether all patients entering the study were included in the analysis, so the risk of bias was unclear.

Selective reporting

Reporting included all important kidney outcomes and adverse effects of medications in 10 studies (CESAR 2010; Dudley 2013; Fuentes 2010; Geng 2021; He 2002; Huber 2004; Jauhola 2010; Mollica 1992; Peratoner 1990; Tarshish 2004). Seven studies were considered at high risk of reporting bias because they did not report all expected outcomes (Du 2016; Mollica 1992; Peratoner 1990; Tian 2015; Zhang 2021a) or provided outcomes in a graphical form that could not be included in meta-analyses (Ronkainen 2006a; Wu 2014c). In the remaining three studies, it was unclear whether important kidney outcomes, including nephrotic syndrome, reduced kidney function, and adverse effects of medications, had not occurred or had not been reported.

Other potential sources of bias

Five studies appeared free of other potential sources of bias (CESAR 2010; Dudley 2013; Huber 2004; Ronkainen 2006a; Wu 2014c). One author in Fuentes 2010 was a consultant for a pharmaceutical company, so this study was judged to be at high risk of bias. Five studies (Du 2016; Fuentes 2010; Islek 1999; Yoshimoto 1987a; Yoshimoto 1987b) were only available as conference abstracts with no full-text reports identified, so they were judged to be at high risk of bias. In the remaining studies, there was insufficient information provided to determine if there were other potential sources of bias.

Effects of interventions

See: [Summary of findings 1](#) Prednisone versus placebo or supportive treatment for preventing persistent kidney disease in patients with IgAV (Henoch- Schönlein Purpura)

Preventing persistent kidney disease

Eleven studies enrolled participants with no kidney involvement or mild degrees of haematuria and proteinuria (Dudley 2013; He 2002; Huber 2004; Islek 1999; Mollica 1992; Peratoner 1990; Ronkainen 2006a; Tian 2015; Wu 2014c; Yoshimoto 1987a; Yoshimoto 1987b).

Prednisone versus placebo or supportive treatment

In children with newly diagnosed IgAV and without significant kidney disease, prednisone treatment compared with placebo or supportive treatment probably makes little or no difference to the risk of any kidney disease ([Analysis 1.1](#) (5 studies, 746 children): RR 0.73, 95% CI 0.43 to 1.24; $I^2 = 44\%$; moderate certainty evidence).

Prednisone compared with placebo or supportive therapy probably makes little or no difference to the risk of development or persistence of kidney involvement at one ([Analysis 1.2.1](#) (4 studies, 655 children): RR 0.80, 95% CI 0.34 to 1.84; $I^2 = 72\%$), three ([Analysis 1.2.2](#) (4 studies, 655 children): RR 0.83, 95% CI 0.46 to 1.52; $I^2 = 44\%$) and six months ([Analysis 1.2.3](#) (3 studies, 379 children): RR 0.51, 95% CI 0.24 to 1.11; $I^2 = 0\%$; moderate certainty evidence). At 12 months, prednisone may make little or no difference to the development or persistence of kidney involvement ([Analysis 1.2.4](#) (3 studies, 455 children): RR 1.06, 95% CI 0.38 to 2.91; $I^2 = 32\%$; low certainty evidence). There was substantial heterogeneity in study outcomes at one, three and 12 months, which was largely due to Mollica 1992. This study, which was at high risk of bias due to inadequate allocation concealment, showed a large benefit of prednisone in contrast to the other three studies. Sensitivity analysis with the exclusion of this study eliminated the heterogeneity except at one month with no change to significance ([Analysis 1.3](#)). Removal of this study from the analysis indicated that prednisone compared with placebo, makes little difference to the risk of development or persistence of kidney involvement at three months or 12 months (high certainty evidence at 3 months;

moderate certainty evidence at 12 months) ([Summary of findings 1](#)).

In [Ronkainen 2006a](#), a post hoc subgroup analysis of 71 children with kidney disease at or within one month of randomisation found that kidney disease may be less common at six months after prednisone therapy compared with placebo ([Analysis 1.4.3](#): RR 0.45, 95% CI 0.21 to 0.98).

Two studies ([Dudley 2013](#); [Ronkainen 2006a](#)) reported the number of children who developed severe kidney disease with nephrotic range proteinuria, hypertension or reduced kidney function. There may be little or no difference in the risk of severe kidney disease between children treated with prednisone or placebo ([Analysis 1.5](#) (2 studies, 418 children): RR 1.58, 95% CI 0.42 to 6.00; $I^2 = 0\%$). However, there was considerable imprecision in the results.

[Islek 1999](#) assessed the duration of haematuria and proteinuria and found that there may be little or no difference in the duration of haematuria ([Analysis 1.6.1](#) (33 children): MD -1.00, 95% CI -10.26 to 8.26) or proteinuria ([Analysis 1.6.2](#) (33 children): MD -1.60, 95% CI -15.62 to 12.42) between treatment groups.

The risk of gastrointestinal involvement requiring hospital admission may not differ between prednisone and placebo or supportive treatment ([Analysis 1.7](#) (3 studies, 517 participants): RR 0.56, 95% CI 0.25 to 1.23; $I^2 = 0\%$). In [Huber 2004](#), two children in the placebo group required surgery for intussusception and were withdrawn from the study. Based on patient diary records in [Ronkainen 2006a](#), children on prednisone may have had less abdominal pain and joint pain based on lower pain severity scores for abdominal or joint pain. They also may have shorter durations of abdominal pain but not joint pain compared with placebo.

[Ronkainen 2006a](#) completed an eight-year follow-up on 138/176 children originally randomised. They reported minor abnormalities after two clinical screenings in 10 children; eight who had received prednisone and two who received placebo. There may be no differences in haematuria ([Analysis 1.8.1](#): RR 4.86, 95% CI 0.24 to 99.39), proteinuria ([Analysis 1.8.2](#): RR 2.92, 95% CI 0.12 to 70.35), haematuria and proteinuria ([Analysis 1.8.3](#): RR 2.92, 95% CI 0.12 to 70.35), hypertension ([Analysis 1.8.4](#): RR 0.97, 95% CI 0.14 to 6.70), and decreased GFR by Schwartz formula ([Analysis 1.8.5](#): RR 4.86, 95% CI 0.24 to 99.39) but there was considerable imprecision in the results.

[Huber 2004](#) and [Ronkainen 2006a](#) reported no serious adverse effects caused by prednisone or placebo. In [Ronkainen 2006a](#), children receiving prednisone had a 1 kg greater increase in weight and a 4 mm Hg increase in diastolic blood pressure during treatment. In [Dudley 2013](#), one child developed behavioural problems, and one had an infection; these were considered related to prednisone therapy, while one child developed abdominal pain in the placebo group. Adverse effects were not recorded in either [Islek 1999](#) or [Mollica 1992](#).

Antiplatelet agents versus supportive treatment

Treatment with antiplatelet agents compared with supportive treatment may make little or no difference to the risk of kidney disease occurring at any time during follow-up in children without kidney disease at entry ([Analysis 2.1.1](#) (2 studies, 101 children): RR 1.16, 95% CI 0.46 to 2.95; $I^2 = 0\%$) or with kidney disease at study

entry ([Analysis 2.1.2](#) (19 children): RR 0.92, 95% CI 0.23 to 3.72) (low certainty evidence).

It is unclear whether aspirin compared with supportive therapy makes any difference to the risk of kidney disease as the certainty of the evidence is very low ([Yoshimoto 1987a](#)) ([Analysis 2.1.3](#) (18 children): RR 0.14, 95% CI 0.01 to 2.42).

Duration of follow-up and adverse effects were not recorded in these studies.

Heparin versus placebo or conventional treatment

Two studies ([He 2002](#); [Tian 2015](#)) reported heparin may reduce proteinuria ([Analysis 3.2.2](#) (2 studies, 317 children): RR 0.47, 95% CI 0.31 to 0.73; $I^2 = 0\%$). However, for the outcomes of any kidney disease ([Analysis 3.1](#) (2 studies, 317 children): RR 0.48, 95% CI 0.15 to 1.54; $I^2 = 89\%$) and haematuria ([Analysis 3.2.1](#) (2 studies, 317 children): RR 0.39, 95% CI 0.07 to 2.21; $I^2 = 82\%$), there was considerable heterogeneity between studies so that it is uncertain whether heparin has any effect on the outcomes. The risk for nephrotic syndrome in [He 2002](#) may not differ between the groups, but event numbers were small, resulting in imprecision ([Analysis 3.2.3](#) (228 children): RR 0.31, 95% CI 0.03 to 2.89). The development of kidney disease may be delayed in the heparin group compared with placebo in [He 2002](#) ([Analysis 3.3](#) (228 children): MD 47.3 days, 95% CI 34.24 to 60.36). No child developed severe bleeding.

Montelukast sodium versus placebo

[Wu 2014c](#) compared montelukast sodium with placebo in 84 children with IgAV without nephritis and in 46 children with IgAV with haematuria and proteinuria. In children with nephritis, the authors reported that montelukast compared with placebo may reduce the severity scale scores for proteinuria and haematuria at four weeks and at the end of treatment at three months, but scores may increase at six months in both groups. The data were shown graphically only, so meta-analysis could not be performed. One child experienced irritability, and one reported dizziness. Clinical features of IgAV may be reduced at one, two and three weeks in children with and without nephritis. Montelukast did not alter the outcome of nephritis at three years in children with nephritis at study onset.

Treating severe kidney disease

Nine studies enrolled participants with more severe kidney disease ([CESAR 2010](#); [Du 2016](#); [Fuentes 2010](#); [Geng 2021](#); [Jauhola 2010](#); [Liu 2019e](#); [Tarshish 2004](#); [Xu 2009](#); [Zhang 2021a](#)).

Cyclophosphamide versus supportive treatment

In [Tarshish 2004](#), a study of children with significant IgAV-associated kidney disease (proteinuria, reduced kidney function, crescents, segmental lesions or both on kidney biopsy) treated within three months of the onset of IgAV, cyclophosphamide compared with supportive treatment may make little difference to the risk of persistent kidney disease of any severity ([Analysis 4.1](#) (56 children): RR 1.07, 95% CI 0.65 to 1.78), severe kidney disease (heavy proteinuria, reduced GFR, ESKD) ([Analysis 4.2](#) (56 children): RR 0.88, 95% CI 0.37 to 2.09), or ESKD ([Analysis 4.3](#) (56 children): RR 0.75, 95% CI 0.18 to 3.05) during follow-up (low certainty evidence). Adverse effects of cyclophosphamide were not reported.

Cyclophosphamide plus steroids versus steroids

CESAR 2010 compared cyclophosphamide and methylprednisolone followed by prednisone with methylprednisolone and prednisone in adults. With treatment, there may be no difference in the number of adults who achieved a BVAS of zero by six months ([Analysis 5.1.1](#) (54 adults): RR 1.16, 95% CI, 0.26 to 5.24) or in the number whose BVAS score improved by six months ([Analysis 5.1.2](#) (54 adults): RR 0.98, 95% CI 0.81 to 1.19). There may be no differences in the secondary outcomes, including hypertension, reduced GFR, proteinuria, improvement in kidney function and ESKD at 12 months ([Analysis 5.2](#)). There may be no difference in death between treatment groups ([Analysis 5.2.7](#) (54 adults): RR 0.19, 95% CI 0.02 to 1.50). These deaths were not considered to be related to the treatments received, and the authors noted that those in the prednisone group had more severe disease at baseline based on BVAS scores. There may be no differences in adverse effects ([Analysis 5.3](#)) (low certainty evidence).

Tacrolimus versus IV cyclophosphamide

Zhang 2021a compared tacrolimus with IV cyclophosphamide, with both groups receiving prednisolone. Since tripterygium glycosides is an excluded intervention for this systematic review, only the data comparing tacrolimus with IV cyclophosphamide were included in this review. The number of children without proteinuria ([Analysis 6.1.1](#) (170 children): RR 1.13, 95% CI 0.99 to 1.30) or haematuria ([Analysis 6.1.2](#) (170 children): RR 1.18, 95% CI 1.02 to 1.37) may be slightly increased at two years following tacrolimus compared with IV cyclophosphamide. At two years, the number of children with recurrence of proteinuria or haematuria may not differ between treatment groups ([Analysis 6.2](#)). Twenty-four-hour urinary protein excretion at three months (end of treatment) ([Analysis 6.3.3](#) (170 children): MD -0.31, 95% CI -0.40 to -0.22) and at six months ([Analysis 6.3.4](#) (170 children): MD -0.20, 95% CI -0.24 to -0.16) may be slightly lower in children treated with tacrolimus compared with cyclophosphamide. Urine red blood cell excretion may be lower in tacrolimus-treated children compared with cyclophosphamide at three months ([Analysis 6.3.5](#)) and six months ([Analysis 6.3.6](#)) (low certainty evidence).

Respiratory infections ([Analysis 6.4.2](#) (68 children): RR 0.55, 95% CI 0.38 to 0.82) and other severe adverse effects (including hypertension, diabetes, ocular hypertension, lipid abnormalities) may be less common with tacrolimus compared with IV cyclophosphamide ([Analysis 6.4.6](#) (170 children): RR 0.45, 95% CI 0.20 to 0.98). Other adverse effects (leucopenia, abnormal liver function tests, urinary tract infections and poor appetite) may not differ between treatment groups ([Analysis 6.4](#)).

Cyclosporin versus methylprednisolone

Jauhola 2010 compared cyclosporin with methylprednisolone in children with severe kidney disease. All seven children treated with cyclosporin compared with 4/8 treated with methylprednisolone were in remission by three months. Because of the small numbers, it is unclear whether the number with remission differs between groups ([Analysis 7.1](#) (15 children): RR 1.88, 95% CI 0.95 to 3.69). At the two-year follow-up, 23 children were assessed, including eight non-randomised children; the remission rate was 70% in children treated with cyclosporin and 58% in children treated with methylprednisolone. At final follow-up at a mean of 6.3 years, it is unclear whether there is any difference in efficacy with 6/7 children treated with cyclosporin compared with 5/8 treated with

methylprednisolone in remission ([Analysis 7.2](#) (15 children): RR 1.37, 95% CI 0.74 to 2.54) (all low certainty evidence). Adverse effects related to cyclosporin and methylprednisolone were not reported separately for the randomised children.

Mycophenolate mofetil versus IV cyclophosphamide

Geng 2021 compared MMF with IV cyclophosphamide. Both groups received prednisone, and most received three pulses of IV methylprednisolone. The number of children with complete remission at three months ([Analysis 8.1.1](#) (68 children): RR 1.20, 95% CI 0.72 to 1.99), six months ([Analysis 8.1.2](#) (68 children): RR 1.02, 95% CI 0.78 to 1.34), and 12 months ([Analysis 8.1.3](#) (68 children): RR 1.06, 95% CI 0.83 to 1.35) may not differ between groups. Similarly, the number of children with complete or partial remission at three months ([Analysis 8.2.1](#)), six months ([Analysis 8.2.2](#)), and 12 months ([Analysis 8.2.3](#)) may not differ between treatment groups. Adverse effects may not differ between treatment groups ([Analysis 8.3](#)) (all low certainty evidence).

Mycophenolate mofetil versus azathioprine

Fuentes 2010 compared AZA versus MMF in children with biopsy-proven IgAV (class I-III); both groups received prednisone. There may be little or no difference in the numbers with protein remission between the two groups ([Analysis 9.1](#) (26 children): RR 1.32, 95% CI 0.86 to 2.02). There may be little or no difference in the number of children with improvement in kidney histology between treatment groups ([Analysis 9.2](#) (26 children): RR 1.24, 95% CI 0.66 to 2.36), in the number with relapse of IgAV ([Analysis 9.3](#)) or the GFR ([Analysis 9.4](#)) between groups (all low certainty evidence).

Mycophenolate mofetil versus leflunomide

Du 2016 compared leflunomide with MMF. Both groups received tapering prednisone and ACEi. There may be little or no difference in proteinuria between groups at three months ([Analysis 10.1.1](#) (19 participants): MD 360 mg/day, 95% CI -43.35 to 763.35) and nine months ([Analysis 10.1.2](#) (19 participants): MD 49.00 mg/day, 95% CI 3.09 to 94.91) (all low certainty evidence).

Double filtration plasmapheresis with cyclophosphamide and methylprednisolone compared with cyclophosphamide and methylprednisolone alone

In **Liu 2019e**, double filtration plasmapheresis with cyclophosphamide and methylprednisolone compared with cyclophosphamide and methylprednisolone alone may make little or no difference to the numbers of children with complete remission ([Analysis 11.1.1](#) (60 participants): RR 1.43, 95% CI 0.63 to 3.25), the numbers with complete and partial remission after three cycles of treatment ([Analysis 11.1.2](#) (60 participants): RR 1.26, 95% CI 0.91 to 1.75) or for to the numbers with complete remission at six months ([Analysis 11.1.3](#) (60 participants) RR 1.13, 95% CI 0.89 to 1.44). Adverse effects were uncommon and may not differ between treatment groups ([Analysis 11.2](#)) (all low certainty evidence).

Fosinopril plus supportive treatment versus supportive treatment alone

Xu 2009 reported fosinopril given for two months may increase the number of children with complete remission of proteinuria compared with supportive treatment ([Analysis 12.1](#) (48 children): RR 5.83, 95% CI 1.50 to 22.74) (low certainty evidence).

Other outcomes

In most studies, the severity of haematuria and proteinuria, the degree of kidney dysfunction and the presence of hypertension were not specified. [Dudley 2013](#) provided information on UPCR, and [Tarshish 2004](#) provided separate information on ESKD.

DISCUSSION

Summary of main results

In this 2023 update of a review first published in 2008 and updated in 2015, we identified 20 studies involving 1963 participants with IgAV.

- Eleven studies examined the efficacy of therapies to prevent persistent kidney disease in IgAV.
- Nine studies examined the efficacy of therapies to treat kidney disease in IgAV.

Prevention of persistent kidney disease in IgAV with prednisone therapy

- Five studies with 746 evaluated children found that prednisone therapy compared with placebo or supportive therapy may have little or no effect in preventing persistent kidney disease at any time up to one year.
- Two of the five studies with 418 evaluated children found that prednisone therapy compared with placebo or supportive therapy may make little or no difference in the number of children with severe kidney disease at one year.

Prevention of persistent kidney disease in IgAV with antiplatelet therapy or heparin

- Three small studies found that antiplatelet therapies (dipyridamole, cyproheptadine, aspirin) compared with supportive therapy may make little or no difference to the number of children with any kidney involvement.
- Two studies suggested that heparin may reduce proteinuria at three months after presentation in children with IgAV and kidney involvement.

Prevention of persistent kidney disease in IgAV with other treatments

- One study reported montelukast sodium compared with placebo may reduce clinical symptoms in IgAV.

Treatment of severe kidney disease in IgAV with immunosuppressive therapy

- One study in children (56 evaluated children) and one study in adults (54 evaluated adults) found that cyclophosphamide compared with corticosteroids may make little or no difference to any persistent kidney disease or ESKD in IgAV.
- One study (170 evaluated children) reported that tacrolimus compared with IV cyclophosphamide may slightly reduce 24-hour urine excretion of protein and red cells at six months, may slightly increase the number with resolution of proteinuria and haematuria by two years, but may not reduce the number with recurrence of disease by two years in IgAV. Adverse effects, particularly infections, may be fewer with tacrolimus.

- It is unclear from one study with only 15 children enrolled whether cyclosporin compared with methylprednisolone makes any difference to the outcomes of IgAV nephritis.
- MMF has been evaluated in three studies compared with IV cyclophosphamide (68 children), azathioprine (26 children) and leflunomide (19 children).
 - MMF compared with IV cyclophosphamide, may make little or no difference to any persistent kidney disease in IgAV.
 - MMF compared with azathioprine may make little or no difference to persistent kidney disease in IgAV.
 - MMF compared with leflunomide may make little or no difference to persistent kidney disease in IgAV.
- One study (60 children) reported that double filtration plasmapheresis with IV cyclophosphamide and methylprednisolone compared with IV cyclophosphamide and methylprednisolone may make little or no difference to persistent kidney disease in IgAV.
- One study reported that proteinuria in IgAV may be reduced with the ACEi (fosinopril) in children with IgAV and kidney involvement.

Overall completeness and applicability of evidence

Twenty studies enrolling 1963 participants were included in this review update; five studies were only available in abstract form, and several studies included small numbers of participants with incomplete outcome data, which could result in incomplete information being included in this systematic review.

Three well-designed, placebo-controlled studies ([Dudley 2013](#); [Huber 2004](#); [Ronkainen 2006a](#)) have provided data from over 400 children. There is no demonstrable benefit of prednisone therapy at six to 12 months in children with IgAV, with no or minor kidney involvement at presentation, in preventing subsequent important kidney involvement. In addition, eight-year follow-up data in [Ronkainen 2006a](#) found no longer-term benefit. Therefore, further RCTs to evaluate prednisone to prevent kidney disease in this group of children presenting with IgAV are unlikely to be justified. In the studies evaluating prednisone therapy, the outcomes reported were poorly defined except for [Dudley 2013](#), which used UPCR as the primary outcome. The potential significance for long-term kidney function of any residual urinary abnormalities could not be assessed in the other studies since they reported the endpoint as the presence of haematuria or proteinuria or both without measuring the degree of proteinuria.

Persistent proteinuria with or without reduction in GFR places the child with IgAV at risk of progression to chronic kidney disease and is considered by paediatric rheumatologists and nephrologists to require treatment with some urgency since delay in treatment may result in more chronic histological changes ([Davin 2011](#); [KDIGO 2021](#)). Observational studies ([Niaudet 1998](#)) support the use of corticosteroids, including IV methylprednisolone followed by oral prednisone, in patients with nephrotic range proteinuria and no reduction in kidney function. The European Consensus Guidelines on IgAV ([Ozen 2019](#)) recommend oral prednisolone for mild IgAV nephritis (normal GFR with mild to moderate proteinuria) and IV methylprednisolone followed by oral prednisone for moderate IgAV nephritis (persistent proteinuria $\pm$ reduced GFR with kidney biopsy evidence of $< 50\%$ crescents). However, there are no adequately powered RCTs that have evaluated corticosteroid

therapy (prednisolone, methylprednisolone) in children with moderate or severe IgAV-associated nephritis.

The European Consensus Guidelines (Ozen 2019) and KDIGO (KDIGO 2021) suggest that children with IgAV and severe kidney involvement (> 50% crescents on kidney biopsy, impaired GFR and nephrotic range proteinuria) should be treated similarly to systemic small vessel vasculitis with kidney involvement using IV cyclophosphamide and high dose corticosteroids. We identified four studies that compared oral or IV cyclophosphamide with supportive therapy (Tarshish 2004), corticosteroids (CESAR 2010), tacrolimus (Zhang 2019b) and MMF (Geng 2021). Tacrolimus or MMF may achieve similar degrees of improvement to IV cyclophosphamide (Geng 2021; Zhang 2021a), suggesting that these agents could be used in preference to IV cyclophosphamide. Small studies found no benefit of cyclosporin over methylprednisolone (Jauhola 2010) or MMF over azathioprine (Fuentes 2010) or leflunomide (Du 2016). Plasma exchange is also used in patients with IgAV and severe kidney involvement. A single study found no increased benefit of plasmapheresis with IV cyclophosphamide and corticosteroids over IV cyclophosphamide and corticosteroids alone (Liu 2019e). Limited reporting revealed small numbers of adverse events in all studies with no significant difference between interventions. No studies were identified that evaluated therapy to prevent or treat persistent kidney disease in participants with recurrent episodes of IgAV.

A single study showed that the ACEi, fosinopril, reduced the number of children with proteinuria associated with IgAV. Both the European Consensus Guidelines (Ozen 2019) and the KDIGO guidelines (KDIGO 2021) state that these children should be treated with ACEi or ARBs.

No RCTs were identified that examined IV immunoglobulin, rituximab, fish oil or ARB. No studies specifically addressing whether therapy reduced the risk of recurrent episodes of IgAV were identified.

Quality of the evidence

Sequence generation and allocation concealment were at low risk of bias in nine studies (45%). This may be attributed to poorer reporting of these parameters in the earlier studies as well as the inclusion of studies only available as abstracts. Blinding of participants and investigators was reported in four studies while seven studies were at low risk of detection bias, reflecting a high risk of bias in the remaining studies since knowledge of treatment groups could influence patient management and reporting. Only nine studies were at low risk of attrition bias. The otherwise robust study of Dudley 2013 was at high risk of attrition bias due to a significant dropout rate, with only 72% (123/171) reporting the primary outcome. Ten studies were at low risk of selective reporting. Studies with a high risk of bias are associated with an increased likelihood of results favouring the study intervention (Schulz 1995; Wood 2008). The exclusion of one study at high risk of bias removed heterogeneity between studies without altering the overall result but reinforcing the strength of the evidence suggesting that short courses of prednisone do not prevent serious kidney disease in children with IgAV (Analysis 1.3).

Only the five studies comparing prednisone with placebo or supportive treatment for the prevention of persistent kidney disease in IgAV could be assessed in a summary of findings table

(Summary of findings 1). The overall certainty of the studies was considered moderate for the persistence of kidney disease at any time after treatment and for the number of children with continuing kidney disease at varying time points. With the removal of one study, which had inadequate allocation concealment and no blinding, the remaining three studies were graded as of high certainty at the three-month interval but only moderate at 12 months because of the small number of events and high loss to follow-up in Dudley 2013. The number developing severe kidney disease was graded as low in two studies as a result of a significant loss to follow-up in the largest included study and small numbers of events.

The remaining studies could not be included in summary of findings tables as they were single studies, or study data could not be included in meta-analyses.

Potential biases in the review process

A thorough search utilising Cochrane Kidney and Transplant's Specialised Register was completed in February 2023.

The Specialised Register includes published studies and conference abstracts with no restriction on language. The omission of eligible studies was therefore minimised. However, 30% of study reports in the Specialised Register have been identified by handsearching of conference proceedings so it remains possible that further studies of therapy to prevent or treat serious kidney disease in IgAV will be identified as conference proceedings from different congresses are searched.

Five (25%) of the included studies were only available in the abstract form, thus limiting information on study methods and outcomes. Four of the studies were published prior to 2000 before the CONSORT checklist (first published in 1996) would influence trial methodology and reporting. Incomplete reporting of these studies may result in incomplete information being included in this systematic review.

Two authors independently undertook all the steps of this review, thereby minimising risks of errors in determining study eligibility, data extraction and risk of bias assessment and data synthesis.

Agreements and disagreements with other studies or reviews

Three earlier systematic reviews assessed the effects of corticosteroid therapy to prevent or alter the course of kidney disease in IgAV (Weiss 2007; Wyatt 2001; Zaffanello 2007). All three included data from RCTs and observational studies. Two reviews determined that it remained unclear whether corticosteroid therapy prevented or altered the course of IgAV-associated kidney disease (Wyatt 2001; Zaffanello 2007). The third review concluded that corticosteroids decreased the likelihood of developing persistent kidney disease but did not prevent kidney disease (Weiss 2007). However, in a further study (Dudley 2013), there was probably no benefit of corticosteroid therapy to prevent the development or persistence of kidney disease in IgAV, so corticosteroids are not recommended in children with IgAV to prevent kidney involvement.

We identified no randomised studies evaluating the use of corticosteroids in treating established kidney involvement of any severity in IgAV though most clinicians will use them

and report some benefit. Despite the lack of supporting data, recent recommendations for the treatment of nephritis in IgAV recommend that corticosteroids should be used without delay in children presenting with moderate or severe nephritis (Delbert 2021; KDIGO 2021; Ozen 2019).

An earlier systematic review also evaluated immunosuppressive and other therapies in IgAV nephritis (Zaffanello 2007). It concluded, based on observational studies, that cyclophosphamide was of value in treating IgAV-associated kidney disease. Studies evaluating cyclophosphamide compared with corticosteroids or supportive treatment in this review may not show any benefit of cyclophosphamide (CESAR 2010; Tarshish 2004), with both interventions associated with improvements in kidney involvement in some participants. While earlier studies comparing cyclosporin with methylprednisolone and MMF with azathioprine were too small to establish whether or not these treatments may be effective (Fuentes 2010; Jauhola 2010), two larger studies found that tacrolimus (Zhang 2021a) or MMF (Geng 2021) may be as effective as IV cyclophosphamide in IgAV associated nephritis. These studies support recent recommendations for the use of MMF or tacrolimus in IgAV nephritis (Delbert 2021). This review did not identify any studies evaluating rituximab. However, rituximab has been reported to be effective in case reports of patients with IgAV nephritis (Delbert 2021).

Guidelines (Delbert 2021; KDIGO 2021; Ozen 2019) recommend treatment of nephritis in IgAV with ACEi or ARB based largely on observational studies. This review identified one small study (Xu 2009), which demonstrated that fosinopril may result in complete remission of proteinuria in children with IgAV.

AUTHORS' CONCLUSIONS

Implications for practice

Prevention of kidney disease in IgAV

- Corticosteroids probably do not prevent serious kidney disease in children with IgAV with or without minor kidney abnormalities at presentation (moderate certainty evidence).
- Antiplatelet agents may not have any benefit in preventing serious kidney disease, but the certainty of the evidence is low.
- Two studies of heparin came to different conclusions. Tian 2015 only partially supported the conclusions of He 2002 that heparin may reduce the risk for kidney disease in IgAV. The certainty of the evidence was low in both studies. The use of such a potentially harmful treatment cannot be justified when only a third of children with IgAV develop kidney disease, and most have spontaneous resolution of their kidney involvement.

Treatment of serious kidney disease in IgAV

- Cyclophosphamide treatment compared with supportive therapy or corticosteroids in children and adults with IgAV and severe kidney disease may not increase the number of participants with improvement in outcomes.
- It remains unclear whether cyclosporin is more effective than methylprednisolone in children with IgAV and severe kidney disease; further studies with longer follow-up are required.

- Tacrolimus and MMF may be as effective as cyclophosphamide in the treatment of severe IgAV-associated kidney disease at two years, but longer follow-up is required to confirm efficacy.
- The addition of plasmapheresis to cyclophosphamide and methylprednisolone may not improve outcomes.
- No studies addressing the management of kidney disease in participants with recurrent episodes of IgAV were identified.

Implications for research

Prevention of serious kidney disease in IgAV

- While short-term prednisone may not prevent the development of IgAV nephritis, it remains possible that prednisone therapy has a role in children with risk factors for developing kidney disease, including older age (Shin 2006), severe abdominal pain (Ronkainen 2006a; Shin 2006), persistent (Ronkainen 2006a; Shin 2006) or recurrent purpura (Shin 2006) so a further RCT in this group of children may be warranted. However, recruitment to a placebo-controlled RCT may be difficult since Ronkainen 2006a has demonstrated that short-course prednisone significantly reduces the severity and duration of abdominal pain in children with IgAV, making it unlikely that clinicians would be prepared to withhold prednisone from children with severe IgAV-associated abdominal pain.

Treatment of serious kidney disease in IgAV

- Further adequately powered and well-designed RCTs with at least five-year follow-up periods are needed in children with IgAV associated with kidney involvement, including nephritic syndrome and/or nephrotic syndrome.
- The study treatment investigated should be adapted according to the severity of kidney disease and should be started early after the onset of IgAV (Davin 2011; Delbert 2021).
- Treatment regimens requiring further evaluation in RCTs include the following in combination with corticosteroids:
 - Calcineurin inhibitors
 - MMF
 - Rituximab and other biological agents.

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REFERENCES

References to studies included in this review

CESAR 2010 {published data only}

* Pillebout E, Alberti C, Guillevin L, Ouslimani A, Thervet E, CESAR study group. Addition of cyclophosphamide to steroids provides no benefit compared with steroids alone in treating adult patients with severe Henoch Schonlein Purpura. *Kidney International* 2010;**78**(5):495-502. [MEDLINE: 20505654]

Pillebout E, Alberti C, Guillevin L, Thervet E, CESAR Study Group. Prospective randomized study of cyclophosphamide and steroid versus steroid alone therapy in adult patients with severe Henoch Schonlein Purpura (HSP) [abstract no: SA275]. In: World Congress of Nephrology; 2009 May 22-26; Milan, Italy. 2009. [CENTRAL: CN-01657237]

Pillebout E, Alberti C, Guillevin L, Thervet E. Prospective randomized study of cyclophosphamide and steroid versus steroid alone therapy in adults patients with severe Henoch Schonlein Purpura (HSP)-CESAR study group [abstract no: TH-PO175]. *Journal of the American Society of Nephrology* 2009;**20**(Abstract Suppl):150A.

Du 2016 {published data only}

Du Y, Zhang Z, Hou L, Qin K, Wang X, Wu Y. Comparison of Leflunomide and Mycophenolate mofetil in children with Henoch-Schonlein nephritis [abstract no: PO-163]. *Pediatric Nephrology* 2016;**31**(10):1817. [EMBASE: 612479523]

Dudley 2013 {published and unpublished data}**ISRCTN71445600**

Dudley J, Smith G, Llewellyn-Edwards A, Tizard EJ. Randomised placebo controlled trial to assess the role of early prednisolone on the development and progression of Henoch-Schonlein Purpura Nephritis [abstract no: 270 (FC)]. *Pediatric Nephrology* 2007;**22**(9):1457. [CENTRAL: CN-00691227]

* Dudley J, Smith G, Llewellyn-Edwards A, Bayliss K, Pike K, Tizard J. Randomised, double-blind, placebo-controlled trial to determine whether steroids reduce the incidence and severity of nephropathy in Henoch-Schonlein Purpura (HSP). *Archives of Disease in Childhood* 2013;**98**(10):756-63. [MEDLINE: 23845696]

Dudley J, Smith GC, Llewellyn-Edwards A, Tizard J. Randomised placebo controlled trial to assess the role of early prednisolone on the development and progression of Henoch-Schonlein Purpura nephritis [abstract no: SA-FC056]. *Journal of the American Society of Nephrology* 2007;**18**(Abstracts):47A. [CENTRAL: CN-01657243]

Fuentes 2010 {published data only}

Endo JA, Enciso S, Fuentes-Velasco Y, Velasquez-Jones L, Hernandez AM, Valverde S, et al. Efficacy of mycophenolate mofetil and prednisone compared to azathioprine and prednisone treatment in children with Henoch-Schonlein purpura nephritis [abstract no: FP-S26-1]. *Pediatric Nephrology* 2016;**31**(10):1753-4. [EMBASE: 612479687]

Fuentes Y, Valverde S, Hernandez AM, Ortiz LM, Medeiros M. Henoch-Schonlein Purpura nephritis: comparison of two therapies (azathioprine vs mycophenolate) and its effect on the urinary excretion of MCP-1 [abstract no: FR-PO1921].

Journal of the American Society of Nephrology 2011;**22**(Abstract Suppl):558A.

Fuentes Y, Valverde S, Velasquez-Jones L, Romero B, Ramón G, Medeiros M. Comparison of azathioprine vs mofetil mycophenolate for Henoch-Schonlein nephritis treatment [abstract no: 39]. *Pediatric Nephrology* 2010;**25**(9):1802. [CENTRAL: CN-01657240]

Geng 2021 {published data only}

Geng HY, Chen CY, Li HR, Tu J, Du PW, Xia H. Efficacy and safety of mycophenolate mofetil versus cyclophosphamide in the treatment of Henoch-Schonlein purpura nephritis with nephrotic-range proteinuria in children: a prospective randomized controlled trial. *Zhongguo Dangdai Erke Zazhi* 2021;**23**(4):338-42. [MEDLINE: 33840404]

He 2002 {published data only}

He YY, Hongmei S, Lihua S, Min W. Preventive value of heparin on the occurrence of nephropathy in Henoch-Schoenlein Purpura: a randomized controlled clinical trial [abstract no: 0364]. In: 23rd International Congress of Pediatrics; 2001 Sep 9-14; Beijing, China. 2001. [CENTRAL: CN-00462028]

* He YY, Pan W, Song HM, Wei M, Zhu CY. Preventive effect of heparin on the development of nephropathy in Henoch-Schoenlein purpura-a randomized controlled clinical trial. *Chinese Journal of Pediatrics* 2002;**40**(2):99-102. [CENTRAL: CN-01657239]

Huber 2004 {published data only}**85109383**

* Huber AM, King J, McLaine P, Klassen T, Pothos M. A randomized, placebo-controlled trial of prednisone in early Henoch Schonlein purpura [ISRCTN85109383]. *BMC Medicine* 2004;**2**:7. [MEDLINE: 15059282]

Islek 1999 {published data only}

Islek I, Sezer T, Totan M, Cakir M, Kucukoduk S. The effect of profilactic prednisolon therapy on renal involvement in Henoch Schonlein vasculitis [abstract]. In: XXXVI Congress of the European Renal Association European Dialysis & Transplant Association; 1999 Sep 5-8; Madrid, Spain. 1999:103. [CENTRAL: CN-00484464]

Jauhola 2010 {published and unpublished data}

Jauhola O, Ronkainen J, Ala-Houhala M, Autio-Harmainen H, Holtta T, Jahnukainen T, et al. Cyclosporine A (CyA) versus MP pulses (MP) in severe Henoch-Schoenlein nephritis (HSN): outcome after 2 year follow-up [abstract no: O05]. *Pediatric Nephrology* 2008;**23**(9):1584. [CENTRAL: CN-01657238]

* Jauhola O, Ronkainen J, Autio-Harmainen H, Koskimies O, Ala-Houhala M, Arikoski P, et al. Cyclosporine A vs. methylprednisolone for Henoch-Schonlein nephritis: a randomized trial [Erratum in: *Pediatr Nephrol*. 2011 Dec;26(12):2263-4]. *Pediatric Nephrology* 2011;**26**(12):2159-66. [MEDLINE: 21626222]

Jauhola O, Ronkainen J, Koskimies O, Ala-Houhala M, Arikoski P, Holtta T, et al. Clinical course of extrarenal symptoms in

Henoch-Schonlein purpura: a 6-month prospective study. *Archives of Disease in Childhood* 2010;**95**(11):871-6. [MEDLINE: 20371584]

Jauhola O, Ronkainen J, Koskimies O, Ala-Houhala M, Arikoski P, Holtta T, et al. Renal manifestations of Henoch-Schonlein purpura in a 6-month prospective study of 223 children. *Archives of Disease in Childhood* 2010;**95**(11):877-82. [MEDLINE: 20852275]

Ronkainen J, Ala-Houhala M, Antkainen M, Jahnukainen T, Koskimies O, Merenmies J, et al. Cyclosporine A (CyA) versus MP pulses (MP) in the treatment of severe Henoch-Schonlein Nephritis (HSN) [abstract no: COD. PP38]. *Pediatric Nephrology* 2006;**21**(10):1531. [CENTRAL: CN-00689778]

Liu 2019e {published data only}

Liu N, Ma ZZ, Yan HF, Li Q, Lyu XQ, Kang WL, et al. Clinical effect of double filtration plasmapheresis combined with glucocorticoid and immunosuppressant in treatment of children with severe Henoch-Schonlein purpura nephritis. *Zhongguo Dangdai Erke Zazhi* 2019;**21**(10):955-9. [MEDLINE: 31642426]

Mollica 1992 {published data only}

Mollica F, Li Volti S, Garozzo R, Russo G. Effectiveness of early prednisone treatment in preventing the development of nephropathy in anaphylactoid purpura. *European Journal of Pediatrics* 1992;**151**(2):140-4. [MEDLINE: 1343079]

Peratoner 1990 {published data only}

Peratoner L, Longo F, Lepore L, Freschi P. Prophylaxis and therapy of glomerulonephritis in the course of anaphylactoid purpura. The results of a polycentric clinical trial. *Acta Paediatrica Scandinavica* 1990;**79**(10):976-7. [MEDLINE: 2264475]

Ronkainen 2006a {published and unpublished data}

Jauhola O, Ronkainen J, Koskimies O, Ala-Houhala M, Arikoski P, Holtta T, et al. Clinical course of extrarenal symptoms in Henoch-Schonlein purpura: a 6-month prospective study. *Archives of Disease in Childhood* 2010;**95**(11):871-6. [MEDLINE: 20371584]

Jauhola O, Ronkainen J, Koskimies O, Ala-Houhala M, Arikoski P, Holtta T, et al. Outcome of Henoch-Schoenlein purpura 8 years after the treatment with placebo or prednisone at disease onset [abstract no: PS2-FRI-493]. *Pediatric Nephrology* 2011;**26**(9):1678-9.

Jauhola O, Ronkainen J, Koskimies O, Ala-Houhala M, Arikoski P, Holtta T, et al. Outcome of Henoch-Schonlein purpura 8 years after treatment with a placebo or prednisone at disease onset. *Pediatric Nephrology* 2012;**27**(6):933-9. [MEDLINE: 22311342]

Jauhola O, Ronkainen J, Koskimies O, Ala-Houhala M, Arikoski P, Holtta T, et al. Renal manifestations of Henoch-Schonlein purpura in a 6-month prospective study of 223 children. *Archives of Disease in Childhood* 2010;**95**(11):877-82. [MEDLINE: 20852275]

* Ronkainen J, Koskimies O, Ala-Houhala M, Antkainen M, Merenmies J, Rajantie J, et al. Early prednisone therapy in

Henoch-Schonlein purpura: a randomized double-blind, placebo-controlled trial. *Journal of Pediatrics* 2006;**149**(2):241-7. [MEDLINE: 16887443]

Tarshish 2004 {published data only}

Tarshish P, Bernstein J, Edelmann C. Henoch-Schonlein purpura (HSP) nephritis: efficacy of cytoxan (C) & natural history. Report of the ISKDC [abstract]. *Pediatric Research* 2001;**49**(4):421A. [CENTRAL: CN-00486109]

Tarshish P, Bernstein J, Edelmann CM Jr. Henoch-Schonlein purpura nephritis: course of disease and efficacy of cyclophosphamide. *Pediatric Nephrology* 2004;**19**(1):51-6. [MEDLINE: 14634864]

Tian 2015 {published data only}

Tian M, Liu C. Heparin calcium treated Henoch-Schonlein purpura nephritis in children through inhibiting hyperfibrinolysis. *Renal Failure* 2015;**37**(7):1100-4. [MEDLINE: 26133741]

Wu 2014c {published data only}

Wu SH, Liao PY, Chen XQ, Yin PL, Dong L. Add-on therapy with montelukast in the treatment of Henoch-Schonlein purpura. *Pediatrics International* 2014;**56**(3):315-22. [MEDLINE: 24299021]

Xu 2009 {published data only}

Xu J, Cong H, Sheng Y. Efficacy of fosinopril on proteinuria in children with Henoch-Schonlein purpura nephritis. *Zhongguo Dangdai Erke Zazhi* 2009;**11**(3):229-30. [MEDLINE: 19292967]

Yoshimoto 1987a {published data only}

Yoshimoto M, Ito H, Shindo S, Yamashita F. Evaluation of the preventive role of dipyridamole and aspirin against renal complication in Schonlein-Henoch purpura [abstract]. *Pediatric Nephrology* 1987;**1**:C47. [CENTRAL: CN-00445875]

Yoshimoto 1987b {published data only}

Yoshimoto M, Ito H, Shindo S, Yamashita F. Evaluation of the preventive role of dipyridamole and aspirin against renal complication in Schonlein-Henoch purpura [abstract]. *Pediatric Nephrology* 1987;**1**:C47. [CENTRAL: CN-00445875]

Zhang 2021a {published data only}

Zhang H, Li X, Xu H, Ran F, Zhao G. Effect and safety evaluation of tacrolimus and tripterygium glycosides combined therapy in treatment of Henoch-Schonlein purpura nephritis. *International Journal of Urology* 2021;**28**(11):1157-63. [MEDLINE: 34378238]

References to studies excluded from this review

Chen 2010 {published data only}

Chen T, Guo ZP, Zhang YH, Gao Y. Effect of MORA bioresonance therapy in the treatment of Henoch-Schonlein purpura and influence on serum antioxidant enzymes. *Journal of Clinical Dermatology* 2010;**39**(5):283-5. [EMBASE: 359206443]

ChiCTR2100050353 {published data only}

ChiCTR2100050353. Efficacy and side effects of different doses of Tripterygium Wilfordii granules combined with traditional Chinese medicine in the treatment of children with recurrent

Henoch-Schonlein Purpura. <https://trialsearch.who.int/Trial2.aspx?TrialID=ChiCTR2100050353> (first received 26 August 2021). [CENTRAL: CN-02450623]

ChiCTR-INR-17013850 {published data only}

ChiCTR-INR-17013850. Effects of laminar flow bed on Henoch-Schonlein Purpura in children. www.chictr.org.cn/showproj.aspx?proj=20982 (first received 11 December 2017).

Cui 2020 {published data only}

Cui Y, Li J, Zhang J, Han T, Gao L, Wang H, et al. Efficacy of Yupingfeng San for assisting in treatment of children with Henoch-Schonlein purpura and its effects on oxidative stress and the immune system. *International Journal of Clinical & Experimental Medicine* 2020;**13**(1):293-9. [EMBASE: 2003657269]

Ding 1995 {published data only}

Ding HL, Cheng JZ. Clinical trial of potenlin in allergic purpura of adults. *Journal of Clinical Dermatology* 1995;**24**(5):300-2. [CENTRAL: CN-00550728]

Ding 2014 {published data only}

Ding D, Yan H, Zhen X. Effects of Chinese herbs in children with Henoch-Schonlein purpura nephritis: a randomized controlled trial. *Journal of Traditional Chinese Medicine* 2014;**34**(1):15-22. [MEDLINE: 25102685]

Ding 2019 {published data only}

Ding Y, Zhang X, Ren X, Zhai W, He L, Liu J, et al. Traditional Chinese medicine versus regular therapy in Henoch-Schonlein purpura nephritis in children: study protocol for a randomized controlled trial. *Trials [Electronic Resource]* 2019;**20**(1):538. [MEDLINE: 31464626]

Ding 2021 {published data only}

Ding Y, Li XJ, Zhang X, Han SS, Xu SS, Dai YL, et al. Clinical study on different dosage of Tripterygium wilfordii Granules combined with Purpura No. I prescription in treating children with recurrent Henoch-Schonlein purpura. *Chinese Traditional & Herbal Drugs* 2021;**52**(24):7569-76. [EMBASE: 2016021484]

Erdogan 1999 {published data only}

Erdogan O, Oner A, Aydin A, Isimer A, Demircin G, Bulbul M, et al. Lack of efficacy of vitamin e supplementation in decreasing the oxidative damage Henoch-Schonlein Purpura: a randomised controlled trial [abstract]. *Nephrology Dialysis Transplantation* 1999;**14**(9):A46. [CENTRAL: CN-00483878]

Fu 2021 {published data only}

Fu Q, Shi MF, Chen Y. Clinical effect of alfalcidol in children with Henoch-Schonlein purpura: a prospective randomized controlled trial. *Zhongguo Dangdai Erke Zazhi* 2021;**23**(8):797-801. [MEDLINE: 34511168]

Fukui 1989 {published data only}

Fukui H, Kamitsuji H, Nagao T, Yamada K, Akatsuka J, Inagaki M, et al. Clinical evaluation of a pasteurized factor XIII concentrate administration in Henoch-Schönlein purpura. Japanese Pediatric Group. *Thrombosis Research* 1989;**56**(6):667-75. [CENTRAL: CN-00067042]

Gao 2018a {published data only}

Gao X, Miao R, Tao Y, Chen X, Wan C, Jia R. Effect of montmorillonite powder on intestinal mucosal barrier in children with abdominal Henoch-Schonlein Purpura: a randomized controlled study. *Medicine* 2018;**97**(39):e12577. [MEDLINE: 30278566]

Hui-Lan 2001 {published data only}

Hui-Lan Y. Treatment of pediatric purpuric nephritis with Tripterygium wilfordii polyglucoside and radix salviac miltiorrhizae [abstract no: P85]. *Pediatric Nephrology* 2001;**16**(8):C79. [CENTRAL: CN-00445801]

Hui-Lan YY. Treatment of purpuric nephritis in children with Tripterygium wilfordii polyglucoside and Radix salviae miltiorrhizae [abstract no: P79]. *Pediatric Nephrology* 2001;**16**(8):C78. [CENTRAL: CN-00445799]

Jia 2001 {published data only}

Jia WC, Zhang YC, Wei AP. Immune functional changes in patients of acute Henoch-Schonlein purpura and regulatory effect of integrated Traditional Chinese and Western medicine on it. *Zhongguo Zhongxiyi Jiehe Zazhi [Chinese Journal of Integrated Traditional & Western Medicine]* 2001;**21**(8):585-7. [MEDLINE: 12575571]

Jiang 2003 {published data only}

Jiang Y. Therapeutic effect of low-dose hepacarin in the childhood anaphylactoid purpura. *Journal of Pediatric Pharmacy* 2003;**9**(4):58-9. [CENTRAL: CN-00583752]

Jin 2003 {published data only}

Jin ZD, Wang SC, Sun YQ. Effect of Danshao granule on serum superoxide dismutase activity and malonyldialdehyde content in children with Henoch-Schonlein purpura nephritis. *Zhongguo Zhongxiyi Jiehe Zazhi [Chinese Journal of Integrated Traditional & Western Medicine]* 2003;**23**(12):905-7. [MEDLINE: 14714360]

Jin 2021 {published data only}

Jin Y, Wang Y, Wang S, Zhao Q, Zhang D, Feng X. The efficacy of Tripterygium glycosides combined with LMWH in treatment of HSPN in children. *Evidence-based Complementary & Alternative Medicine: eCAM* 2021:7223613. [MEDLINE: 34721642]

Kim 1987 {published data only}

Kim PK, Jeong HJ, Kim KS, Lee JK, Lee JS, Choi IJ. Rifampin therapy in Henoch-Schonlein Purpura nephritis accompanied by the nephrotic syndrome [abstract]. In: 10th International Congress of Nephrology; 1987 Jul 26-31; London, UK. 1987:18. [CENTRAL: CN-01657236]

Li 2022 {published data only}

Li C, Wang ZB. Clinical application of compound Glycyrrhizin tablets in the treatment of patients with simplex Henoch-Schonlein Purpura and its effect on immune function. *Pakistan Journal of Medical Sciences* 2022;**38**(1):271-5. [MEDLINE: 35035438]

Liu 2004a {published data only}

Liu CS, Gong ZZ, Xiang SH. Clinical study on integrative Chinese and Western medicine in treating allergic purpura and

preventing renal impairment. *Zhongguo Zhongxiyi Jiehe Zazhi [Chinese Journal of Integrated Traditional & Western Medicine]* 2004;**24**(8):701-3. [MEDLINE: 15366592]

Liu 2008 {published data only}

Liu QC, Wu WH, Wu DY, Feng XW, Ma YH, Li JY, et al. Clinical observation on the treatment of childhood refractory idiopathic thrombocytopenic purpura with Dihuang Zhixue capsule. *Chinese Journal of Integrative Medicine* 2008;**14**(2):132-6. [MEDLINE: 18679605]

NCT02939573 {published data only}

Micheletti R, Pagnoux C. A randomized multicenter study for isolated skin vasculitis (ARAMIS). clinicaltrials.gov/show/NCT02939573 (first received 20 October 2016).

NCT03647852 {published data only}

Sun L, Huang W. Clinical study on strategy for refractory Henoch-Schönlein Purpura. clinicaltrials.gov/show/NCT03647852 (first received 27 August 2018).

NCT04008316 {published data only}

Pillebout E. Efficacy of colchicine to prevent skin relapses in adult's IgA vasculitis (COLCHIVAS). clinicaltrials.gov/show/NCT04008316 (first received 5 July 2019).

Shao 2021 {published data only}

Shao KF, Guan FJ, Dong C. Clinical effect and mechanism of total glucosides of paeony in the adjuvant therapy for children with Henoch-Schonlein purpura nephritis: a prospective randomized controlled study. *Zhongguo Dangdai Erke Zazhi [Chinese Journal of Contemporary Pediatrics]* 2021;**23**(1):49-54. [MEDLINE: 33476537]

Wang 2011 {published data only}

Wang ZW, Lu Y, Zhen XF. Effect of herbs in early intervention of children with Henoch-Schonlein purpura nephritis. *Zhongguo Zhongxiyi Jiehe Zazhi [Chinese Journal of Integrated Traditional & Western Medicine]* 2011;**31**(4):504-7. [MEDLINE: 21608223]

Wang 2021a {published data only}

Wang L, Yin C, Zhang M, Mao H, Hao H, Hu X, et al. A randomized controlled trial on the effect of dietary guidance on the treatment of Henoch-Schonlein purpura in children. *Journal of Investigative Medicine* 2021;**69**(8):1464-72. [MEDLINE: 34362779]

Wu 2013b {published data only}

Wu L, Mao J, Jin X, Fu H, Shen H, Wang J, et al. Efficacy of triptolide for children with moderately severe Henoch-Schonlein purpura nephritis presenting with nephrotic range proteinuria: a prospective and controlled study in China. *BioMed Research International* 2013;**2013**:292865. [MEDLINE: 24455682]

Xia 2016 {published data only}

Xia LP, Chen X, Jiang Y. Clinical effect of gamma globulin pulse therapy for abdominal Henoch-Schonlein purpura in children. *Zhongguo Dangdai Erke Zazhi* 2016;**18**(10):988-90. [MEDLINE: 27751217]

Xie 2009 {published data only}

Xie XL, Kou SR, Xu YH, Li CY. Effect of composite salvia injection on platelet parameters in children with anaphylactoid purpura. *Chinese Journal of Integrative Medicine* 2009;**15**(2):149-51. [MEDLINE: 19407955]

Xiong 2019 {published data only}

Xiong LJ, Shang LH, Ou XQ, Li Y, Xie XL. Clinical effect of alanyl-glutamine-enriched nutritional support in the treatment of children with abdominal Henoch-Schonlein purpura. *Zhongguo Dangdai Erke Zazhi* 2019;**21**(2):168-71. [MEDLINE: 30782281]

Yang 2010 {published data only}

Yang HJ, Zhuang KS, Bu TW, Mu LQ. Controlled study on therapeutic effect of vessel pricking therapy and western medication for treatment of Henoch-Schonlein purpura nephritis. *Zhongguo Zhenjiu* 2010;**30**(6):449-52. [MEDLINE: 20578379]

Yi 2007 {published data only}

Yi H. Effect of Shenyankangfu tablet on urinary IL-6 and its therapeutic effect in children with Henoch-Schonlein purpura nephritis. *Zhongguo Dangdai Erke Zazhi* 2007;**9**(2):153-4. [MEDLINE: 17448314]

Zhang 1984 {published data only}

Zhang WZ, Zhou MJ. Clinical observation of anisodamine on treatment of anaphylactoid purpura by infused intravenously. *Chinese Journal of Practical Internal Medicine* 1984;**4**(1):26. [CENTRAL: CN-00583439]

Zhang 2019b {published data only}

Zhang X, Jiang D, Liu R, Fang G, Guo Z. Therapeutic effect of Vitamin D on Henoch-Schonlein purpura in children and its influence on immune mechanism. *Pharmaceutical Care & Research* 2019;**19**(3):192-5. [EMBASE: 628873404]

Zhao 2009 {published data only}

Zhao H, Dou ZY, Li YQ. Preventive effect of integrative medical therapy on children Henoch-Schonlein purpura with renal impairment. *Zhongguo Zhongxiyi Jiehe Zazhi [Chinese Journal of Integrated Traditional & Western Medicine]* 2009;**29**(4):351-3. [MEDLINE: 19526764]

Zhu 2016 {published data only}

Zhu TF, Chu ZF, Li JH. Effect of tripterygium glycosides and Danshen injection on blood coagulation mechanism in children with allergic purpura nephritis. *Zhongguo Zhongyao Zazhi [China Journal of Chinese Materia Medica]* 2016;**41**(11):2162-7. [MEDLINE: 28901116]

References to studies awaiting assessment

NCT00301613 {published data only}

Liu ZH. Mycophenolate mofetil (MMF) versus intravenous CTX pulses in the treatment of adult severe HSPN. www.clinicaltrials.gov/ct2/show/NCT00301613 (first received 13 March 2006).

References to ongoing studies

NCT02532777 {published data only}

Zhang A. The research of standard diagnosis and treatment for HSPN in children [The research of standard diagnosis and treatment for Henoch-Schonlein Purpura nephritis in children]. [clinicalTrials.gov/show/NCT02532777](https://clinicaltrials.gov/show/NCT02532777) (first received 26 August 2015).

NCT02532790 {published data only}

Zhang A. The research of standard diagnosis and treatment for HSPN with mild proteinuria in children [The research of standard diagnosis and treatment for Henoch-Schonlein Purpura nephritis with mild proteinuria in children]. [clinicalTrials.gov/show/NCT02532790](https://clinicaltrials.gov/show/NCT02532790) (first received 26 August 2015).

NCT02540720 {published data only}

Zhang A. The research of standard diagnosis and treatment for severe HSP in children [The research of standard diagnosis and treatment for severe Henoch-Schonlein Purpura in children]. [ClinicalTrials.gov/show/NCT02540720](https://clinicaltrials.gov/show/NCT02540720) (first received 4 September 2015).

Additional references

Bergstein 1998

Bergstein J, Leiser J, Andreoli SP. Response of crescentic Henoch-Schoenlein purpura nephritis to corticosteroid and azathioprine therapy. *Clinical Nephrology* 1998;**49**(1):9-14. [MEDLINE: 9491279]

Davin 2011

Davin JC. Henoch-Schonlein purpura nephritis: pathophysiology, treatment and future strategy. *Clinical Journal of The American Society of Nephrology: CJASN* 2011;**6**(3):679-89. [MEDLINE: 21393485]

Delbert 2021

Delbert JD, Parmentier C, Herbez Rea C, Mouche A, Ulinks T. Management of IgA vasculitis with nephritis. *Pediatric Drugs* 2021;**23**(5):425-35. [MEDLINE: 34396492]

Du 2012

Du Y, Hou L, Zhao C, Han W, Wu Y. Treatment of children with Henoch-Schonlein purpura nephritis with mycophenolate mofetil. *Pediatric Nephrology* 2012;**27**(5):765-71. [MEDLINE: 22081165]

Flynn 2001

Flynn JT, Smoyer WE, Bunchman TE, Kershaw DB, Sedman AB. Treatment of Henoch-Schonlein Purpura glomerulonephritis in children with high-dose corticosteroids plus oral cyclophosphamide. *American Journal of Nephrology* 2001;**21**(2):128-33. [MEDLINE: 11359020]

Foster 2000

Foster BJ, Bernard C, Drummond KN, Sharma AK. Effective therapy for severe Henoch-Schonlein purpura nephritis with prednisone and azathioprine: a clinical and histopathologic

study. *Journal of Pediatrics* 2000;**136**(3):370-5. [MEDLINE: 10700695]

Goldstein 1992

Goldstein AR, White RH, Akuse R, Chantler C. Long-term follow-up of childhood Henoch Schonlein nephritis. *Lancet* 1992;**339**(8788):280-2. [MEDLINE: 1346291]

GRADE 2008

Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ* 2008;**336**(7650):924-6. [MEDLINE: 18436948]

GRADE 2011

Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *Journal of Clinical Epidemiology* 2011;**64**(4):383-94. [MEDLINE: 21195583]

Higgins 2003

Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003;**327**(7414):557-60. [MEDLINE: 12958120]

Higgins 2022

Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al. Cochrane Handbook for Systematic Reviews of Interventions version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook.

Iijima 1998

Iijima R, Ito-Kariya S, Nakamura H, Yoshikawa N. Multiple combined therapy for severe Henoch Schonlein nephritis in children. *Pediatric Nephrology* 1998;**12**(3):244-8. [MEDLINE: 9630047]

Jennette 2013

Jennette JC, Falk RJ, Bacon PA, Basu N, Cid MC, Ferrario F, et al. 2012 revised International Chapel Hill Consensus Conference Nomenclature of Vasculitides. *Arthritis & Rheumatism* 2013;**65**(1):1-11. [MEDLINE: 23045170]

Kawasaki 2004

Kawasaki Y, Suzuki J, Suzuki H. Efficacy of methylprednisolone and urokinase pulse therapy combined with or without cyclophosphamide in severe Henoch-Schoenlein nephritis: a clinical and histopathological study. *Nephrology Dialysis Transplantation* 2004;**19**(4):858-64. [MEDLINE: 15031341]

KDIGO 2021

Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group. KDIGO 2021 Clinical practice guideline for the management of glomerular diseases. *Kidney International* 2021;**100**(4S):S115-27. [MEDLINE: 34556256]

Narchi 2005

Narchi H. Risk of long term renal impairment and duration of follow up recommended for Henoch-Schonlein purpura with normal or minimal urinary findings: a systematic review.

Archives of Disease in Childhood 2005;**90**(9):916-20. [MEDLINE: 15871983]

Niaudet 1998

Niaudet P, Habib R. Methylprednisolone pulse therapy in the treatment of severe forms of Schonlein-Henoch purpura nephritis. *Pediatric Nephrology* 1998;**12**(3):238-43. [MEDLINE: 9630046]

Ozen 2019

Ozen S, Marks SD, Brogan P, Groot N, de Graeff N, Avcin T, et al. European consensus-based recommendations for diagnosis and treatment of immunoglobulin A vasculitis-the SHARE initiative. *Rheumatology* 2019;**58**(9):1607-16. [MEDLINE: 30879080]

Piram 2017

Piram M, Maldini C, Biscardi S, De Suremain N, Orzechowski C, Georget E, et al. Incidence of IgA vasculitis in children estimated by four-source capture-recapture analysis: a population-based study. *Rheumatology* 2017;**56**(8):1358-66. [MEDLINE: 28444335]

Ronkainen 2003

Ronkainen J, Ala-Houhala M, Huttunen NP, Jahnukainen T, Koskimies O, Ormala T, et al. Outcome of Henoch-Schoenlein nephritis with nephrotic-range proteinuria. *Clinical Nephrology* 2003;**60**(2):80-4. [MEDLINE: 12940608]

Ruperto 2010

Ruperto N, Ozen S, Pistorio A, Dolezalova P, Brogan P, Cabral DA, et al. EULAR/PRINTO/PRES criteria for Henoch-Schonlein purpura, childhood polyarteritis nodosa, childhood Wegener granulomatosis and childhood Takayasu arteritis: Ankara 2008. Part I: overall methodology and clinical characterisation [Erratum in: *Ann Rheum Dis*. 2011 Feb;**70**(2):397 Note: Paut, Isabelle Kone [corrected to Kone-Paut, Isabella]]. *Annals of the Rheumatic Diseases* 2010;**69**(5):790-7. [MEDLINE: 20388738]

Saulsbury 1999

Saulsbury FT. Henoch-Schonlein Purpura in children. Report of 100 patients and review of the literature. *Medicine* 1999;**78**(6):395-409. [MEDLINE: 10575422]

Schulz 1995

Schulz KF, Chalmers I, Hayes RJ, Altman DG. Empirical evidence of bias. Dimensions of methodological quality associated with estimates of treatment effects in controlled trials. *JAMA* 1995;**273**(5):408-12. [MEDLINE: 7823387]

Schunemann 2022a

Schunemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022. www.training.cochrane.org/handbook.

Schunemann 2022b

Schunemann HJ, Vist GE, Higgins JP, Santesso N, Deeks JJ, Glasziou P, et al. Chapter 15: Interpreting results and

drawing conclusions. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022. Available from www.training.cochrane.org/handbook.

Shenoy 2007

Shenoy M, Ognjanovic MV, Coulthard MG. Treating severe Henoch-Schonlein and IgA nephritis with plasmapheresis alone. *Pediatric Nephrology* 2007;**22**(8):1167-71. [MEDLINE: 17530298]

Shin 2006

Shin JI, Park JM, Shin YH, Hwang DH, Kim JH, Lee JS. Predictive factors for nephritis, relapse, and significant proteinuria in childhood Henoch-Schonlein purpura. *Scandinavian Journal of Rheumatology* 2006;**35**(1):56-60. [MEDLINE: 16467044]

Weiss 2007

Weiss PF, Feinstein JA, Luan X, Burnham JM, Feudtner C. Effects of corticosteroid on Henoch-Schonlein purpura: a systematic review. *Pediatrics* 2007;**120**(5):1079-87. [MEDLINE: 17974746]

Wood 2008

Wood L, Egger M, Gluud LL, Schulz KF, Juni P, Altman DG, et al. Empirical evidence of bias in treatment effect estimates in controlled trials with different interventions and outcomes: meta-epidemiological study. *BMJ* 2008;**336**(7644):601-5. [MEDLINE: 18316340]

Wyatt 2001

Wyatt RJ, Hogg RJ. Evidence-based assessment of treatment options for children with IgA nephropathies. *Pediatric Nephrology* 2001;**16**(2):156-67. [MEDLINE: 11261686]

Zaffanello 2007

Zaffanello M, Brugnara M, Franchini M. Therapy for children with Henoch-Schonlein Purpura nephritis: a systematic review. *TheScientificWorldJournal* 2007;**7**:20-30. [MEDLINE: 17221139]

References to other published versions of this review

Chartapisak 2005

Chartapisak W, Opastirakul S, Hodson EM, Willis NS, Craig JC. Interventions for preventing and treating renal disease in Henoch-Schönlein Purpura (HSP). *Cochrane Database of Systematic Reviews* 2005, Issue 1. Art. No: CD005128. [DOI: [10.1002/14651858.CD005128](https://doi.org/10.1002/14651858.CD005128)]

Chartapisak 2008

Chartapisak W, Opastirakul O, Willis NS, Craig JC, Hodson EM. Prevention and treatment of renal disease in Henoch-Schönlein Purpura: a systematic review. *Archives of Disease in Childhood* 2008;**94**(2):132-7. [MEDLINE: 18701559]

Chartapisak 2009

Chartapisak W, Opastirakul S, Hodson EM, Willis NS, Craig JC. Interventions for preventing and treating kidney disease in Henoch-Schönlein Purpura (HSP). *Cochrane Database of Systematic Reviews* 2009, Issue 3. Art. No: CD005128. [DOI: [10.1002/14651858.CD005128.pub2](https://doi.org/10.1002/14651858.CD005128.pub2)]

Hahn 2015

Hahn D, Hodson EM, Willis NS, Craig JC. Interventions for preventing and treatment of kidney disease in Henoch-Schönlein Purpura (HSP). *Cochrane Database of*

Systematic Reviews 2015, Issue 8. Art. No: CD005128. [DOI: [10.1002/14651858.CD005128.pub3](https://doi.org/10.1002/14651858.CD005128.pub3)]

* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES
Characteristics of included studies [ordered by study ID]

CESAR 2010
Study characteristics

Methods	Study design - Parallel, open-label RCT Time frame - Study duration: September 2002 to September 2006 - Duration of follow-up: 12 months
Participants	Study characteristics - Country: France - Setting: multicentre (31 sites) - Inclusion criteria: aged 18 to 84 years; biopsy-proven diagnosis of IgAV associated with severe involvement of one organ (IgAV diagnosis was based on the Chapel Hill Nomenclature conference) criteria - Exclusion criteria: patients with other causes of purpura (thrombocytopenia, bacterial or other forms of vasculitis); hepatitis B or hepatitis C or HIV infection; receiving immunosuppressants or steroids in the prior two weeks; pregnant or breastfeeding women Baseline characteristics - Number (enrolled/analysed): intervention group (25/17); control group (29/19) - Mean age $\pm$ SD (years): intervention group (52.8 $\pm$ 18.5); control group (60.7 $\pm$ 11.0) - Sex (M/F): intervention group (17/8); control group (17/12)
Interventions	Intervention group - CPA (IV): 0.6 mg/m² on days 1 and 15 and at weeks 4, 8, 12 and 16 (6 doses). Maximum dose 1200 mg with dose reduced according to kidney dysfunction - Methylprednisolone (IV): 7.5 mg/kg/day for 3 days - Prednisone (oral): 1 mg/kg/day from day 4 for 1 week, then tapering to 0.4 mg/kg/day at end of the 1st month; 0.25 mg/kg/day at end of the 2nd month, and stopped at end of the 6th month Control group - Methylprednisolone (IV): 7.5 mg/kg/day for 3 days - Prednisone (oral): 1 mg/kg/day from day 4 for 1 week, then tapering to 0.4 mg/kg/day at end of the 1st month; 0.25 mg/kg/day at end of the 2nd month, and stopped at end of the 6th month
Outcomes	Primary outcome - Complete disease remission at month 6: BVAS score of 0, with no persisting or new clinical, biological or both vasculitis activity Secondary outcomes at 12 months - GFR: < 60 mL/min - Proteinuria: > 1 g/day

CESAR 2010 (Continued)

- BP: > 125/75
- Kidney function improvement: > 50%
- ESKD
- Adverse events: DM and infection
- Death

Notes	Additional information
	• Funding source: Supported by a research grant from the Département a la Recherche Clinique et au Développement, Assistance Publique–Hôpitaux de Paris, which also sponsored the study (PHRCAOM01034P011014)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central randomisation with computer generated random allocation sequence
Allocation concealment (selection bias)	Low risk	Central randomisation with computer generated random allocation sequence
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding and outcome is likely to be influenced by lack of blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding and outcome is likely to be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	All patients accounted for: loss to follow-up: intervention group (8) (death (1), lack of efficacy (1); patient choice (3); adverse event (1); not evaluated (2)); control group (10) (death (6); lack of efficacy (2); patient choice (1); not evaluated (1)
Selective reporting (reporting bias)	Low risk	Reported all outcomes
Other bias	Low risk	Funded by Département a la Recherche Clinique et au Développement

Du 2016
Study characteristics

Methods	Study design • Parallel RCT Time frame • Study duration: not reported • Duration of follow-up: 9 months
Participants	Study characteristics • Country: China

Du 2016 (Continued)

- Setting: single centre
- Inclusion criteria: children with IgAV nephritis, nephrotic syndrome and GFR > 60 mL/min/1.73 m² unresponsive to corticosteroids
- Exclusion criteria: not reported

Baseline characteristics

- Number: intervention group 1 (8); intervention group 2 (10)
- Mean age ± SD (years): not reported
- Sex (M/F): not reported

Interventions	Intervention group 1 • Leflunomide: dose not reported Intervention group 2 • MMF: dose not reported Co-interventions • ACEi • Tapering prednisone dose
Outcomes	Outcomes relevant to this review • 24-hour urine protein concentration • Serum albumin • SCr • Liver function test • Full blood count
Notes	Additional information • Abstract-only publication • Funding: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote: "were randomly given with LEF (LEF group; n=8) or MMF (MMF group; n=10)".
Allocation concealment (selection bias)	Unclear risk	Quote: "were randomly given with LEF (LEF group; n=8) or MMF (MMF group; n=10)".
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label study
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Outcomes laboratory based and unlikely to be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Insufficient information to permit judgement

Du 2016 (Continued)

Selective reporting (re-reporting bias)	High risk	Not all expected outcomes were reported
Other bias	High risk	No information on funding provided; no full-text publication identified

Dudley 2013

Study characteristics

Methods	Study design - Parallel RCT Time frame - Study duration: January 2001 to January 2005 - Duration of follow-up: 1 year
Participants	Study characteristics - Country: UK - Setting: multicentre (24 paediatric departments in secondary/tertiary hospital) - Inclusion criteria: < 18 years enrolled/randomised within 7 days of onset of IgAV rash and no kidney involvement at presentation - IgAV defined as palpable purpura with arthritis, kidney disease, gut involvement - Kidney disease defined as UPCr > 20 mg/mmol; positive dipstick for blood or protein; patients with haematuria, proteinuria or both at study entry were included - Exclusion criteria: already on steroids or immunosuppressives; pre-existing kidney disease; hypertension; immunodeficiency; systemic infection; contraindications to steroids; characteristic purpuric rash > 7 days Baseline characteristics - Number - Intervention group: 181 entered; 180 in study at baseline; 171 in study at 12 months; 150 attended 12-month visit; 123 analysed for primary endpoint (No specimen for UPC at 12 months in 27 of 150) - Control group: 171 entered; 170 in study at baseline; 165 in study at 12 months; 146 attended 12-month visit; 124 analysed for primary endpoint (No specimen for UPC at 12 months in 22 of 146) - Median age, range (years): intervention group (6.34, 1 to 15.7); control group (6.12, 0.5 to 13.9) - Sex (M/F): intervention group (93/88); control group (100/72)
Interventions	Intervention group - Prednisolone (oral): 2 mg/kg/day for 7 days (max dose 80 mg), then 1 mg/kg/day for 7 days Control group - Placebo: given in the same regimen Co-interventions - Additional treatment for gut or kidney disease as steroids (prednisolone (4); placebo (9)); ACEi or ARB (2 prednisolone (2); placebo (3)); other antihypertensives (prednisolone (3); placebo (0))
Outcomes	Primary outcome Proteinuria at 12 months, defined as UPCr > 20 mg/mmol

Dudley 2013 (Continued)

Secondary outcomes

- Possible trial medication-induced toxicity defined as reporting of hypertension, abdominal pain, nausea and or vomiting or adverse effects before the end of the week 4 visit
- Need for additional therapy
- Proteinuria or haematuria at 4 weeks, 3 months and 12 months
- UPCR > 200 mg/mmol at 12 months

Notes

Additional information

- Exclusions post-randomisation but pre-intervention: 1 in each group
- Funding source: Wales Office for Research and Development

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer generated random number sequence using random number generator
Allocation concealment (selection bias)	Low risk	Identical bottles for active and placebo medications coded centrally with trial numbers
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Trial medications (active and placebo) supplied by the same company in identical bottles. Patients, parents, paediatricians and all investigators were blind to treatment management
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Patients, parents, paediatricians and all investigators were blind to outcome assessment
Incomplete outcome data (attrition bias) All outcomes	High risk	Lost to follow-up: prednisone group (10), placebo group (6) Primary outcome analysed in 71% (123/171) in prednisone group and 75% (124/165) in placebo group Secondary outcome of haematuria or proteinuria analysed in 82% (141/171) in prednisone group and 83% (137/165) in placebo group
Selective reporting (reporting bias)	Low risk	Primary outcome pre-specified as UPC in National Research Register of NHS in UK. Information also provided on dipstick analysis of haematuria and proteinuria. Adverse events reported
Other bias	Low risk	Appears to be free of other biases Funded by Wales Office for Research and Development

Fuentes 2010

Study characteristics

Methods

Study design

- Parallel RCT

Time frame

Fuentes 2010 (Continued)

	- Study duration: not reported - Duration of follow-up (months): MMF group (68 ± 12), AZA group (64 ± 24)
Participants	Study characteristics - Country: Mexico - Setting: single centre - Inclusion criteria: children with biopsy-proven IgAV (ISKDC histology class 1 (1); class 2 (6); class 3 (10)) - Exclusion criteria: ISKDC histological classes IV and V excluded Baseline characteristics - Number: intervention group 1 (13); intervention group 2 (13) - Mean age $\pm$ SD (years): intervention group 1 (6.8 ± 2.1); intervention group 2 (6.9 ± 2.2) - Sex (M/F): intervention group 1 (10/3); intervention group 2 (7/6)
Interventions	Intervention group 1 - MMF: $1000 \text{ mg/m}^2/\text{day}$ for 12 months - Prednisone: given for 6 months; dosage not reported Intervention group 2 - AZA: 2 mg/kg/day for 12 months - Prednisone: given for 6 months; dosage not reported
Outcomes	Primary outcomes - Regression of histological lesion - Improvement in proteinuria Other outcomes - SCr - Serum albumin 24-hour urinary protein - eGFR
Notes	Additional information - Two abstracts only available - Further information received from authors: anaemia in one patient in the MMF arm - One author consultant for Novartis, Mexico

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer generated random numbers. Information provided by authors
Allocation concealment (selection bias)	Low risk	Computer generated. Information provided by authors
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding and outcome is likely to be influenced by lack of blinding

Fuentes 2010 (Continued)

Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding and outcome is likely to be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	All patients accounted for
Selective reporting (reporting bias)	Low risk	Data on kidney outcomes and adverse effects provided
Other bias	High risk	One author a consultant for Novartis; no full-text publication after 5 years

Geng 2021
Study characteristics

Methods	Study design - Parallel RCT Time frame - Study duration: August 2016 to November 2019 - Duration of follow-up: 12 months after treatment completed (total of 15 months)
Participants	Study characteristics - Country: China - Setting: single centre - Inclusion criteria: 68 children with IgAV nephritis, 27/33 in group 1 and 27/35 in group 2 had kidney biopsies. - Intervention group 1: ISKDC histopathology grade 2 (5); grade 3a (3); grade 3b (18), grade 4a (1) - Intervention group 2: ISKDC histopathology grade 2 (1), grade 3a (2), grade 3b (20), grade 4a (1), grade 4b (3) - Exclusion criteria: WBC count $< 3 \times 10^9/L$; Hb < 90 g/L; liver function impairment; patients with active infections; kidney biopsy results showing moderate or higher levels of renal tubular atrophy and/or interstitial fibrosis; previous history of using immunosuppressive medications other than prednisone Baseline characteristics - Number: intervention group 1 (33); intervention group 2 (35) - Median age, IQR (years): intervention group 1 (8.3, 7.0, 11.6); intervention group 2 (8.1, 6.8, 11.2) - Sex (M/F): intervention group 1 (17/16) intervention group 2 (21/14)
Interventions	Intervention group 1 - MMF (oral): 15 to 30 mg/kg dose, twice a day (maximum dose of 2 g/day) for 3 months - If treatment effective, MMF continued to 6 months and then the dose reduced - If urine protein levels did not decrease by 50% by 3 months, MMF was replaced with a different immunosuppressive drug - Prednisone (oral): 1.5 to 2 mg/kg/day for 4 to 6 weeks - Methylprednisolone (IV): 3 pulses given if no reduction in urine protein at 4 weeks of oral prednisone followed by further course of oral prednisone Intervention group 2

Geng 2021 (Continued)

- CPA (IV): 8 to 12 mg/kg/day for 2 days every 2 weeks for 6 treatments in 3 months
 - If treatment effective, regimen was changed to 2 days every 1 month till maximum dose of 168 mg/kg reached
- Prednisone (oral): 1.5 to 2 mg/kg/day for 4 to 6 weeks
- Methylprednisolone (IV): 3 pulses given if no reduction in urine protein at 4 weeks of oral prednisone followed by further course of oral prednisone

Co-interventions

- ACEi

Outcomes	Outcomes relevant to this review • Complete remission: 24-hour urine protein was negative, 24-hour urine protein < 150 mg • Partial remission: 24-hour urinary protein was negative, but 24-hour urinary protein levels decreased by > 50% • Ineffective: 24-hour urinary protein levels decrease by < 50% • Adverse effects
Notes	Additional information • Study (methods and results) translated by Dr Rujan Shrestha • Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Computer generated randomisation with blocks of four"
Allocation concealment (selection bias)	Unclear risk	No additional information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding of participants, care givers or investigators
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Primary outcome was 24-hour protein excretion. As a laboratory measure, this is unlikely to be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	All participants accounted for Excluded/loss to follow-up was 3/72 (4.2%)
Selective reporting (reporting bias)	Low risk	Expected outcomes reported
Other bias	Unclear risk	No information on any funding available

He 2002

Study characteristics

He 2002 (Continued)

Methods	Study design - Parallel RCT Time frame - Duration of study: not reported - Duration of follow-up: 6 months
Participants	Study characteristics - Country: China - Setting: single centre - Inclusion criteria: children at onset or relapse of HSP but presumed to be without kidney disease - Exclusion criteria: not reported Baseline characteristics - Number: intervention group (119); control group (109) - Mean age $\pm$ SD (years): not reported - Sex (M/F): not reported
Interventions	Intervention group - Heparin: sodium heparin 120 to 150 IU/kg/day (IV) for 5 days or calcium heparin 10 IU/kg (SC) twice/day for 7 days given at onset or relapse of HSP Control group - Placebo injection only Co-interventions - Vitamin C, vitamin P
Outcomes	Outcomes relevant to this review - Total number of children with kidney disease after 3 months or more (not defined) - Number with haematuria and proteinuria after 3 months or more - Number with nephrotic syndrome after 3 months or more - Time to development of kidney disease - Bleeding
Notes	Additional information - Abstract only available - Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information to permit judgement
Allocation concealment (selection bias)	Unclear risk	Insufficient information to permit judgement

He 2002 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Control group given IV vehicle but unclear whether investigators aware of which patients received heparin or vehicle
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Control group given IV vehicle but unclear whether investigators aware of which patients received heparin or vehicle
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No information provided although all patients appeared to have been followed. Minimum follow-up 6 months. Limited information on adverse effects provided
Selective reporting (reporting bias)	Low risk	Information provided on numbers with haematuria alone, haematuria and proteinuria, haematuria with nephrotic syndrome and adverse effects
Other bias	Unclear risk	Insufficient information to permit judgement Unclear as to whether same patient could enter the trial twice (at onset or at recurrence)

Huber 2004

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: September 1996 to January 2000 - Duration of follow-up: 1 year
Participants	Study characteristics - Country: Canada - Setting: single centre - Inclusion criteria: 2 to 15 years within 7 days of onset of IgAV - IgAV defined as palpable purpura and 1 or more of arthritis, kidney disease or gut involvement - Kidney disease defined as haematuria 5 or more RBC/HPF or RBC casts, proteinuria 0.3g/L or more, hypertension 90th percentile for age/sex or above - Study included 4 children in prednisone group and 2 in placebo group with kidney disease at study entry - Exclusion criteria: known underlying systemic vasculitis; steroids in previous month; underlying kidney, GI or immunodeficiency illness; active infection; a life-threatening complication of IgAV Baseline characteristics - Number: intervention group (21); control group (19) - Median age, range (years): intervention group (5, 2 to 11); control group (6.1, 3 to 15) - Sex (M/F): intervention group (13/8); control group (7/12)
Interventions	Intervention group Prednisone (oral): 2 mg/kg/day for 7 days; reducing dose over 7 days

Huber 2004 (Continued)

	Control group
	Placebo: given in same regimen
	Co-interventions
	Not reported
Outcomes	Outcomes relevant to this review
	- New or persistent kidney disease at 1 year - GI involvement
Notes	Additional information
	Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer generated sequence of random numbers
Allocation concealment (selection bias)	Low risk	Plain sealed numbered envelopes opened at subject randomisation by research pharmacist. Individuals directly involved in the study had no access to these envelopes
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Prednisone and placebo groups received identical number of pills and followed the same schedule Prednisone and placebo tablets placed in opaque tasteless gelatin capsules The subjects and all individuals involved in the enrolment and assessment of study participants were blinded to the study group
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The subjects and all individuals involved in the enrolment and assessment of study participants were blinded to the study group
Incomplete outcome data (attrition bias) All outcomes	Low risk	One subject enrolled declined randomisation Three children were withdrawn from the placebo group due to complications but included in the analysis (intussusception (2); severe rash (1))
Selective reporting (reporting bias)	Low risk	Reported that children with persistent kidney disease had haematuria, proteinuria or both and did not have hypertension or kidney insufficiency Reported adverse effects
Other bias	Low risk	The study appears to be free of other sources of bias

Islek 1999

Study characteristics

Methods	Study design
	Parallel RCT

Interventions for preventing and treating kidney disease in IgA vasculitis (Review)

Islek 1999 (Continued)

	Time frame - Duration of study: September 1996 to January 2000 - Duration of follow-up: unclear
Participants	Study characteristics - Country: Turkey - Setting: single centre - Inclusion criteria: 9.2 ± 2.7 years with HSP without haematuria/proteinuria on admission defined as non-thrombocytopenic purpura, arthritis and arthralgia, abdominal pain, GI haemorrhage - Exclusion criteria: kidney disease Baseline characteristics - Number: intervention group (70); control group (50) - Mean age $\pm$ SD (years): not reported - Sex (M/F): 69/48
Interventions	Intervention group - Prednisolone: 1 mg/kg/day for 10 days; tapered over 1 week and withdrawn Control group - No intervention Co-interventions - Not reported
Outcomes	Outcomes relevant to this review Haematuria, proteinuria or both: no definitions provided
Notes	Additional information - Abstract only available - Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information to permit judgement
Allocation concealment (selection bias)	Unclear risk	Insufficient information to permit judgement
Blinding of participants and personnel (performance bias) All outcomes	High risk	No placebo tablets administered in the control group. No information provided on whether outcome assessors were blinded
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No information provided on whether outcome assessors were blinded. Abstract only

Islek 1999 (Continued)

Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unclear whether all eligible patients entered and completed the trial and whether there was any missing data. All reported patients appeared to have completed study
Selective reporting (reporting bias)	Unclear risk	Only outcomes reported were haematuria and proteinuria. No reports separately of more severe kidney disease (acute nephritic syndrome, nephrotic syndrome, hypertension)
Other bias	High risk	Insufficient information available to determine

Jauhola 2010
Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: December 1999 to April 2006 - Duration of follow-up: 2.9 years (range 1.1 to 5.6)
Participants	Study characteristics - Country: Finland - Setting: multicentre (number of sites not reported) - Inclusion criteria: kidney biopsy diagnosis of crescentic HSP-associated kidney disease of ISKDC grade III or IV or grade II with nephrotic syndrome - Exclusion criteria: medication known to interact with CSA Baseline characteristics - Number: intervention group (7); control group (8); 9 patients non-randomised - Mean age, range (years): intervention group (9.2, 2 to 18); control group (7.9, 4.0 to 14.8) - Sex (M/F): intervention group (5/2); control group (6/2)
Interventions	Intervention group - CSA: 5 mg/kg/day for 12 months, titrated according to CSA level Control group - Methylprednisolone (IV): 30 mg/kg 3 times in 1 week - Prednisone 30 mg/m²/day for 1 month and tapered over 3 months - Intermediate days and 1 month after methylprednisolone patients received oral prednisone 30mg/m² in 2 daily doses, medication Co-interventions - ACEi
Outcomes	Outcomes relevant to this review - Remission (UPCR < 200 or urine protein < 40 mg/m²/hour) at 3 months - Remission at last follow-up
Notes	Additional information

Jauhola 2010 (Continued)

- Funding source: OJ and JR received a grant from the Alma and KA Snellman Foundation, Oulu, Finland and from the Foundation for Paediatric Research for writing this report

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomised by a central office
Allocation concealment (selection bias)	Low risk	Central randomisation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label study comparing an orally administered agent with an IV administered agent
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label study comparing an orally administered agent with an IV administered agent
Incomplete outcome data (attrition bias) All outcomes	High risk	No SD provided with means of urinary protein and SCr at last follow-up. Duration of study not defined
Selective reporting (reporting bias)	Low risk	Data on kidney outcomes and adverse effects provided
Other bias	Unclear risk	Insufficient information to permit judgement

Liu 2019e
Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: January 2014 to March 2018 - Duration of follow-up: unclear, but at least 6 months
Participants	Study characteristics - Country: China - Setting: single centre - Inclusion criteria: < 18 years with HSPN; nephrotic syndrome with > 2 g/day proteinuria: biopsy grade IIIB HSP or more severe; diagnosis of HSP based on rash ± GI symptoms and/or joint symptoms - Exclusion criteria: coagulopathy; previously treated with the treatment regimen used in this study; severe infections Baseline characteristics - Number: intervention group (30); control group (30) - Mean age ± SD (years): intervention group (10.4 ± 1.9); control group (9.8 ± 1.6)

Liu 2019e (Continued)

- Sex (M/F): intervention group (18/12); control group (21/9)
- Pathology
 - Intervention group: histology grade IIIB (7), IV (19), V (4)
 - Control group: histology grade IIIB (9), IV (15), V (6)
- Kidney function
 - Intervention group: SCr > 300 µmol/L in 19 patients
 - Control group: SCr > 300 µmol/L in 22 patients

Interventions	Intervention group • Double filtration plasmapheresis: 2 hours/day, every 3 days for 4 to 5 treatments started 3 to 5 days after pulse methylprednisolone/CPA, though exact timing unclear • Methylprednisolone (IV): 15 to 20 mg/kg for 3 days; 3 courses at 5- to 7-day intervals, then prednisone (oral) 1 mg/kg/day. Unclear how long this was continued • CPA (IV): 8 to 10 mg/kg/day for 2 days; repeated for a total of 6 to 7 courses. Maximum dose CPA 120 to 130 mg/kg Control group • Methylprednisolone (IV): 15 to 20 mg/kg for 3 days; 3 courses at 5- to 7-day intervals, then prednisone (oral) 1 mg/kg/day. Unclear how long this was continued • CPA (IV): 8 to 10 mg/kg/day for 2 days; repeated for a total of 6 to 7 courses. Maximum dose CPA 120 to 130 mg/kg Co-interventions • Anti-rheumatic drugs, antihistamine drugs, calcium channel blockers agents, and anticoagulant drugs • Treated symptomatically when adverse reactions occur
Outcomes	Outcomes relevant to this review • Complete remission: haematuria disappears, 24-hour urine protein < 150 mg, urine series micro-proteins normal; kidney function normal • Partial remission: haematuria is significantly improved, 24-hour urine protein levels and urine series micro-proteins had decreased by more than 50% compared to level before treatment; kidney function close to normal • No response to treatment: haematuria, proteinuria, and urine series of micro-proteins had not improved or had worsened; kidney function remained abnormal
Notes	Additional information • Article translated from Chinese before data extraction • Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote: "Randomly divided according to the random table method"
Allocation concealment (selection bias)	Unclear risk	Quote: "Randomly divided according to the random table method"
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding

Liu 2019e (Continued)

Blinding of outcome assessment (detection bias) All outcomes	Low risk	Outcomes measured by 24 urine protein and kidney function so unlikely to be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	All participants accounted for
Selective reporting (reporting bias)	Low risk	Expected outcomes reported
Other bias	Unclear risk	No information on how study was funded were provided

Mollica 1992

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: October 1978 to September 1987 - Duration of follow-up: 24 to 36 months
Participants	Study characteristics - Country: Italy - Setting: single centre - Inclusion criteria: unselected children with HSP without kidney disease at study entry - Exclusion criteria: haematuria, proteinuria or both on initial presentation Baseline characteristics - Number: intervention group (84); control group (84) - Mean age $\pm$ SD (years): not reported - Sex (M/F): not reported
Interventions	Intervention group - Prednisolone (oral): 1 mg/kg for 2 weeks Control group - No intervention Co-interventions - Not reported
Outcomes	Outcomes relevant to this review - Number of patients who develop HSP kidney disease with 2 or more of the following: - Proteinuria $\geq$ 4 mg/m²/hour - Haematuria $\geq$ 10 RBC/HPF - BP $\geq$ 2 SD above normal for age - BUN $\geq$ 54 mg/dL

Mollica 1992 (Continued)

- $Cr \geq 0.8 \text{ mg/dL/m}^2$

Notes	Additional information	
	Funding source: not reported	
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information to permit judgement
Allocation concealment (selection bias)	High risk	Each patient on entry to the trial was alternatively assigned to 1 of the 2 treatment groups
Blinding of participants and personnel (performance bias) All outcomes	High risk	Control group did not receive placebo medications
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Insufficient information to permit judgement
Incomplete outcome data (attrition bias) All outcomes	High risk	19 patients excluded because of insufficient or inadequate follow-up
Selective reporting (reporting bias)	High risk	Information on the numbers with proteinuria, haematuria, hypertension and reduced kidney function provided; no report on adverse effects
Other bias	Unclear risk	Insufficient information to permit judgement

Peratoner 1990

Study characteristics	
Methods	Study design • Parallel RCT Time frame • Duration of study: not reported • Duration of follow-up: 1 year
Participants	Study characteristics • Country: Italy • Setting: multicentre (12 sites) • Inclusion criteria: 2 to 14 years with IgAV • Exclusion criteria: not reported Baseline characteristics • Number/kidney disease at entry: intervention group (60/13); control group (41/6)

Peratoner 1990 (Continued)

- Mean age $\pm$ SD (years): not reported
- Sex (M/F): not reported

Interventions	Intervention group • Dipyridamole (oral): 4 mg/kg/day in 3 doses • Cyproheptadine (oral): 0.5 mg/kg/day in 3 doses • Salicylates (oral): 10 mg/kg/day in 1 dose for 8 weeks Control group • No intervention Co-interventions • Antipyretics, antibiotics
Outcomes	Outcomes relevant to this review • Kidney disease during 1 year of follow-up, assessed using the following scoring system: ◦ Haematuria > 5 RBC/mm³ (score 0 to 2) ◦ Cylinduria (0-1) ◦ Hypertension (0-1) ◦ Reduced CrCl (0-2)
Notes	Additional information • Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information to permit judgement
Allocation concealment (selection bias)	Unclear risk	Insufficient information to permit judgement
Blinding of participants and personnel (performance bias) All outcomes	High risk	No placebo given to control group
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Insufficient information to permit judgement
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unclear whether all entered patients completed trial and were included in the results
Selective reporting (reporting bias)	High risk	Information on type of residual kidney disease provided in text. No report of adverse effects
Other bias	Unclear risk	Insufficient information to permit judgement

Ronkainen 2006a

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: 1999 to 2005 - Duration of follow-up: mean 7.7 years (range 4.6 to 11.4 years)
Participants	Study characteristics - Country: Finland - Setting: multicentre (14 sites) - Inclusion criteria: ≤ 16 years; clinical diagnosis of newly diagnosed HSP (purpura, petechiae $\pm$ gut/joint pain) - Exclusion criteria: established kidney disease (haematuria > 10 RBC/HPF or proteinuria > 300 mg/L on initial presentation); thrombocytopenia; systemic vasculitis; prednisone contraindicated Baseline characteristics - Number (randomised/analysed): intervention group (87/84); control group (89/87) - Mean age, range (years): intervention group (6.8, 2.0 to 15.2); control group (7.3, 1.7 to 15.6) - Sex (M/F): intervention group (49/35); control group (44/43) - Duration of disease at entry (mean days, range): intervention group (4.7, 0 to 28); control group (6.4, 0 to 63)
Interventions	Intervention group - Prednisone (oral): 1 mg/kg/day for 14 days; 0.5 mg/kg/day for 7 days; 0.5 mg/kg on alternate days for 7 days Control group - Placebo tablets Co-interventions - Paracetamol for pain
Outcomes	Outcomes relevant to this review - Haematuria: urinary > 5 RBC/HPF at 1, 3, 6 months - Proteinuria: urinary protein > 200 mg/L or albumin > 30 mg/L at 1, 3, 6 months - Severity and duration of abdominal pain during 1st month - Severity and duration of joint pain during 1st month - Adverse effects: weight; BP
Notes	Additional information - Exclusions post-randomisation but pre-intervention: 3 excluded from prednisone group and 2 from placebo group after randomisation - Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
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Ronkainen 2006a (Continued)

Random sequence generation (selection bias)	Low risk	Block randomisation scheme with block size of 6
Allocation concealment (selection bias)	Low risk	Observers and subjects were unaware of randomisation scheme. Pharmia Ltd packed drugs, labelled containers, performed randomisation and retained key to randomisation till end of study
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Prednisone (5 mg tablets) and placebo were similar in size and supplied in lots of 200 tablets in similar containers marked with sequential numbers
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Assessors blinded until end of trial
Incomplete outcome data (attrition bias) All outcomes	Low risk	Three (2 dropped out; 1 protocol violation) excluded from prednisone group. Two (both dropped out) excluded from placebo group. These unlikely to influence final results 138/176 (21%) screened on long-term follow-up
Selective reporting (reporting bias)	High risk	Data on kidney outcomes extrapolated from graphs. Data on adverse effects included
Other bias	Low risk	Appears to be free of other biases

Tarshish 2004

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: 1973 to 1980 - Duration of follow-up: 6.93 ± 3.32 years in patients who recovered; 6.57 ± 4.1 years in group with persistent abnormalities; 3.71 ± 2.14 years in patients progressing to ESKD
Participants	Study characteristics - Countries: Europe, USA, Canada - Setting: multicentre (14 sites) - Inclusion criteria: > 12 weeks to 16 years; HSP: purpura plus urticaria with 1 or more of the following: joint pain and swelling, kidney disease, abdominal pain and intestinal bleeding; eGFR > 35 mL/min/1.73 m²; proteinuria > 40 mg/m²/hr for > 1 month; histopathology: crescents/segmental lesions (ISKDC classification) - Exclusion criteria: HSP present > 3 months; prior use of immunosuppressive or cytotoxic therapy other than steroids; concurrent or pre-existing kidney disease Baseline characteristics - Number: intervention group (28); control group (28) - Mean age ± SD (years): not reported

Tarshish 2004 (Continued)

- Sex (M/F): not reported

Interventions	Intervention group • CPA (oral): 90 mg/m²/day for 42 days Control group • No therapy Co-interventions • Diet modification, ion exchange resins, vitamins, diuretics
Outcomes	Outcomes relevant to this review • Total number of patients with any persistent kidney disease • Number with severe kidney disease (decreased GFR, severe proteinuria, ESKD) • Number of patients with ESKD
Notes	Additional information • Funding source: "Financial support by National Institutes of Health Research Grant 1-RO1-AM18234, the National Kidney Foundation of New York, the Kidney Disease Institute of the State of New York, the William Beaumont Hospital Pathology Projects Fund, the John Rath Foundation, the National Kidney Research Foundation (United Kingdom), and the Kidney Foundation of the Netherlands."

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information to permit judgement
Allocation concealment (selection bias)	Low risk	Random allocation at central office
Blinding of participants and personnel (performance bias) All outcomes	High risk	No placebo therapy used. While most outcome measures were laboratory measurements and unlikely to be influenced by lack of blinding, endpoints in some children were judged on dipstick protein urinalyses, which could be either doctor or patient-reported
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No information provided on outcome assessors
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Duration of follow-up variable and unclear whether all patients completed follow-up
Selective reporting (reporting bias)	Low risk	Data on persistent abnormalities, severe abnormalities and ESKD provided though detailed information on GFR and urinary protein excretion at follow-up not provided
Other bias	Unclear risk	Insufficient information provided on study design

Tian 2015

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: January 2007 to December 2012 - Duration of follow-up: 8 weeks
Participants	Study characteristics - Country: China - Setting: single centre - Inclusion criteria: children with HSPN (based on the classification from the Chinese Medical Association (Pediatric Nephrology group) - Exclusion criteria: hormone therapy; severe infection; severe bleeding tendency; leucopenia; pre-existing shock; HSP patients Baseline characteristics - Number: intervention group (44); control group (45) - Mean age $\pm$ SD (years): intervention group (7.94 ± 2.90); control group (8.46 ± 2.61) - Sex (M/F): intervention group (25/19); control group (29/16)
Interventions	- Intervention group - Low-molecular weight heparin (calcium): 100 U/kg/day for 4 weeks, then 50 U/kg/day for 4 weeks - Conventional treatment (cimetidine, vitamin C, antihistamines) - Control group - Conventional treatment only
Outcomes	Outcomes relevant to this review - Clinical symptoms - Proteinuria - Haematuria
Notes	Additional information - Failure of treatment defined as persistence of clinical symptoms, proteinuria and haematuria - Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote: "Eighty-nine patients were randomized into control group (n=45) and treatment group (n=44)". No other information provided
Allocation concealment (selection bias)	Unclear risk	Quote: "Eighty-nine patients were randomized into control group (n=45) and treatment group (n=44)". No other information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label study

Tian 2015 (Continued)

Blinding of outcome assessment (detection bias) All outcomes	High risk	Not blinded and assessment of clinical outcomes could be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	All participants included in assessment
Selective reporting (reporting bias)	High risk	No report of adverse effects and limited reporting of kidney outcomes
Other bias	Unclear risk	No information on funding provided

Wu 2014c
Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: March 2006 to June 2011 - Duration of follow-up: detailed follow-up for 6 months; further follow-up to 1.5 to 6 years (median 3.8 years)
Participants	Study characteristics - Country: China - Setting: multicentre (4 sites) - Inclusion criteria - Group 1: previously healthy children aged 2 to 14 years with IgAV without nephritis - Group 2: previously healthy children aged 2 to 14 years with IgA nephritis (proteinuria and/or hematuria without rapidly progressive nephritis) - Exclusion criteria: not reported Baseline characteristics - Group 1 - Number: intervention group (42); control group (42) - Mean age $\pm$ SD (years): intervention group (6.6 ± 1.7); control group (6.8 ± 1.5) - Sex (M/F): intervention group (26/16); control group (27/15) - Group 2 - Number: intervention group (23); control group (23) - Mean age $\pm$ SD (years): intervention group (7.2 ± 2.0); control group (7.3 ± 1.8) - Sex (M/F): intervention group (13/10); control group (15/8)
Interventions	Groups 1 and 2 - Intervention group - Montelukast sodium: chewable tablets for 3 months, 4 mg/day (ages 2 to 5 years), 5 mg/day (ages 6 to 14 years) - Control group - Placebo in same dosages as montelukast Group 2

Wu 2014c (Continued)

- Mild IgAV nephritis (16 participants with histological grade IIa or proteinuria < 25 mg/kg/day): captopril and dipyridamole for 3 months
- Moderate IgAV nephritis (22 participants with histological grade IIb, IIIa or proteinuria 25 to 50 mg/kg/day): captopril, dipyridamole for 3 months; prednisone for 6 months
- Severe IgAV nephritis (8 participants with histological grade IIIb, IV, V or proteinuria > 50 mg/kg/day): captopril, dipyridamole for 3 months; prednisone for 6 months; MMF for 6 months

Co-interventions (both groups)

- Avoidance of possible allergens
- Cimetidine, prednisone for abdominal pain with or without GI bleeding for 1 week, and then decreased dose of prednisone for 2 weeks
- Oral aspirin and prednisone for joint pain and swelling for 1 week, and then decreased dose of prednisone for 2 weeks

Outcomes	Outcomes relevant to this review • SSS for clinical features of HSP at 1, 2 and 4 weeks • SSS for haematuria and proteinuria at 1, 2 and 4 weeks, 3 and 6 months • Relapse rate during follow-up to > 3 years in both groups • Clinical status at follow-up to > 3 years in group 2 • Adverse effects of montelukast
Notes	Additional information • Chinese Clinical Trial Registry (http://www.chictr.org; registration no. ChiCTR-TRC-12002671) • Funding source: support from Medical Emphasis Grant (No. 56RC2002056) from Government of Jiangsu Province, PR China

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "In this four-center, double-blind, placebo-controlled, randomised, prospective, parallel paired comparative study, every pair of patients with the same severity of disease in each large group was randomly divided into two subgroups according to a computer-generated randomised allocation schedule"
Allocation concealment (selection bias)	Low risk	Quote: "In this four-center, double-blind, placebo-controlled, randomised, prospective, parallel paired comparative study, every pair of patients with the same severity of disease in each large group was randomly divided into two subgroups according to a computer-generated randomised allocation schedule"
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "The color, smell and external appearance of the placebo and montelukast sodium were identical"
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	SSS is based on clinical assessment; unclear whether assessors were blinded to treatment groups
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Follow-up data provided as % so unclear how many participants were followed up at each time period

Wu 2014c (Continued)

Selective reporting (reporting bias)	High risk	Data only provided graphically so could not be included in meta-analyses
Other bias	Low risk	Support from Medical Emphasis Grant (No. 56RC2002056) from Government of Jiangsu Province, PR China

Xu 2009
Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: January 2007 to March 2008 - Duration of follow-up: 2 months
Participants	Study characteristics - Country: China - Setting: single centre - Inclusion criteria: 4 to 14 years with HSP; proteinuria ≥ 150 mg/day; medications ceased 2 weeks prior to study - Exclusion criteria: not reported Baseline characteristics - Number: intervention group (27); control group (21) - Mean age $\pm$ SD (years): not reported - Sex (M/F): 28/20
Interventions	Intervention group - Fosinopril: 5 mg/day for 8 weeks (< 5 years), 10 mg/day for 8 weeks (5 to 12 years) - Standard therapy: penicillin, claratyne, fraxiparine, dipyridamole, nifedipine, low salt diet Control group - Standard therapy: penicillin, claratyne, fraxiparine, dipyridamole, nifedipine, low salt diet Co-interventions - Not reported
Outcomes	Outcomes relevant to this review - Proteinuria < 150 mg - Adverse effects
Notes	Additional information Funding source: not reported

Risk of bias

Xu 2009 (Continued)

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised but no method mentioned
Allocation concealment (selection bias)	Unclear risk	Said to be randomised but no further information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding and outcome is likely to be influenced by lack of blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding and outcome is likely to be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	All enrolled patients accounted for
Selective reporting (reporting bias)	Low risk	Reported all expected outcomes
Other bias	Unclear risk	Insufficient information to permit judgement

Yoshimoto 1987a

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: October 1984 to September 1985 - Duration of follow-up: not reported
Participants	Study characteristics - Country: Japan - Setting: single centre - Inclusion criteria: children admitted with HSP without kidney disease aged 3 to 10 years - Exclusion criteria: known underlying systemic vasculitis; steroids in previous month; underlying kidney, GI or immunodeficiency illness; active infection; a life-threatening complication of HSP Baseline characteristics - Number: intervention group (9); control group (9) - Mean age $\pm$ SD (years): not reported - Sex (M/F): 13/15
Interventions	Intervention group Aspirin: 5 mg/kg/day for 5 weeks

Yoshimoto 1987a (Continued)

	Control group - Vitamin pills for 5 weeks Co-interventions - Not reported
Outcomes	Outcomes relevant to this review Number of patient who has kidney disease (not defined); time of involvement not specified
Notes	Additional information - Abstract only available - Same study as Yoshimoto 1987b; 2 treatment arms, 1 control - Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information to permit judgement
Allocation concealment (selection bias)	Unclear risk	Insufficient information to permit judgement
Blinding of participants and personnel (performance bias) All outcomes	High risk	Control group received vitamin pills. Outcome measures not defined
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Insufficient information to permit judgement
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Criteria for kidney disease not defined
Selective reporting (reporting bias)	Unclear risk	Insufficient information to permit judgement
Other bias	High risk	Published only as abstract and no full text publication identified

Yoshimoto 1987b

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: October 1984 to September 1985
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Yoshimoto 1987b (Continued)

- Duration of follow-up: not reported

Participants	Study characteristics • Country: Japan • Setting: single centre • Inclusion criteria: children admitted with HSP without kidney disease aged 3 to 10 years • Exclusion criteria: known underlying systemic vasculitis; steroids in previous month; underlying kidney, GI or immunodeficiency illness; active infection; a life-threatening complication of HSP Baseline characteristics • Number: intervention group (10); control group (9) • Mean age $\pm$ SD (years): not reported • Sex (M/F): 13/15
Interventions	Intervention group • Dipyridamole: mg/kg/day for 5 weeks Control group: • Vitamin pills for 5 weeks Co-interventions • Not reported
Outcomes	Outcomes relevant to this review • Number of patient who has kidney disease (not defined); time of involvement not specified
Notes	Additional information • Abstract only available • Same study as Yoshimoto 1987a; 2 treatment arms, 1 control • Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Insufficient information to permit judgement
Allocation concealment (selection bias)	Unclear risk	Insufficient information to permit judgement
Blinding of participants and personnel (performance bias) All outcomes	High risk	Control group received vitamin pills. Outcome measures not defined
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Insufficient data to permit judgement
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Criteria for kidney disease not defined

Yoshimoto 1987b (Continued)

Selective reporting (re-reporting bias)	Unclear risk	Insufficient information to permit judgement
Other bias	Unclear risk	Published only as abstract and no full text publication identified

Zhang 2021a

Study characteristics

Methods	Study design - Parallel RCT Time frame - Duration of study: 1 January 2015 to 1 January 2020 - Duration of follow-up: 2 years
Participants	Study characteristics - Country: China - Setting: single centre - Inclusion criteria: children 4 to 18 years admitted with IgAV for at least 2 months and kidney disease (24-hour proteinuria ≥ 50 mg/kg and urine RBC count ≥ 10/HPF) for at least 1 week - Exclusion criteria: known underlying systemic vasculitis; steroids in previous month; underlying kidney, GI or immunodeficiency illness; active infection; a life-threatening complication of IgAV Baseline characteristics - Number (enrolled/evaluated): intervention group 1 (93/87); intervention group 2 (93/85); control group (93/83) - Mean age $\pm$ SD (years): intervention group 1 (8.52 ± 3.05); intervention group 2 (8.69 ± 3.01); control group (8.74 ± 2.79) - Sex (M/F): intervention group 1 (43/44); intervention group 2 (43/42); control group (47/36)
Interventions	Intervention group 1 - Tacrolimus (oral): 0.1 to 0.15 mg/kg/day to achieve blood levels of 7 to 10 ng/mL. Dose reduced by 20% to 25% every 7 to 10 days when blood level exceeded range or nephrotoxicity developed; duration 6 months - Prednisolone: 1.5 to 2 mg/kg/day for 4 to 8 weeks; then 1.5 to 2 mg/kg/day every 3 days for 4 to 8 weeks and reduced gradually and ceased after 6 months Control group - CPA (IV): 500 to 750 mg²/kg monthly for 6 months. Maximum total dose < 150 mg/kg - Prednisolone: 1.5 to 2 mg/kg/day for 4 to 8 weeks; then 1.5 to 2 mg/kg/day every 3 days for 4 to 8 weeks and reduced gradually and ceased after 6 months Intervention group 2 (participants were excluded from analyses as the treatment included Chinese medicines, which were excluded from this review) - Tacrolimus (oral): 0.1 to 0.15 mg/kg/day to achieve blood levels of 4 to 7 ng/mL; duration 6 months - Tripterygium glycosides (oral) 2 mg/kg/day for 15 days, then reduced to 1.5 mg/kg/day for 15 days and reduced to 1 mg/kg/day for the next 3 months - Prednisolone: 1.5 to 2 mg/kg/day for 4 to 8 weeks; then 1.5 to 2 mg/kg/day every 3 days for 4 to 8 weeks and reduced gradually and ceased after 6 months

Zhang 2021a (Continued)

Outcomes	Outcomes relevant to this review • Resolution of proteinuria, haematuria by 2 years • Recurrence of proteinuria, haematuria by 2 years • SCr and urea at 3 and 6 months • 24-hour urine protein at 3 and 6 months • Urine RBC count at 3 and 6 months • Adverse effects
Notes	Additional information • Intervention group 2 not included in participant numbers or used in analyses as studies including Tripterygium glycosides were excluded from this review • Information requested from authors on the definitions of resolution and recurrence of proteinuria and haematuria and of the serious adverse effects but no information was received • Funding source: not reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "The children were randomly divided into three groups by the Chinese Central Randomization System at 1:1:1 ratio. The randomization code in the system was kept blind until the basic clinical features were tested"
Allocation concealment (selection bias)	Low risk	Quote: "The children were randomly divided into three groups by the Chinese Central Randomization System at 1:1:1 ratio. The randomization code in the system was kept blind until the basic clinical features were tested"
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding of participants, care givers or physicians
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Primary outcome was laboratory measure of 24 hr urine protein excretion and unlikely to be influenced by lack of blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	All participants accounted for; 94%, 91%, 89% completed and analysed each group
Selective reporting (reporting bias)	High risk	Incomplete reporting of adverse effects and patient centred outcomes
Other bias	Unclear risk	Authors declared no conflicts of interest but no information provided on funding sources

ACEi: angiotensin-converting enzyme inhibitor; ARB: angiotensin receptor blocker; AZA: azathioprine; BP: blood pressure; BUN: blood urea nitrogen; BVAS: Birmingham Vascular Activity Score; CPA: cyclophosphamide; Cr: creatinine; CrCl: creatinine clearance; CSA: cyclosporin; DM: diabetes mellitus; eGFR: estimated glomerular filtration rate; ESKD: end-stage kidney disease; GFR: glomerular filtration rate; GI: gastrointestinal; Hb: haemoglobin; HIV: human immunodeficiency virus; HPF: high power field; HSP: Henoch Schonlein Purpura; IgAV: IgAV; ISKDC: International Study of Kidney Disease in Children; IV: intravenous; M/F: male/female; MMF: mycophenolate mofetil; RBC: red blood cells; RCT: randomised controlled trial; SC: subcutaneous; SCr: serum creatinine; SD: standard deviation; SSA: Severity Scale Score; UPCR: urinary protein:creatinine ratio; WBC: white blood cell

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Chen 2010	Wrong intervention: effect of MORA bioresonance therapy
ChiCTR2100050353	Wrong intervention: Traditional Chinese Medicines
ChiCTR-INR-17013850	Wrong intervention: laminar flow
Cui 2020	Wrong intervention: Traditional Chinese Medicines
Ding 1995	Wrong intervention: Traditional Chinese Medicines
Ding 2014	Wrong intervention: Traditional Chinese Medicines
Ding 2019	Wrong intervention: Traditional Chinese Medicines
Ding 2021	Wrong intervention: Traditional Chinese Medicines
Erdogan 1999	Wrong intervention: vitamin E supplementation
Fu 2021	Wrong intervention: vitamin D analogue
Fukui 1989	Study design: unclear whether study is RCT
Gao 2018a	Wrong population and intervention: montmorillonite powder for abdominal disease
Hui-Lan 2001	Wrong intervention: Traditional Chinese Medicines
Jia 2001	Wrong intervention: Traditional Chinese Medicines
Jiang 2003	Study design: unclear whether study is RCT
Jin 2003	Wrong intervention: Danshao granules
Jin 2021	Wrong intervention: Traditional Chinese Medicines
Kim 1987	Wrong intervention: antibiotic (rifampin)
Li 2022	Wrong intervention: Traditional Chinese Medicines
Liu 2004a	Wrong intervention: integrative Chinese and Western medicine
Liu 2008	Wrong population: idiopathic thrombocytopenic purpura
NCT02939573	Wrong population: children with isolated cutaneous vasculitis without kidney disease
NCT03647852	Wrong population: children with abdominal vasculitis not kidney disease
NCT04008316	Wrong population: colchicine for skin vasculitis not kidney disease
Shao 2021	Wrong intervention: Traditional Chinese Medicines
Wang 2011	Wrong intervention: Traditional Chinese Medicines

Study	Reason for exclusion
Wang 2021a	Wrong intervention: dietary guidance
Wu 2013b	Wrong intervention: Traditional Chinese Medicines
Xia 2016	Wrong population: children with IgAV and abdominal involvement
Xie 2009	Wrong intervention: composite saliva injection
Xiong 2019	Wrong population: nutritional support for abdominal disease in IgAV
Yang 2010	Wrong intervention: vessel pricking therapy with Western medicines
Yi 2007	Wrong intervention: Traditional Chinese Medicines
Zhang 1984	Wrong intervention: anisodamine
Zhang 2019b	Wrong intervention: Vitamin D
Zhao 2009	Wrong intervention: integrative Chinese therapy with Western therapy
Zhu 2016	Wrong intervention: Tripterygium glycosides and Danshen injection

IgAV: IgA vasculitis; RCT: randomised controlled trial

Characteristics of studies awaiting classification *[ordered by study ID]*

NCT00301613

Methods	Study design Parallel RCT
Participants	60 participants aged 16 to 50 years with - Biopsy-proved HSP - Proteinuria ≥ 3.0 g/24 hours - SCr < 5.0 mg/dL
Interventions	Intervention group 1 - MMF: 1 g/day Intervention group 2 - IV CPA pulses
Outcomes	To compare the efficacy, safety, tolerability and relapse of MMF vs CPA in the treatment of severe HSP nephritis (time frame: 12 months)
Notes	Additional information - Study completed in 2006 so results unlikely to be published. No update since 2010 - Study chair: Zhi-Hong Liu, M.D; Research Institute of Nephrology, Jinling Hospital, Nanjing University School of Medicine

CPA: cyclophosphamide; HSP: Henoch Schonlein Purpura; IV: intravenous; MMF: mycophenolate mofetil; SCr: serum creatinine

Characteristics of ongoing studies [ordered by study ID]

NCT02532777

Study name	The research of standard diagnosis and treatment for Henoch-Schonlein Purpura nephritis in children
Methods	Study design - Parallel RCT - Participants masked
Participants	100 children aged 2 to 16 years with HSP nephritis on biopsy and proteinuria ≥ 50 mg/kg/day
Interventions	Regimen 1 - Prednisone: 2 mg/kg/day (maximum dose 60 mg) for 6 to 8 weeks, then two-thirds of the two days dose given on alternate days - CPA: 0.1 mg/kg - ACEi: 0.2 to 0.3 mg/kg/day Regimen 2 - Prednisone: 2 mg/kg/day (maximum dose 60 mg) for 6 to 8 weeks, then two-thirds of the two days dose given on alternate days - MMF: 25 mg/kg/day bid (maximum dose 5 g/day) - ACEi: 0.2 to 0.3 mg/kg/day Regimen 3 - Prednisone: 2 mg/kg/day (maximum dose 60 mg) for 6 to 8 weeks, then two-thirds of the two days dose given on alternate days - Leflunomide: 1 mg/kg/day (maximum dose 40 mg/kg) for 3 days, then 0.5mg/kg/day - ACEi: 0.2 to 0.3 mg/kg/day
Outcomes	Planned outcomes - Resolution of proteinuria to < 150 mg/day - Resolution of haematuria to < 3/hpf - Normal GFR
Starting date	August 2015; completion expected July 2020
Contact information	Aihua Zhang, M.D. (bszah@163.com)
Notes	

NCT02532790

Study name	The research of standard diagnosis and treatment for Henoch-Schonlein Purpura nephritis with mild proteinuria in children
Methods	Study design Parallel RCT

NCT02532790 (Continued)

	Participants masked
Participants	100 children aged 2 to 16 years with HSP with proved HSP nephritis (ISKDC class II) and proteinuria < 25 mg/kg/day
Interventions	Intervention group - Prednisone: starting dose 1.5mg/kg/day for 4 to 6 weeks, 1.5mg/kg/every other day for 4 weeks, then taper Control group - ACEi: 0.2 to 0.3mg/kg/day
Outcomes	Planned outcomes - Resolution of proteinuria to < 150 mg/day - Resolution of haematuria to < 3/hpf - Normal GFR
Starting date	August 2015; completion expected July 2020
Contact information	Aihua Zhang, M.D. (bszah@163.com)
Notes	

NCT02540720

Study name	The research of standard diagnosis and treatment for severe Henoch-Schonlein Purpura in children
Methods	Study design - Parallel RCT - Participants masked
Participants	30 children aged 2 to 16 years with severe HSP resistant to dexamethasone 0.5 mg/kg/day
Interventions	Regimen 1 - Dexamethasone (IV): 0.5 mg/kg/day Regimen 2 - Dexamethasone (IV): 0.5 mg/kg/day - Gamma globulin (IV): daily for 3 days (total dose 2 g/kg) Regimen 3 - Dexamethasone (IV): 0.5 mg/kg/day - Haemoperfusion: ≥ 3 times in 5 days
Outcomes	Planned outcomes - Resolution of abdominal symptoms - Arthralgia - Haematuria

NCT02540720 (Continued)

- Proteinuria

Starting date	August 2015; completion expected July 2020
Contact information	Aihua Zhang, M.D. (bszah@163.com)
Notes	

ACEi: angiotensin-converting enzyme inhibitors; CPA: cyclophosphamide; GFR: glomerular filtration rate; hpf: high power field; HSP: Henoch Schonlein Purpura; IV: intravenous; MMF: mycophenolate mofetil; RCT: randomised controlled trial

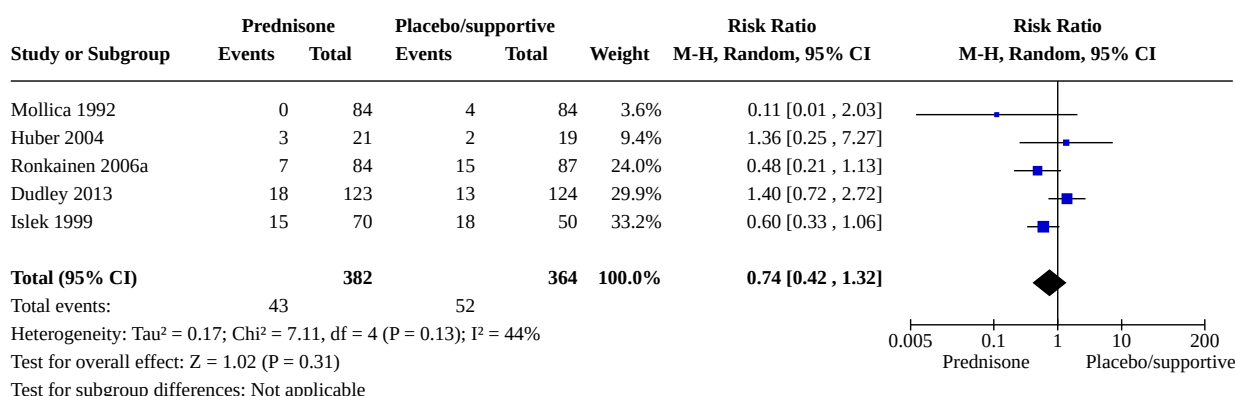
DATA AND ANALYSES

Comparison 1. Prednisone versus placebo/supportive treatment for preventing persistent kidney disease

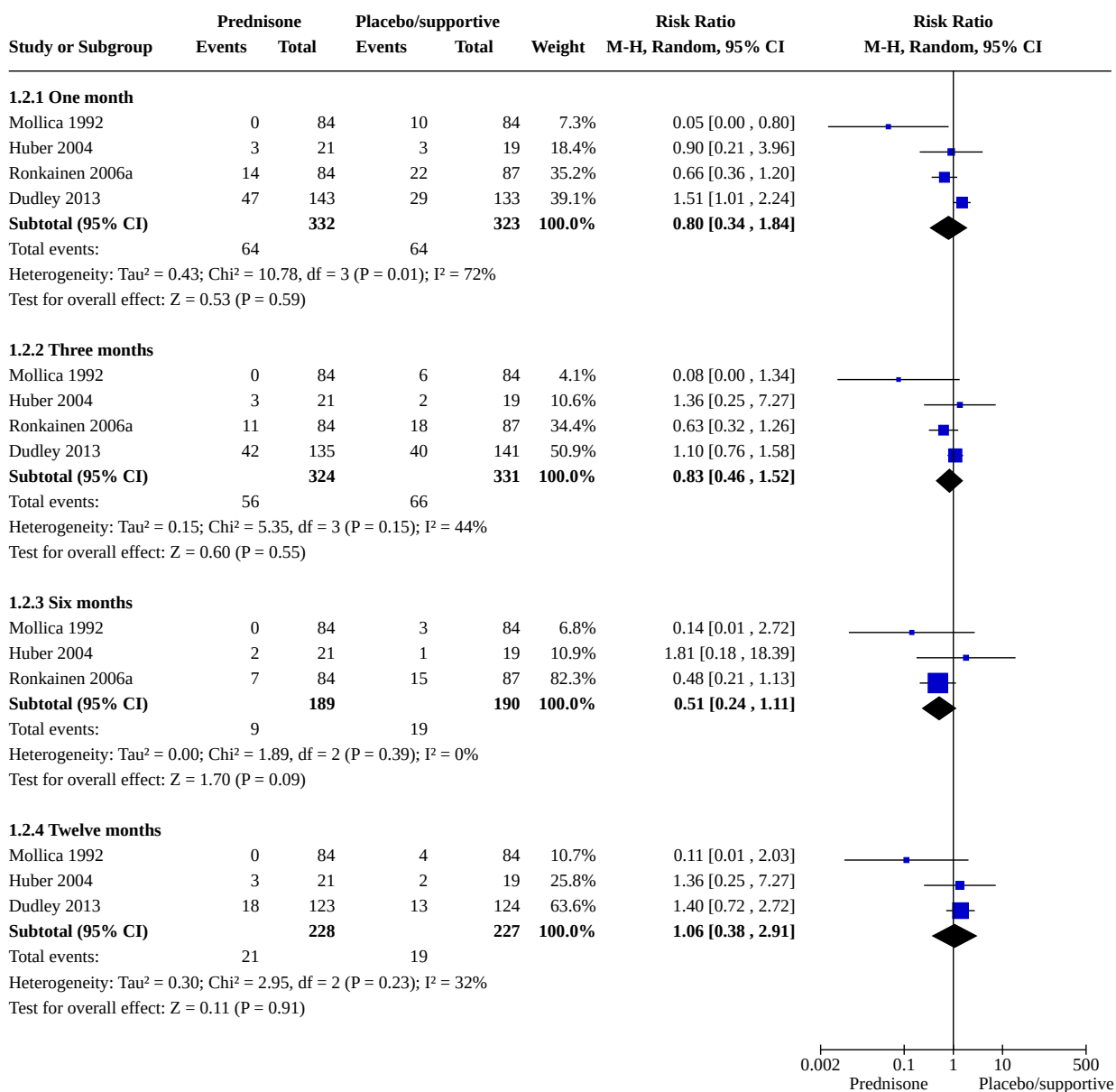
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Persistent kidney disease at any time after treatment	5	746	Risk Ratio (M-H, Random, 95% CI)	0.74 [0.42, 1.32]
1.2 Number of children with any continuing kidney disease at different time points	4		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.2.1 One month	4	655	Risk Ratio (M-H, Random, 95% CI)	0.80 [0.34, 1.84]
1.2.2 Three months	4	655	Risk Ratio (M-H, Random, 95% CI)	0.83 [0.46, 1.52]
1.2.3 Six months	3	379	Risk Ratio (M-H, Random, 95% CI)	0.51 [0.24, 1.11]
1.2.4 Twelve months	3	455	Risk Ratio (M-H, Random, 95% CI)	1.06 [0.38, 2.91]
1.3 Any continuing kidney disease at different time points (study with high risk of bias excluded)	3		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.3.1 One month	3	487	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.54, 1.93]
1.3.2 Three months	3	487	Risk Ratio (M-H, Random, 95% CI)	0.98 [0.70, 1.36]
1.3.3 Six months	2	211	Risk Ratio (M-H, Random, 95% CI)	0.59 [0.23, 1.50]
1.3.4 Twelve months	2	287	Risk Ratio (M-H, Random, 95% CI)	1.39 [0.75, 2.59]
1.4 Number of children with kidney disease in first month/with kidney disease at follow-up	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.4.1 One month	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.4.2 Three months	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.4.3 Six months	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.5 Number developing severe kidney disease	2	418	Risk Ratio (M-H, Random, 95% CI)	1.58 [0.42, 6.00]
1.6 Duration of kidney disease	1		Mean Difference (IV, Random, 95% CI)	Totals not selected
1.6.1 Haematuria	1		Mean Difference (IV, Random, 95% CI)	Totals not selected
1.6.2 Proteinuria	1		Mean Difference (IV, Random, 95% CI)	Totals not selected
1.7 Gastrointestinal complications requiring hospital admission	3	517	Risk Ratio (M-H, Random, 95% CI)	0.56 [0.25, 1.23]
1.8 Eight-year outcomes	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.8.1 Haematuria	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.8.2 Proteinuria	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.8.3 Haematuria and proteinuria	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.8.4 Hypertension	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.8.5 Decreased GFR (Schwartz formula)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

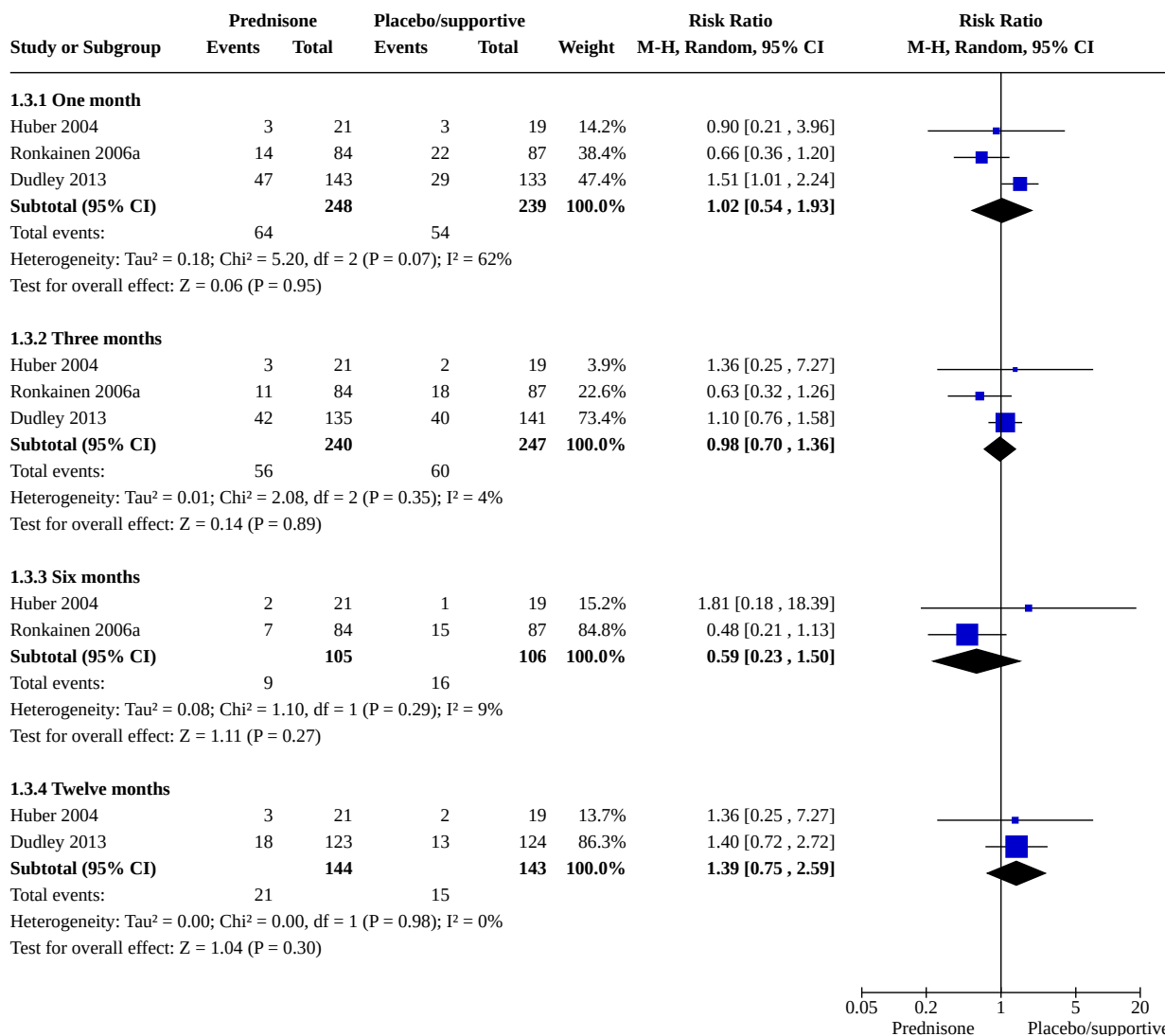
Analysis 1.1. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 1: Persistent kidney disease at any time after treatment



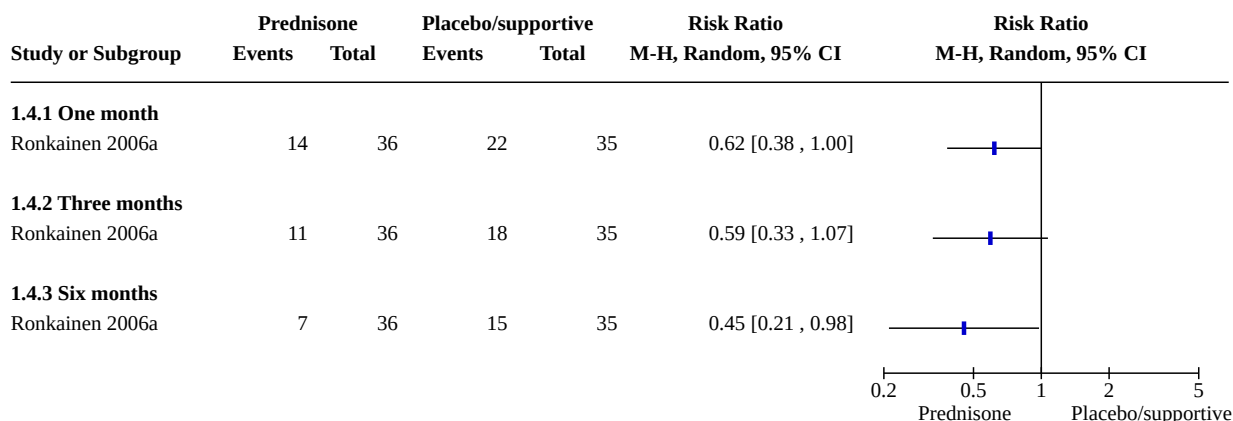
Analysis 1.2. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 2: Number of children with any continuing kidney disease at different time points



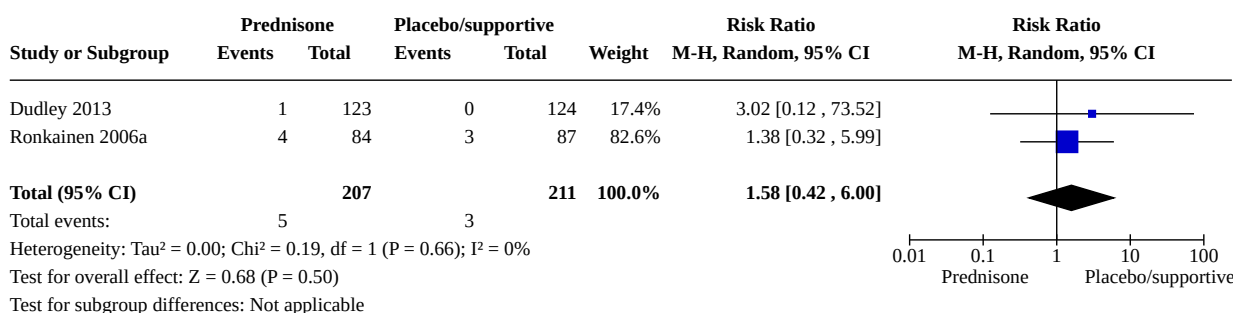
Analysis 1.3. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 3: Any continuing kidney disease at different time points (study with high risk of bias excluded)



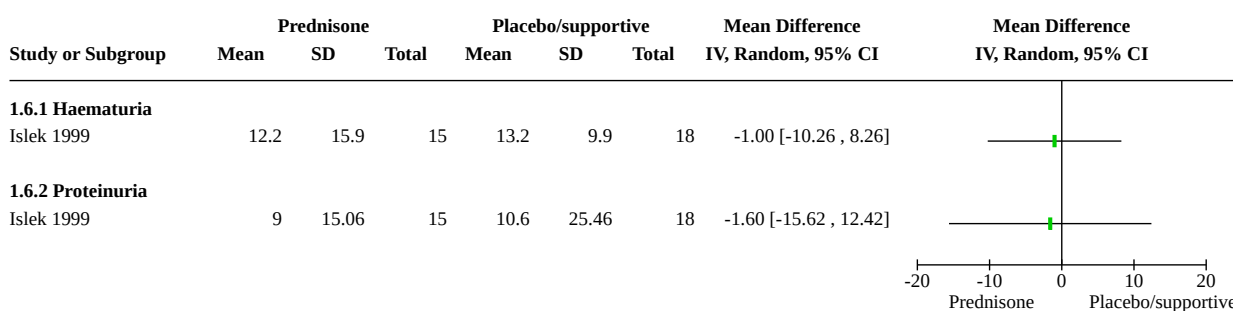
Analysis 1.4. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 4: Number of children with kidney disease in first month/with kidney disease at follow-up

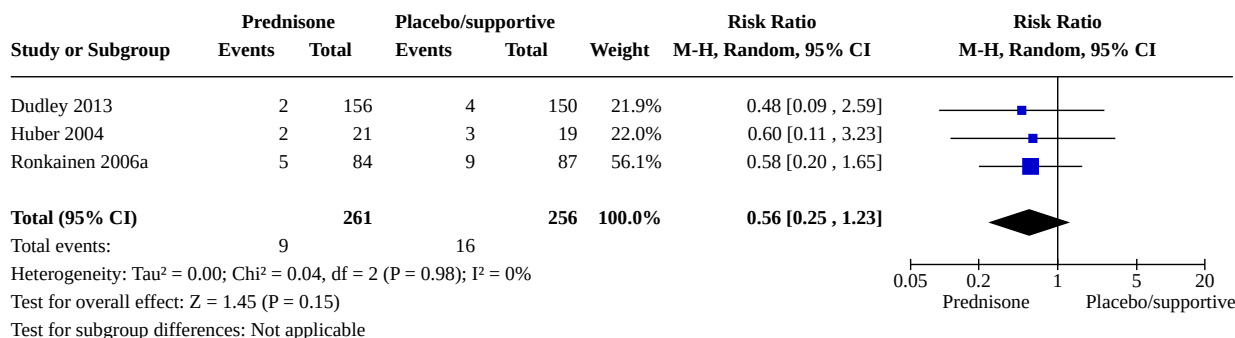
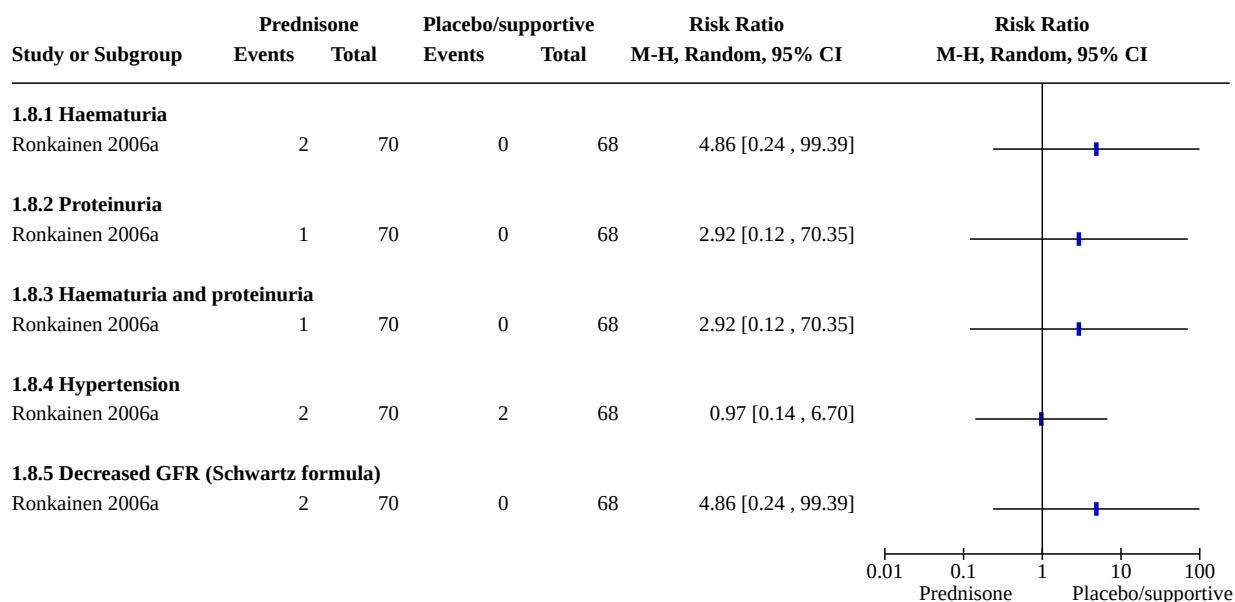


Analysis 1.5. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 5: Number developing severe kidney disease

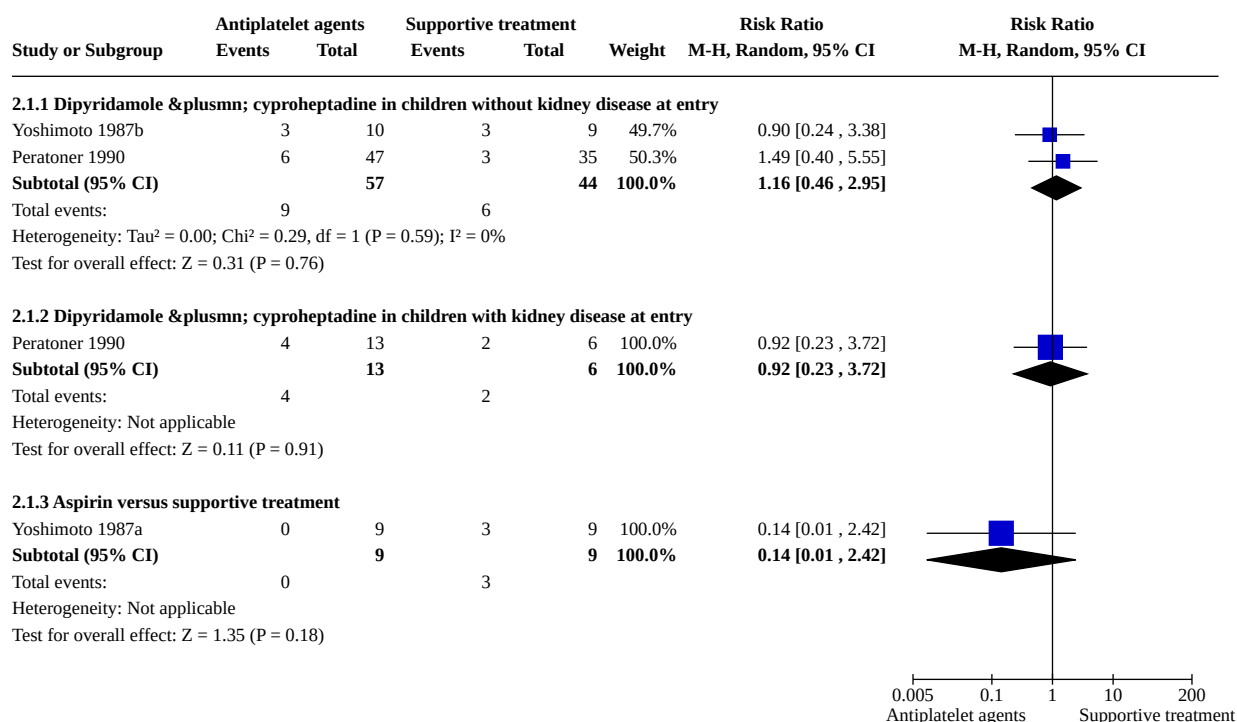


Analysis 1.6. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 6: Duration of kidney disease

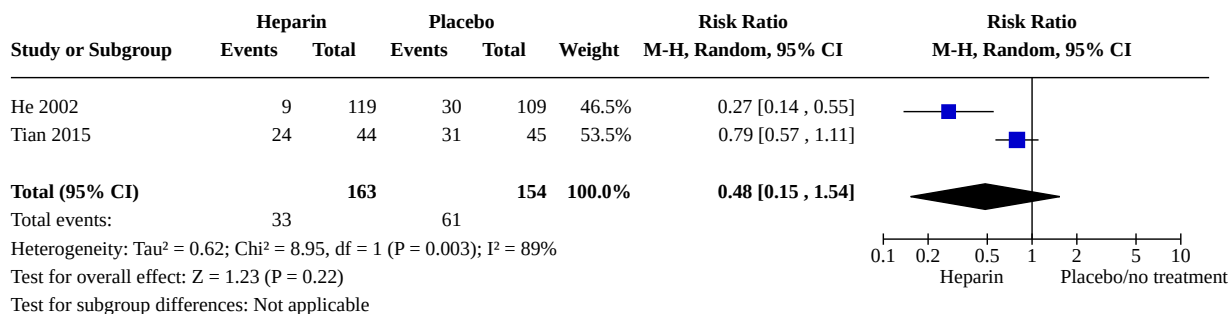
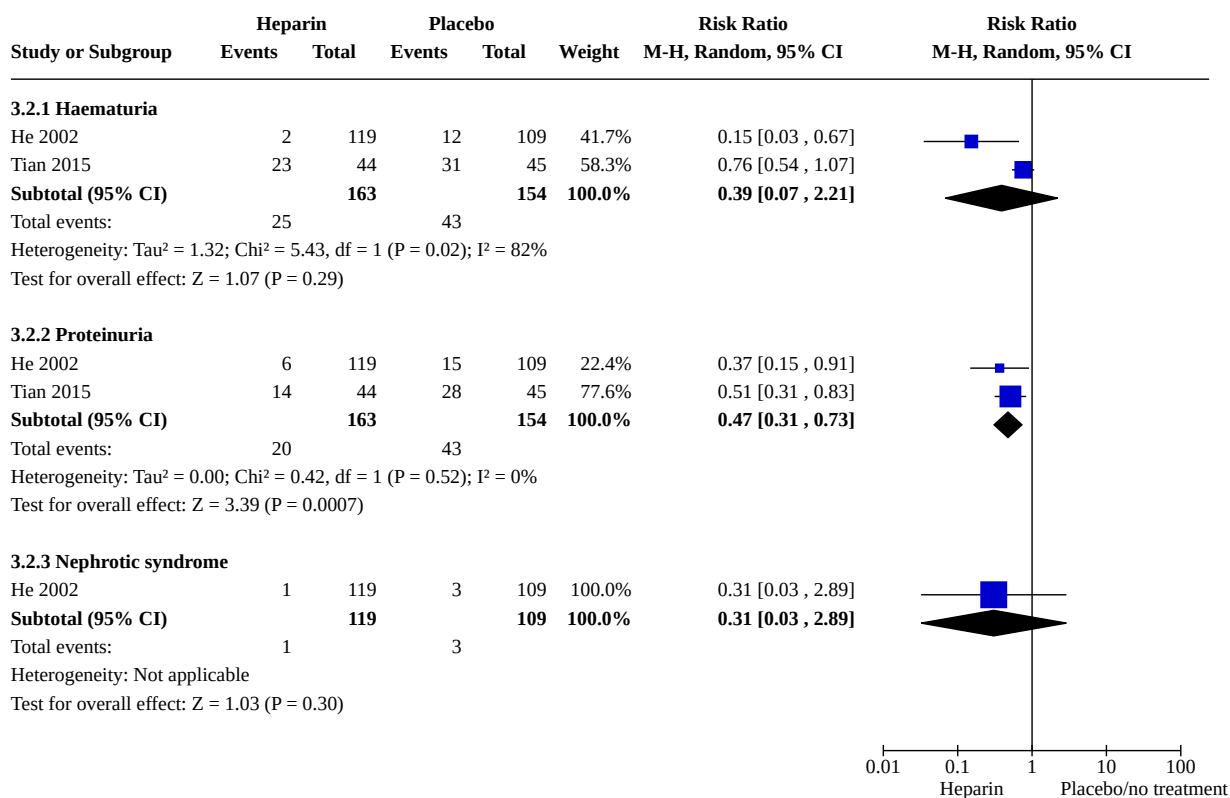
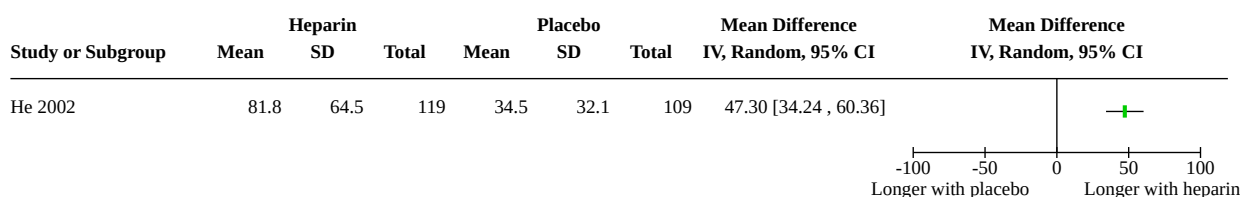


Analysis 1.7. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 7: Gastrointestinal complications requiring hospital admission**Analysis 1.8. Comparison 1: Prednisone versus placebo/supportive treatment for preventing persistent kidney disease, Outcome 8: Eight-year outcomes****Comparison 2. Antiplatelet agents versus supportive treatment for preventing persistent kidney disease**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Kidney disease at any time	3		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.1.1 Dipyridamole ± cyproheptadine in children without kidney disease at entry	2	101	Risk Ratio (M-H, Random, 95% CI)	1.16 [0.46, 2.95]
2.1.2 Dipyridamole ± cyproheptadine in children with kidney disease at entry	1	19	Risk Ratio (M-H, Random, 95% CI)	0.92 [0.23, 3.72]
2.1.3 Aspirin versus supportive treatment	1	18	Risk Ratio (M-H, Random, 95% CI)	0.14 [0.01, 2.42]

Analysis 2.1. Comparison 2: Antiplatelet agents versus supportive treatment for preventing persistent kidney disease, Outcome 1: Kidney disease at any time**Comparison 3. Heparin versus placebo for preventing persistent kidney disease**

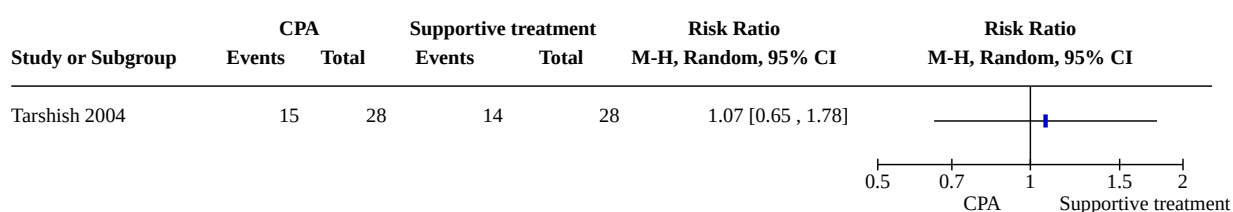
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1 Any kidney disease at 2 to 3 months after onset or relapse	2	317	Risk Ratio (M-H, Random, 95% CI)	0.48 [0.15, 1.54]
3.2 Type of kidney disease at 2 to 3 months or more after onset or relapse	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
3.2.1 Haematuria	2	317	Risk Ratio (M-H, Random, 95% CI)	0.39 [0.07, 2.21]
3.2.2 Proteinuria	2	317	Risk Ratio (M-H, Random, 95% CI)	0.47 [0.31, 0.73]
3.2.3 Nephrotic syndrome	1	228	Risk Ratio (M-H, Random, 95% CI)	0.31 [0.03, 2.89]
3.3 Time to development of kidney disease	1		Mean Difference (IV, Random, 95% CI)	Subtotals only

Analysis 3.1. Comparison 3: Heparin versus placebo for preventing persistent kidney disease, Outcome 1: Any kidney disease at 2 to 3 months after onset or relapse**Analysis 3.2. Comparison 3: Heparin versus placebo for preventing persistent kidney disease, Outcome 2: Type of kidney disease at 2 to 3 months or more after onset or relapse****Analysis 3.3. Comparison 3: Heparin versus placebo for preventing persistent kidney disease, Outcome 3: Time to development of kidney disease**

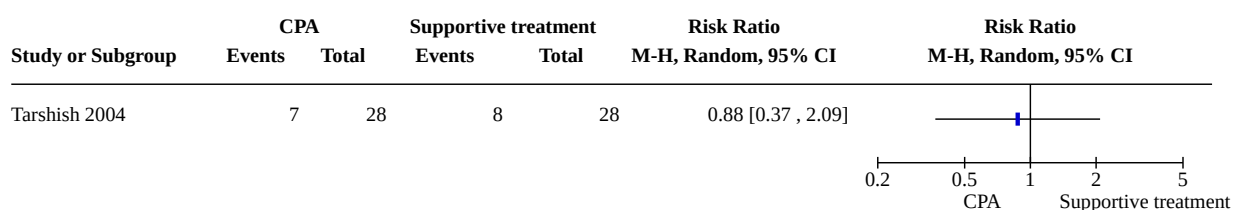
Comparison 4. Cyclophosphamide versus supportive treatment for treating severe kidney disease

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.1 Persistent kidney disease	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
4.2 Persistent severe kidney disease	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
4.3 ESKD	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

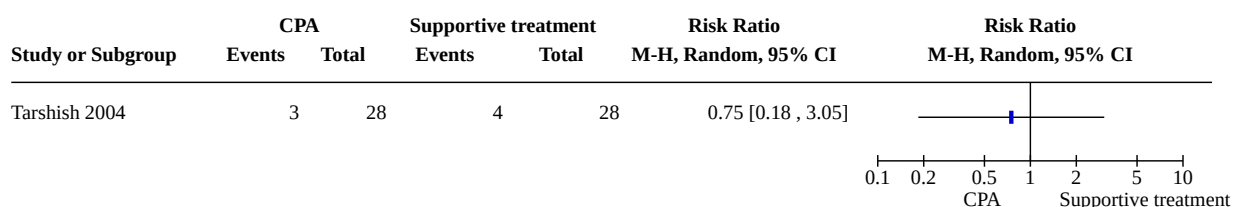
Analysis 4.1. Comparison 4: Cyclophosphamide versus supportive treatment for treating severe kidney disease, Outcome 1: Persistent kidney disease



Analysis 4.2. Comparison 4: Cyclophosphamide versus supportive treatment for treating severe kidney disease, Outcome 2: Persistent severe kidney disease



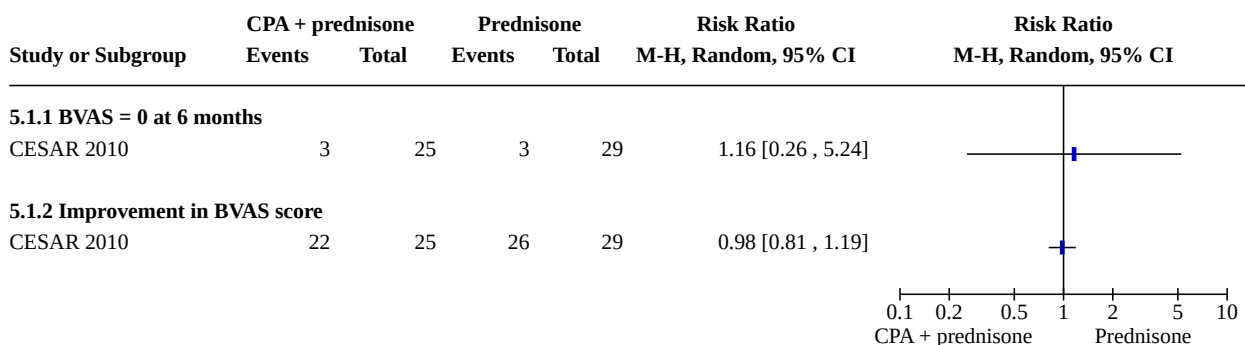
Analysis 4.3. Comparison 4: Cyclophosphamide versus supportive treatment for treating severe kidney disease, Outcome 3: ESKD



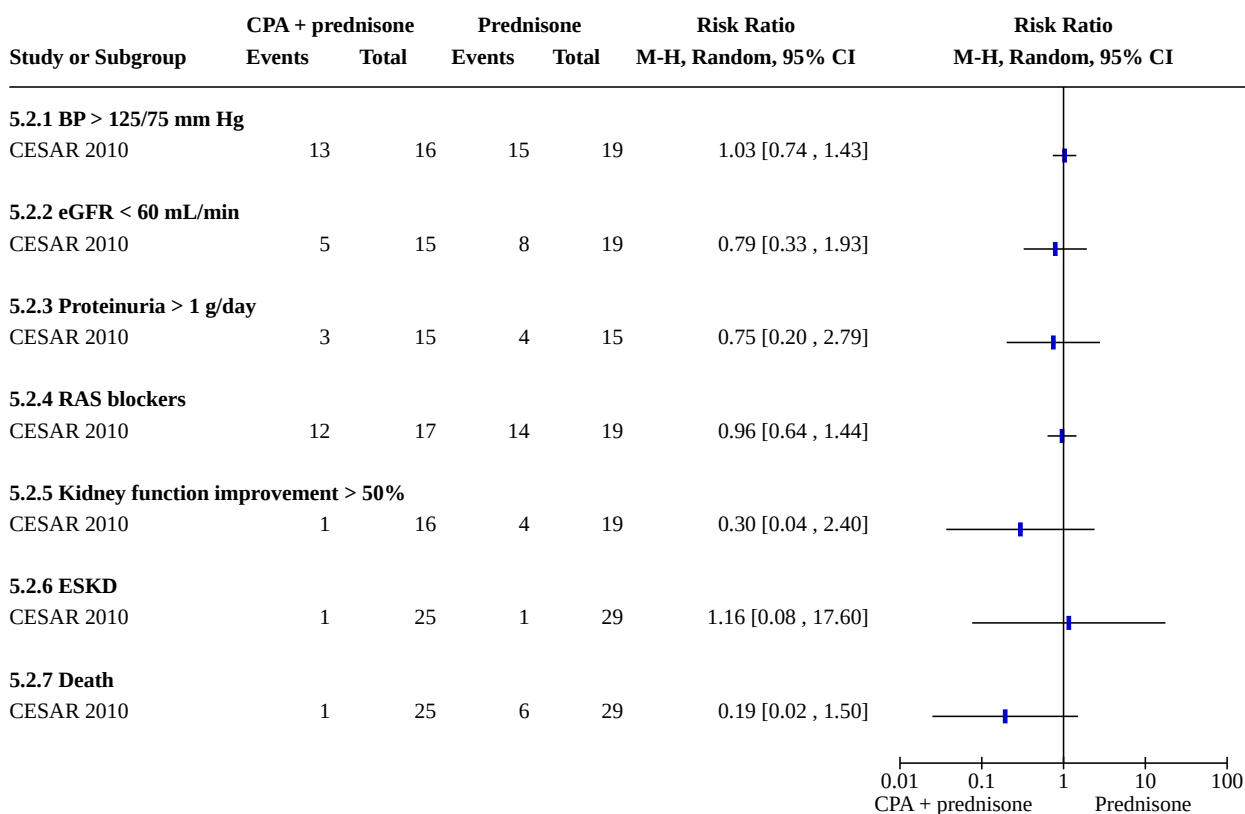
Comparison 5. Cyclophosphamide + steroids versus steroids for treating severe kidney disease

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.1 Primary outcome: BVAS at 6 months	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.1.1 BVAS = 0 at 6 months	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.1.2 Improvement in BVAS score	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2 Secondary endpoints at 12 months	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2.1 BP > 125/75 mm Hg	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2.2 eGFR < 60 mL/min	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2.3 Proteinuria > 1 g/day	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2.4 RAS blockers	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2.5 Kidney function improvement > 50%	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2.6 ESKD	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.2.7 Death	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.3 Adverse effects	1		Risk Difference (M-H, Random, 95% CI)	Totals not selected
5.3.1 infection	1		Risk Difference (M-H, Random, 95% CI)	Totals not selected
5.3.2 Newly diagnosed or deterioration in existing diabetes	1		Risk Difference (M-H, Random, 95% CI)	Totals not selected
5.3.3 Depression/anxiety	1		Risk Difference (M-H, Random, 95% CI)	Totals not selected
5.3.4 Alopecia	1		Risk Difference (M-H, Random, 95% CI)	Totals not selected
5.3.5 Insomnia	1		Risk Difference (M-H, Random, 95% CI)	Totals not selected

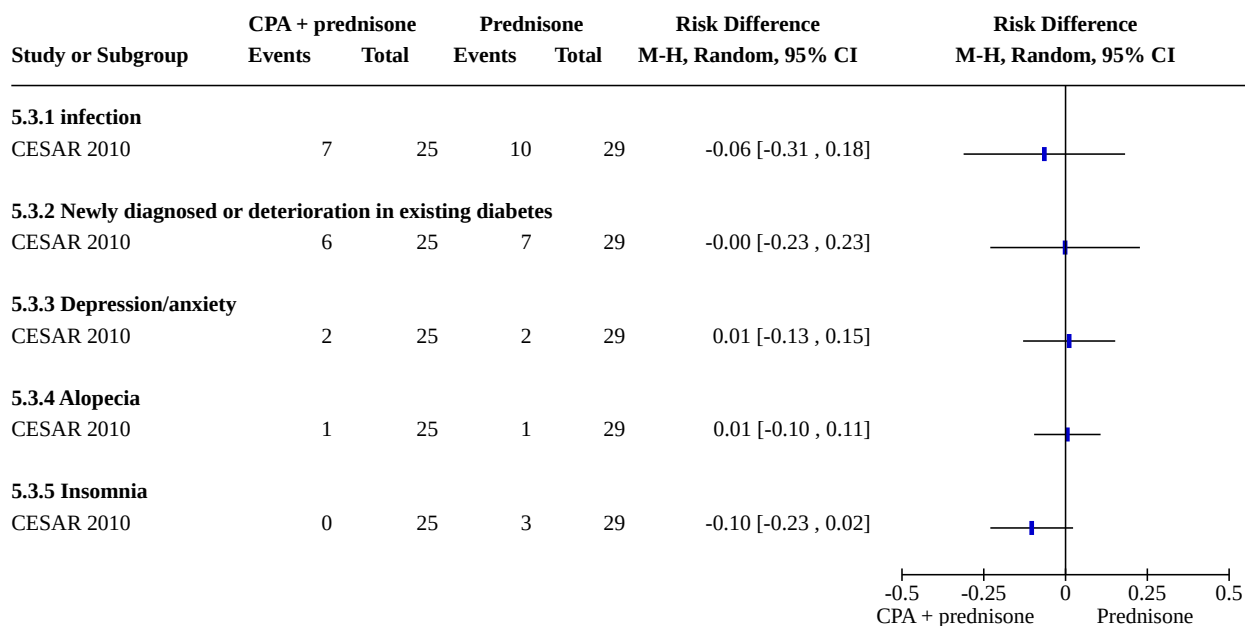
Analysis 5.1. Comparison 5: Cyclophosphamide + steroids versus steroids for treating severe kidney disease, Outcome 1: Primary outcome: BVAS at 6 months



Analysis 5.2. Comparison 5: Cyclophosphamide + steroids versus steroids for treating severe kidney disease, Outcome 2: Secondary endpoints at 12 months



Analysis 5.3. Comparison 5: Cyclophosphamide + steroids versus steroids for treating severe kidney disease, Outcome 3: Adverse effects

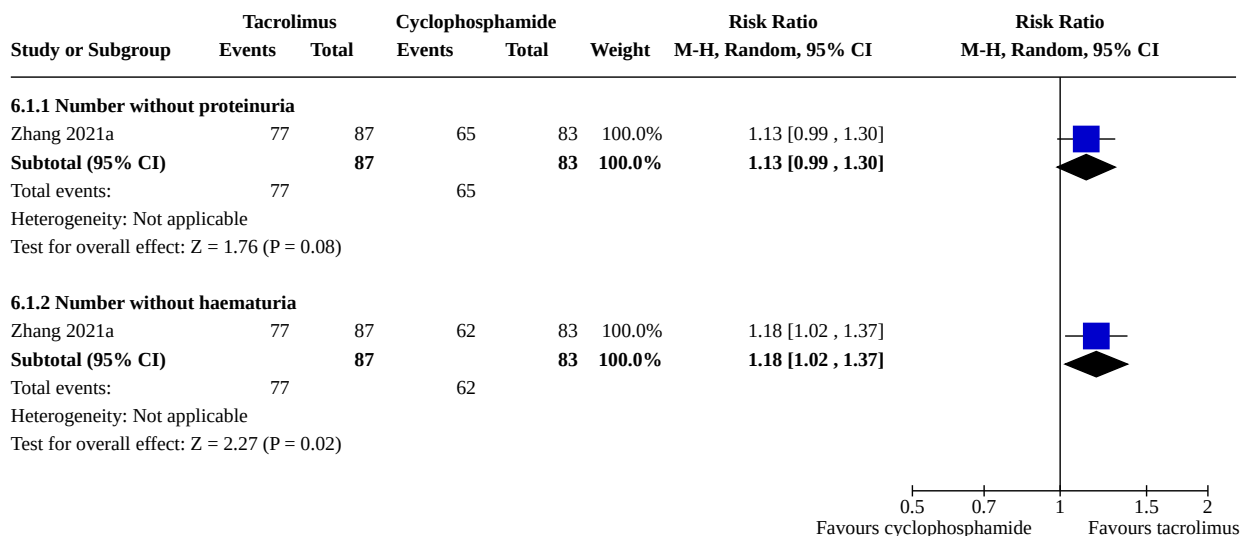


Comparison 6. Tacrolimus versus IV cyclophosphamide for treating severe kidney disease

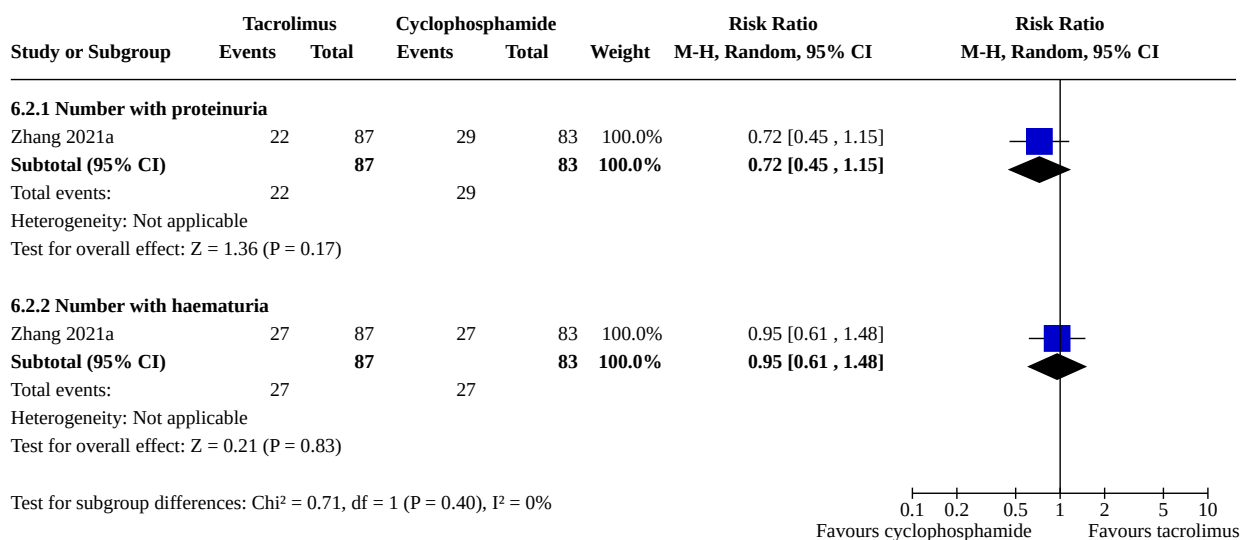
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.1 Number with resolution of proteinuria and haematuria at 2 years	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
6.1.1 Number without proteinuria	1	170	Risk Ratio (M-H, Random, 95% CI)	1.13 [0.99, 1.30]
6.1.2 Number without haematuria	1	170	Risk Ratio (M-H, Random, 95% CI)	1.18 [1.02, 1.37]
6.2 Number with recurrence of proteinuria and haematuria at 2 years	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
6.2.1 Number with proteinuria	1	170	Risk Ratio (M-H, Random, 95% CI)	0.72 [0.45, 1.15]
6.2.2 Number with haematuria	1	170	Risk Ratio (M-H, Random, 95% CI)	0.95 [0.61, 1.48]
6.3 Laboratory parameters	1		Mean Difference (IV, Random, 95% CI)	Subtotals only
6.3.1 Serum creatinine at 3 months	1	170	Mean Difference (IV, Random, 95% CI)	-1.00 [-2.80, 0.80]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.3.2 Serum creatinine at 6 months	1	170	Mean Difference (IV, Random, 95% CI)	-1.00 [-2.06, 0.06]
6.3.3 24-hour urinary protein at 3 months	1	170	Mean Difference (IV, Random, 95% CI)	-0.31 [-0.40, -0.22]
6.3.4 24-hour urinary protein at 6 months	1	170	Mean Difference (IV, Random, 95% CI)	-0.20 [-0.24, -0.16]
6.3.5 Urine RBC/HPF at 3 months	1	170	Mean Difference (IV, Random, 95% CI)	-2.62 [-3.79, -1.45]
6.3.6 Urine RBC/HPF at 6 months	1	170	Mean Difference (IV, Random, 95% CI)	-2.45 [-2.90, -2.00]
6.4 Adverse effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
6.4.1 Leucopenia	1	170	Risk Ratio (M-H, Random, 95% CI)	0.64 [0.19, 2.17]
6.4.2 Respiratory infections	1	170	Risk Ratio (M-H, Random, 95% CI)	0.55 [0.38, 0.82]
6.4.3 Abnormal liver function tests	1	170	Risk Ratio (M-H, Random, 95% CI)	0.92 [0.58, 1.45]
6.4.4 Urinary tract infection	1	170	Risk Ratio (M-H, Random, 95% CI)	0.56 [0.27, 1.15]
6.4.5 Poor appetite	1	170	Risk Ratio (M-H, Random, 95% CI)	0.82 [0.47, 1.42]
6.4.6 Other serious adverse effect	1	170	Risk Ratio (M-H, Random, 95% CI)	0.45 [0.20, 0.98]

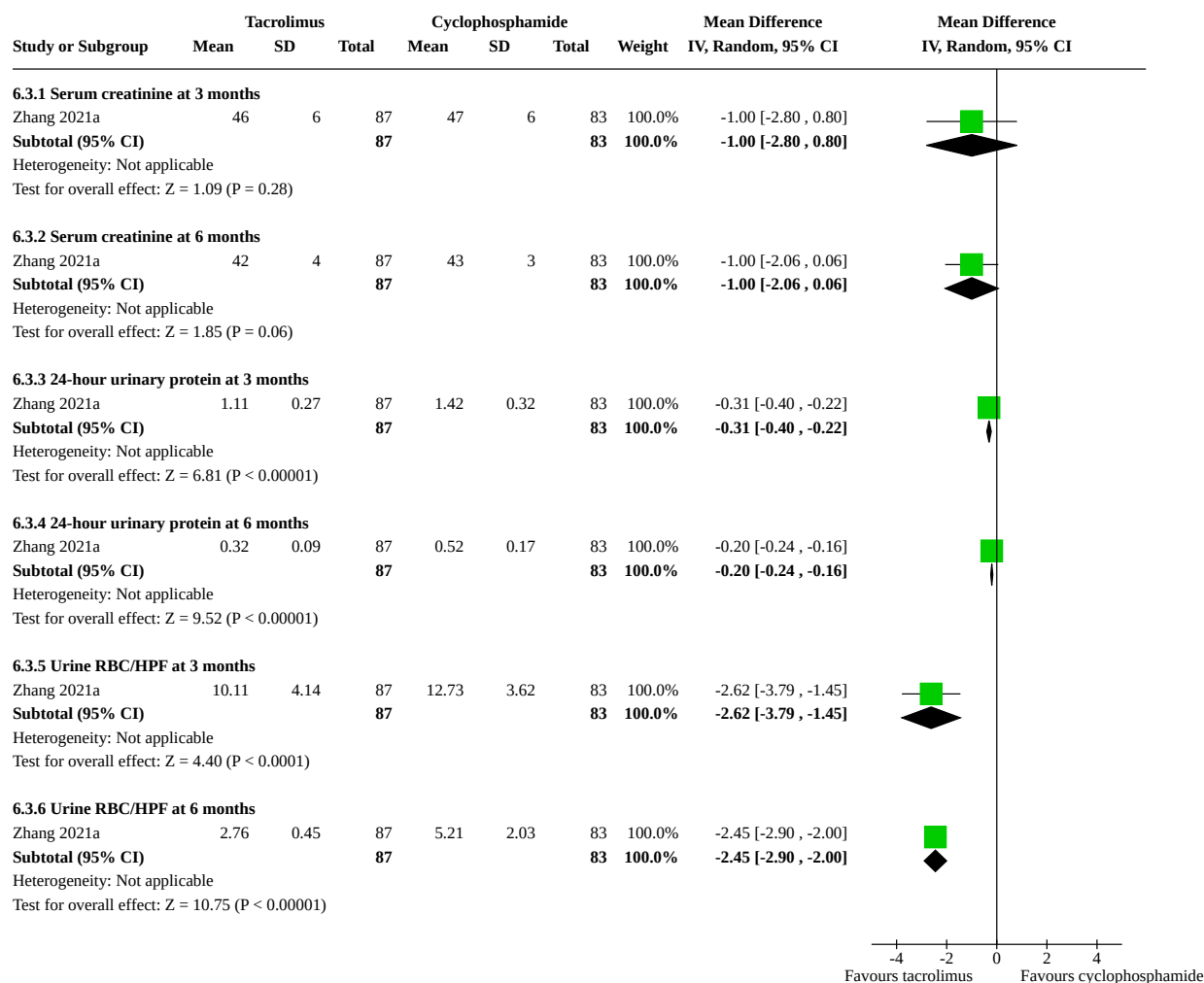
Analysis 6.1. Comparison 6: Tacrolimus versus IV cyclophosphamide for treating severe kidney disease, Outcome 1: Number with resolution of proteinuria and haematuria at 2 years

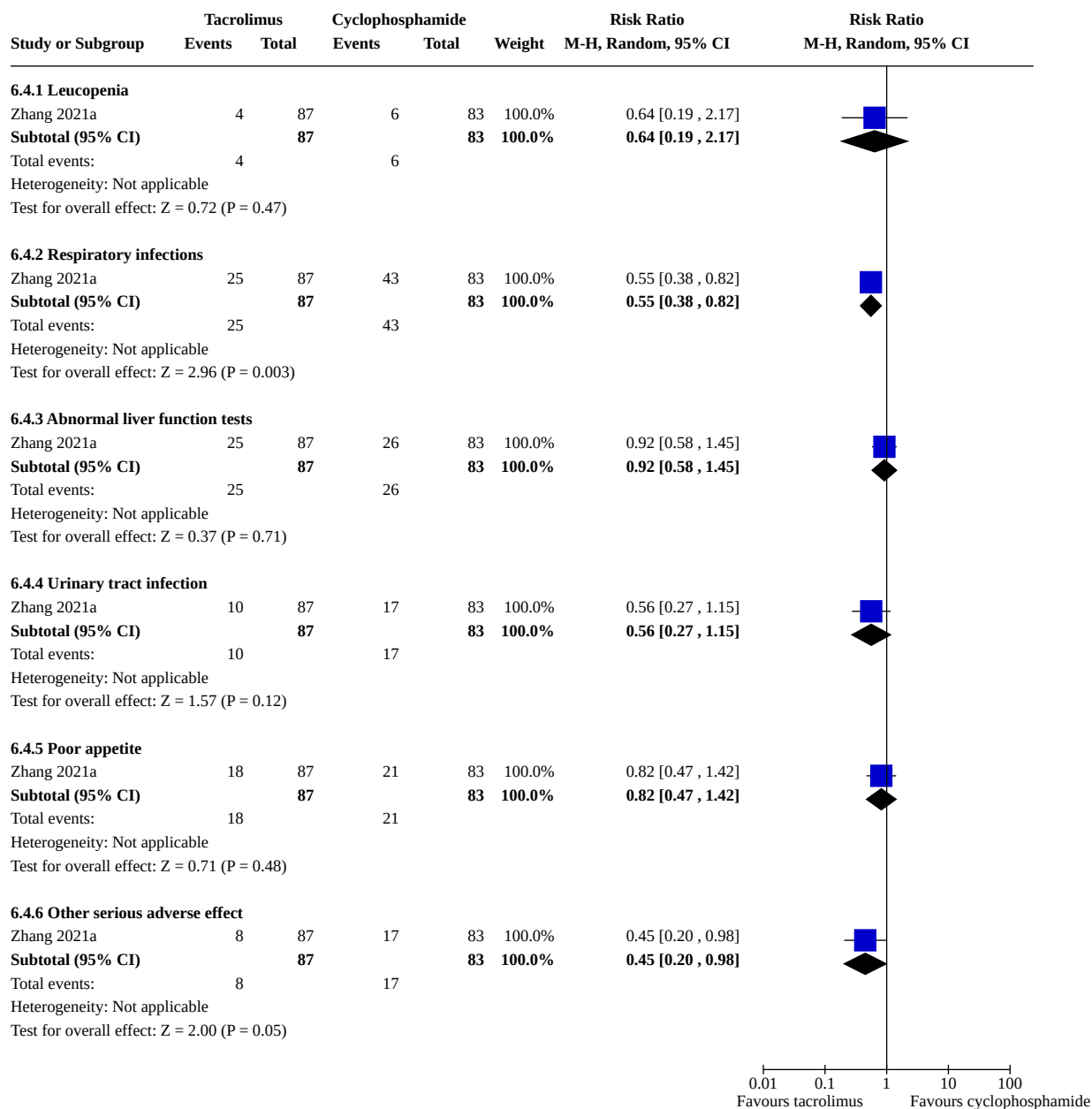


Analysis 6.2. Comparison 6: Tacrolimus versus IV cyclophosphamide for treating severe kidney disease, Outcome 2: Number with recurrence of proteinuria and haematuria at 2 years



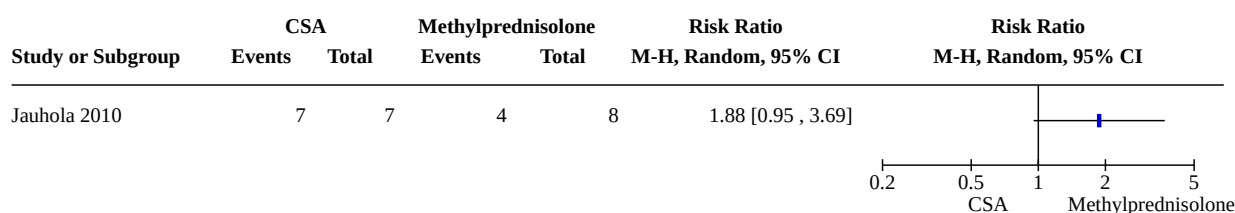
Analysis 6.3. Comparison 6: Tacrolimus versus IV cyclophosphamide for treating severe kidney disease, Outcome 3: Laboratory parameters



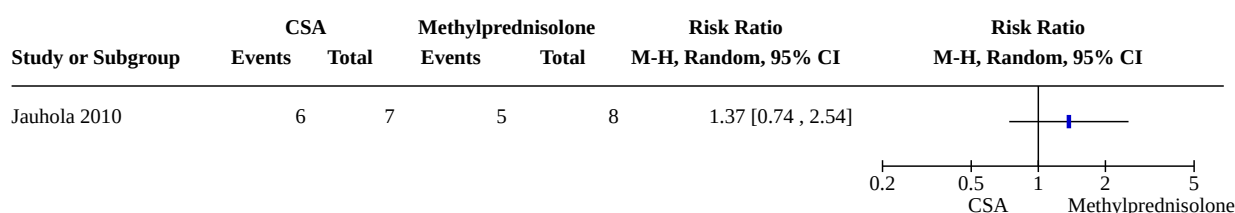
Analysis 6.4. Comparison 6: Tacrolimus versus IV cyclophosphamide for treating severe kidney disease, Outcome 4: Adverse effects**Comparison 7. Cyclosporin versus methylprednisolone for treating severe kidney disease**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
7.1 Number with remission at 3 months	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
7.2 Number with remission at last follow-up (mean 6.3 years)	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

Analysis 7.1. Comparison 7: Cyclosporin versus methylprednisolone for treating severe kidney disease, Outcome 1: Number with remission at 3 months



Analysis 7.2. Comparison 7: Cyclosporin versus methylprednisolone for treating severe kidney disease, Outcome 2: Number with remission at last follow-up (mean 6.3 years)

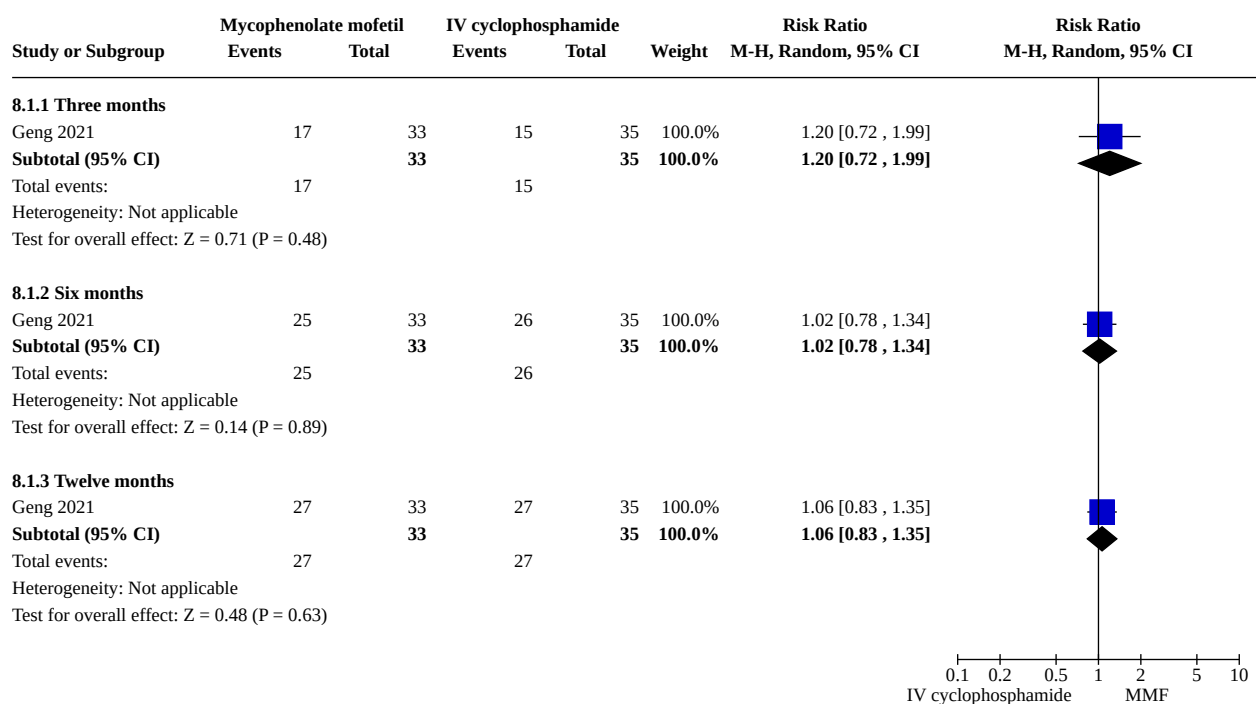


Comparison 8. Mycophenolate mofetil versus IV cyclophosphamide for treating severe kidney disease

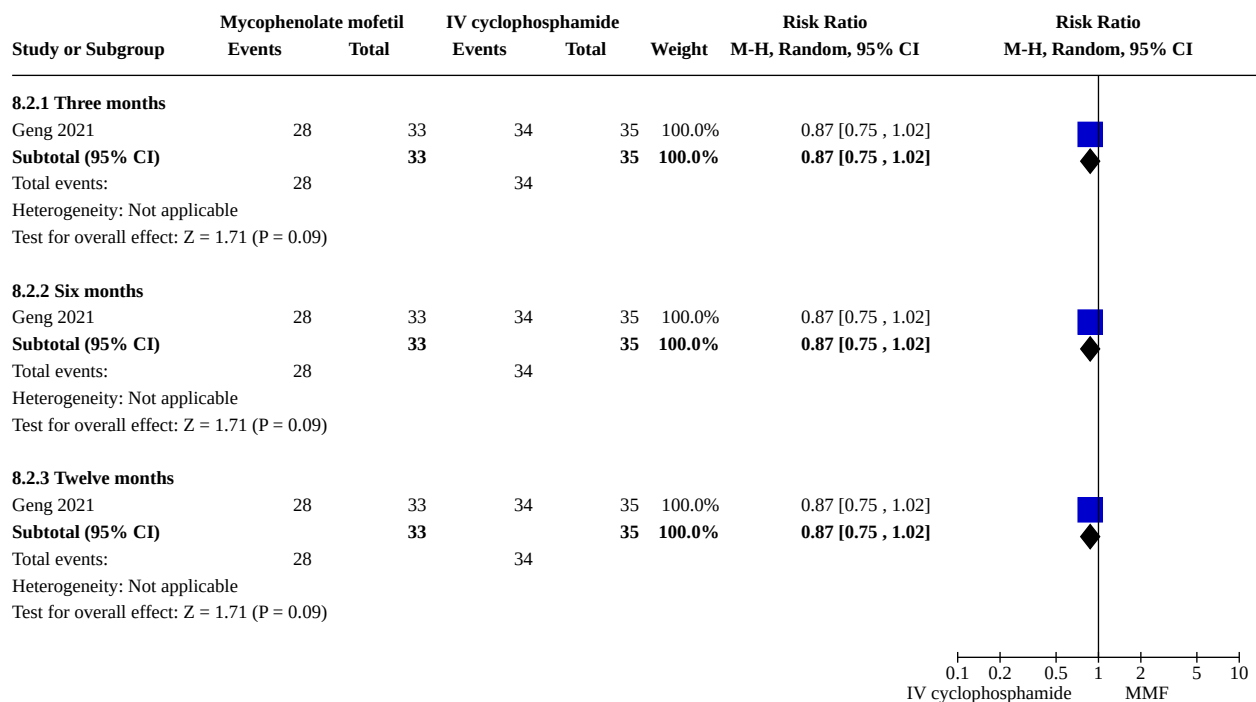
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.1 Number with complete remission	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
8.1.1 Three months	1	68	Risk Ratio (M-H, Random, 95% CI)	1.20 [0.72, 1.99]
8.1.2 Six months	1	68	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.78, 1.34]
8.1.3 Twelve months	1	68	Risk Ratio (M-H, Random, 95% CI)	1.06 [0.83, 1.35]
8.2 Number with complete or partial remission	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
8.2.1 Three months	1	68	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.75, 1.02]
8.2.2 Six months	1	68	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.75, 1.02]
8.2.3 Twelve months	1	68	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.75, 1.02]
8.3 Adverse effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
8.3.1 Cerebral abscess	1	68	Risk Ratio (M-H, Random, 95% CI)	0.35 [0.01, 8.37]
8.3.2 Abnormal LFTS requiring treatment change	1	68	Risk Ratio (M-H, Random, 95% CI)	5.29 [0.26, 106.33]
8.3.3 Infection	1	68	Risk Ratio (M-H, Random, 95% CI)	0.71 [0.37, 1.35]

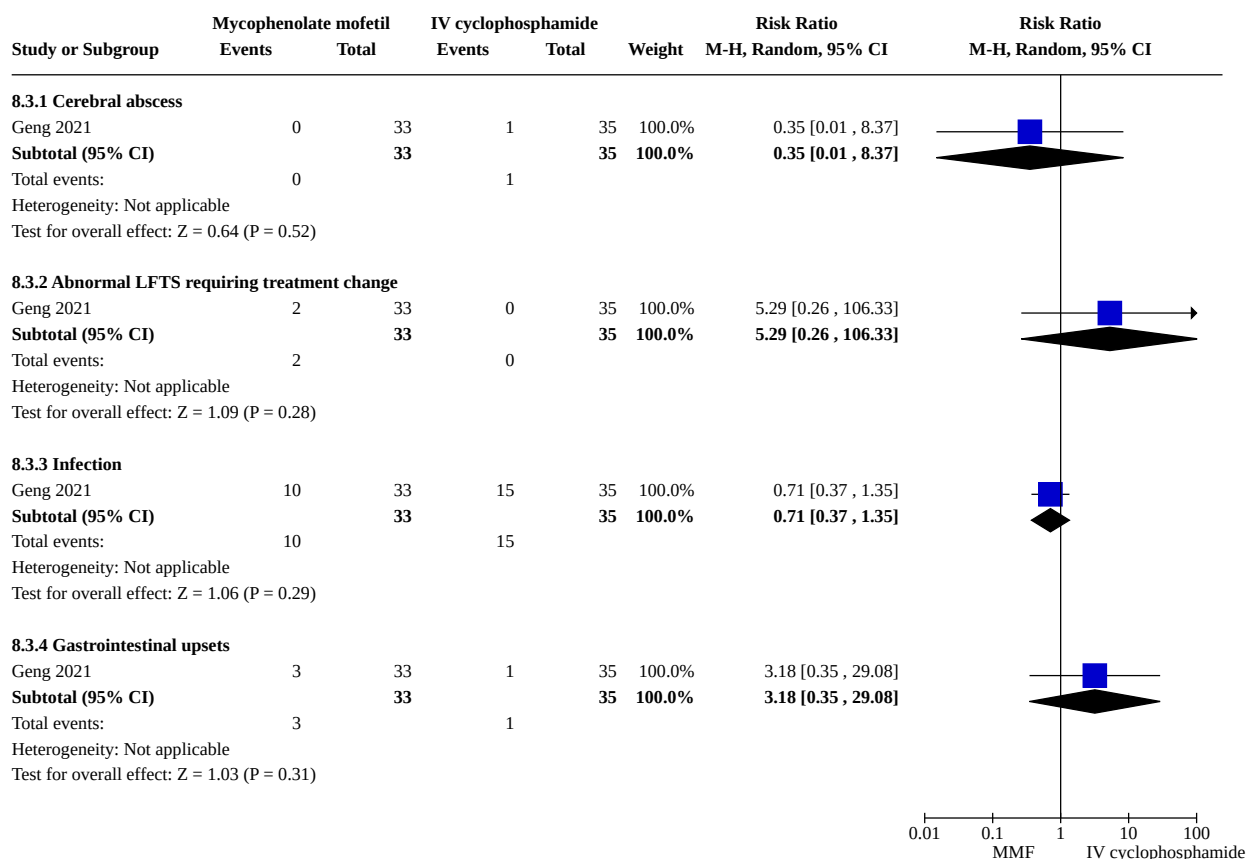
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.3.4 Gastrointestinal upsets	1	68	Risk Ratio (M-H, Random, 95% CI)	3.18 [0.35, 29.08]

Analysis 8.1. Comparison 8: Mycophenolate mofetil versus IV cyclophosphamide for treating severe kidney disease, Outcome 1: Number with complete remission

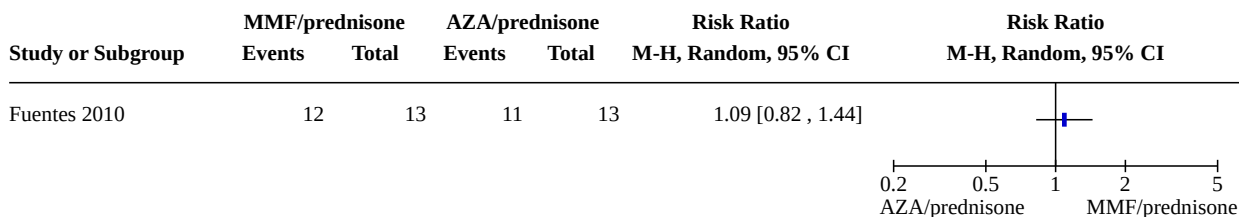
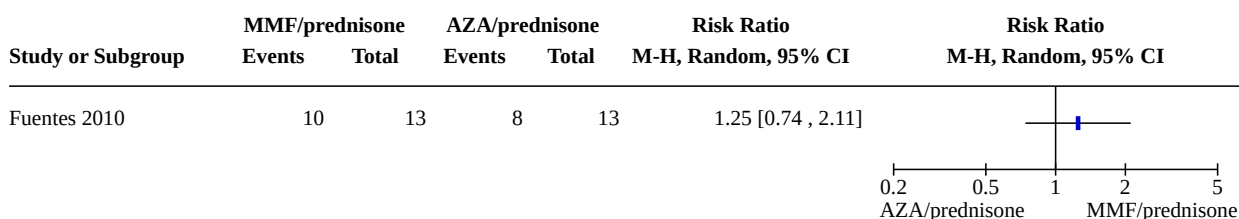
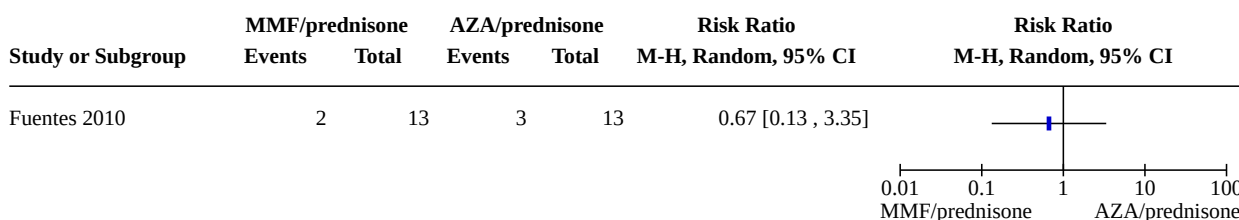
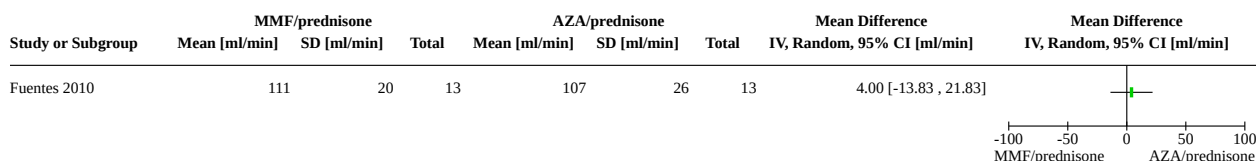


Analysis 8.2. Comparison 8: Mycophenolate mofetil versus IV cyclophosphamide for treating severe kidney disease, Outcome 2: Number with complete or partial remission



Analysis 8.3. Comparison 8: Mycophenolate mofetil versus IV cyclophosphamide for treating severe kidney disease, Outcome 3: Adverse effects**Comparison 9. Mycophenolate mofetil versus azathioprine for treating severe kidney disease**

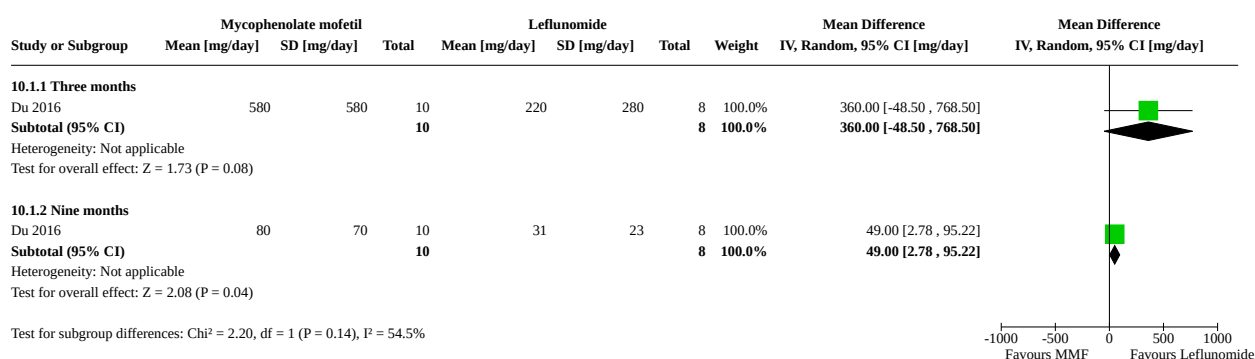
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
9.1 Remission of proteinuria at 1 year	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
9.2 Regression of histological lesions at 1 year	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
9.3 Relapse of IgAV	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
9.4 Glomerular filtration rate	1		Mean Difference (IV, Random, 95% CI)	Totals not selected

Analysis 9.1. Comparison 9: Mycophenolate mofetil versus azathioprine for treating severe kidney disease, Outcome 1: Remission of proteinuria at 1 year**Analysis 9.2. Comparison 9: Mycophenolate mofetil versus azathioprine for treating severe kidney disease, Outcome 2: Regression of histological lesions at 1 year****Analysis 9.3. Comparison 9: Mycophenolate mofetil versus azathioprine for treating severe kidney disease, Outcome 3: Relapse of IgAV****Analysis 9.4. Comparison 9: Mycophenolate mofetil versus azathioprine for treating severe kidney disease, Outcome 4: Glomerular filtration rate****Comparison 10. Mycophenolate mofetil versus leflunomide for treating severe kidney disease**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
10.1 24-hour urine proteinuria	1		Mean Difference (IV, Random, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
10.1.1 Three months	1	18	Mean Difference (IV, Random, 95% CI)	360.00 [-48.50, 768.50]
10.1.2 Nine months	1	18	Mean Difference (IV, Random, 95% CI)	49.00 [2.78, 95.22]

Analysis 10.1. Comparison 10: Mycophenolate mofetil versus leflunomide for treating severe kidney disease, Outcome 1: 24-hour urine proteinuria

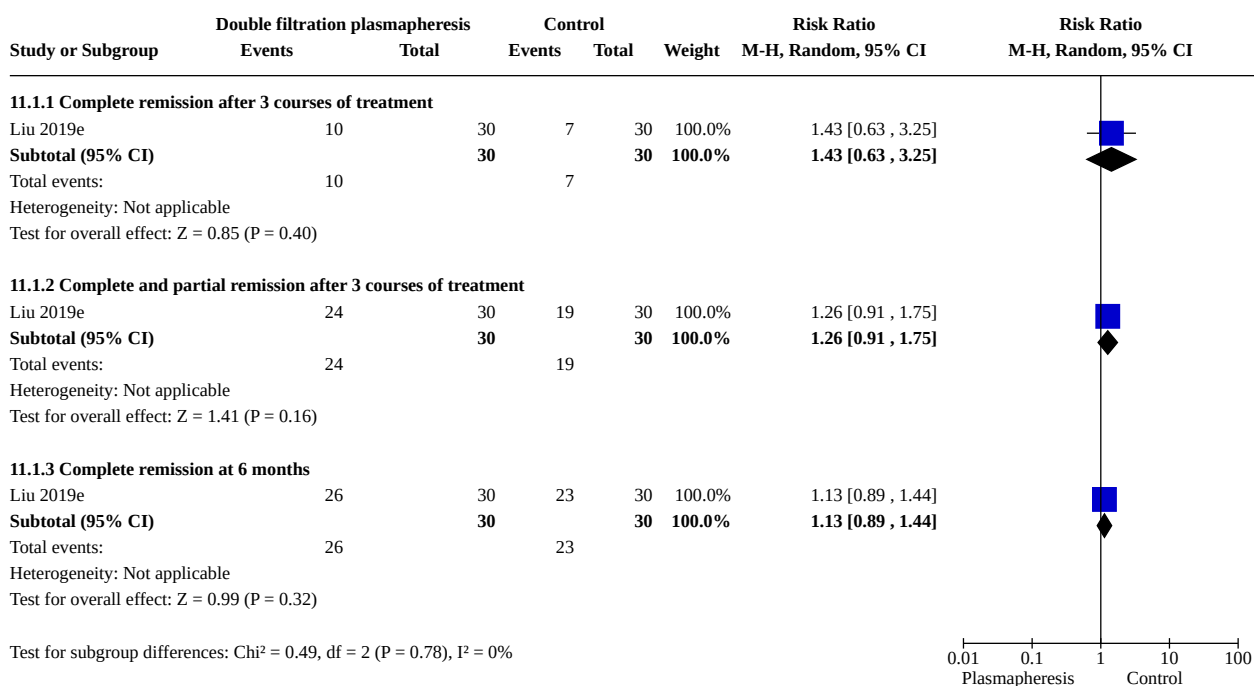


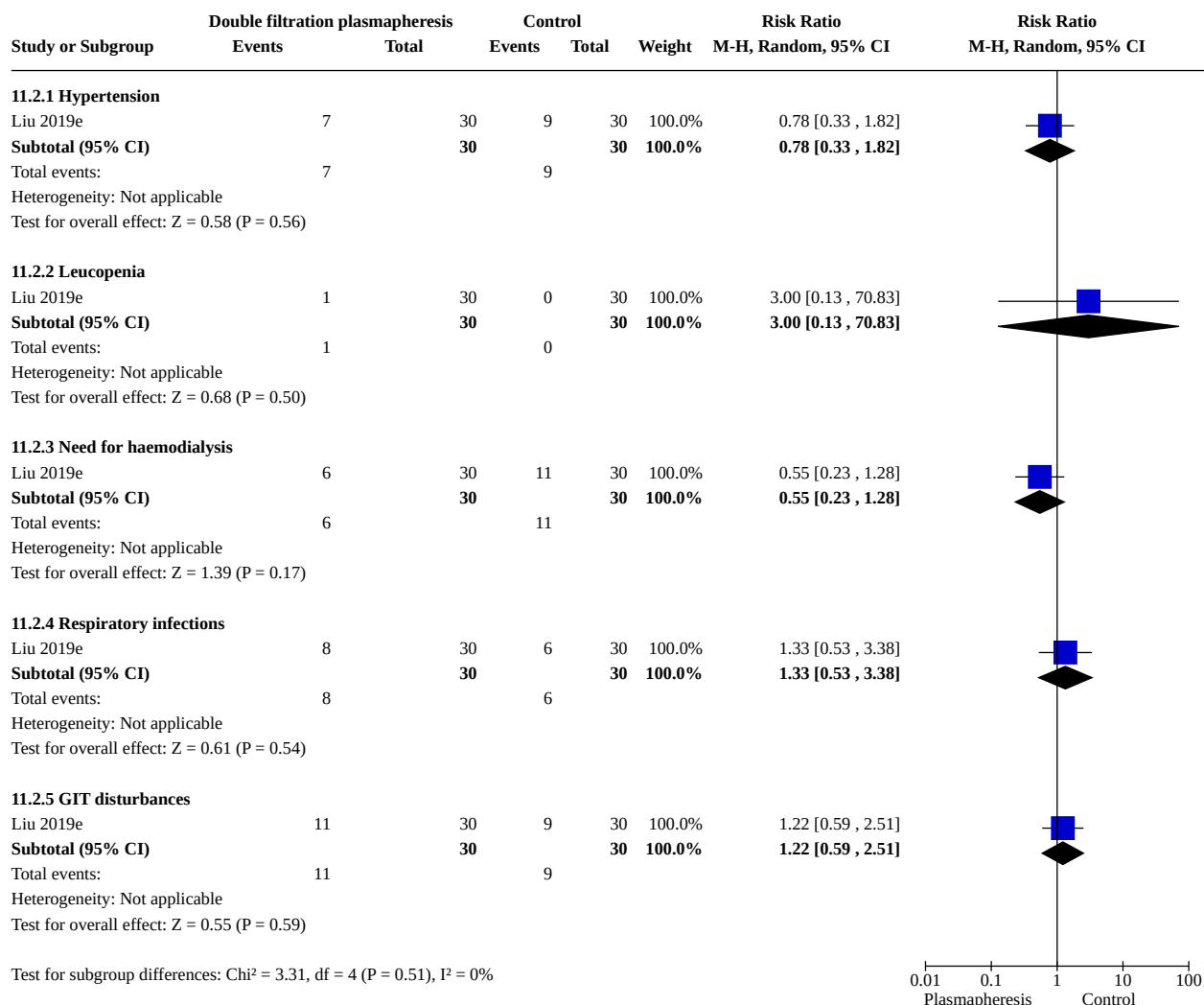
Comparison 11. Double filtration plasmapheresis versus no plasmapheresis for treating severe kidney disease

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
11.1 Remission	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
11.1.1 Complete remission after 3 courses of treatment	1	60	Risk Ratio (M-H, Random, 95% CI)	1.43 [0.63, 3.25]
11.1.2 Complete and partial remission after 3 courses of treatment	1	60	Risk Ratio (M-H, Random, 95% CI)	1.26 [0.91, 1.75]
11.1.3 Complete remission at 6 months	1	60	Risk Ratio (M-H, Random, 95% CI)	1.13 [0.89, 1.44]
11.2 Adverse effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
11.2.1 Hypertension	1	60	Risk Ratio (M-H, Random, 95% CI)	0.78 [0.33, 1.82]
11.2.2 Leucopenia	1	60	Risk Ratio (M-H, Random, 95% CI)	3.00 [0.13, 70.83]
11.2.3 Need for haemodialysis	1	60	Risk Ratio (M-H, Random, 95% CI)	0.55 [0.23, 1.28]

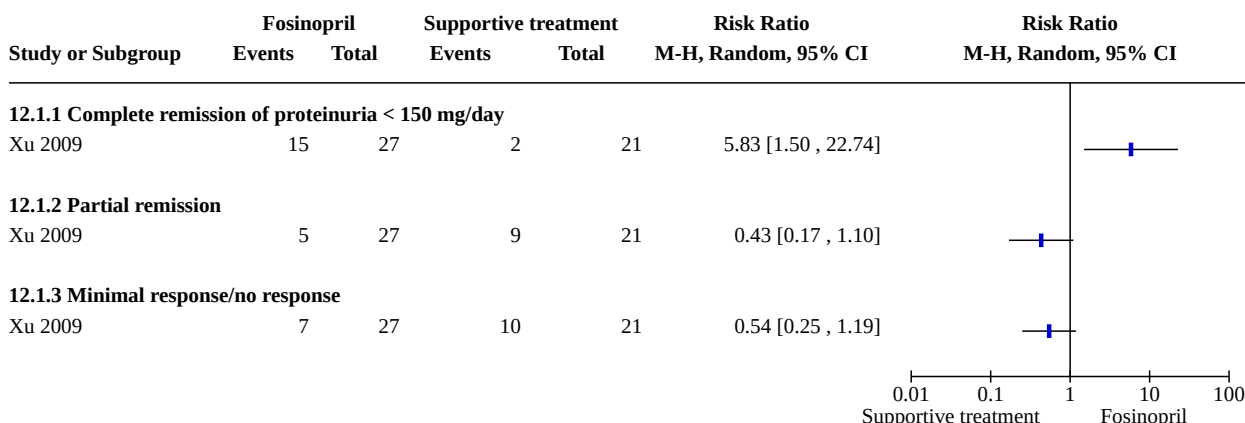
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
11.2.4 Respiratory infections	1	60	Risk Ratio (M-H, Random, 95% CI)	1.33 [0.53, 3.38]
11.2.5 GIT disturbances	1	60	Risk Ratio (M-H, Random, 95% CI)	1.22 [0.59, 2.51]

Analysis 11.1. Comparison 11: Double filtration plasmapheresis versus no plasmapheresis for treating severe kidney disease, Outcome 1: Remission



Analysis 11.2. Comparison 11: Double filtration plasmapheresis versus no plasmapheresis for treating severe kidney disease, Outcome 2: Adverse effects**Comparison 12. Fosinopril + supportive treatment versus supportive treatment for treating proteinuria in IgAV**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
12.1 Proteinuria	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
12.1.1 Complete remission of proteinuria < 150 mg/day	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
12.1.2 Partial remission	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
12.1.3 Minimal response/no response	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Analysis 12.1. Comparison 12: Fosinopril + supportive treatment versus supportive treatment for treating proteinuria in IgAV, Outcome 1: Proteinuria**APPENDICES****Appendix 1. Electronic search strategies**

Database	Search terms
CENTRAL	1. MeSH descriptor Purpura, Schoenlein-Henoch, this term only 2. henoch next scho*nlein next purpura:ti,ab,kw 3. allergic next purpura:ti,ab,kw 4. anaphylactoid next purpura:ti,ab,kw 5. henoch next purpura:ti,ab,kw 6. nonthrombocytopenic next purpura:ti,ab,kw 7. non next thrombocytopenic next purpura:ti,ab,kw 8. leukocytoclastic next vasculitis:ti,ab,kw 9. peliosis next rheumatica:ti,ab,kw 10.purpura next rheumatica:ti,ab,kw 11.rheumatoid next purpura:ti,ab,kw 12.scho*nlein next disease:ti,ab,k 13.(#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12)
MEDLINE	1. Purpura, Schoenlein-Henoch/ 2. henoch scho?nlein purpura.tw. 3. scho?nlein henoch purpura.tw. 4. allergic purpura.tw. 5. anaphylactoid purpura.tw. 6. henoch purpura.tw. 7. (nonthrombocytopenic purpura or non thrombocytopenic purpura).tw. 8. leukocytoclastic vasculitis.tw. 9. peliosis rheumatica.tw. 10.purpura rheumatica.tw. 11.rheumatoid purpura.tw. 12.scho?nlein disease.tw.

(Continued)

13.or/1-12

EMBASE

1. Anaphylactoid Purpura/
2. henoch scho?nlein.tw.
3. scho?nlein henoch.tw.
4. allergic purpura.tw.
5. anaphylactoid purpura.tw.
6. henoch purpura.tw.
7. nonthrombocytopenic purpura.tw.
8. non thrombocytopenic purpura.tw.
9. leukocytoclastic purpura.tw.
- 10.leukocytoclastic vasculitis.tw.
- 11.peliosis rheumatica.tw.
- 12.purpura rheumatica.tw.
- 13.rheumatoid purpura.tw.
- 14.scho?nlein disease.tw.
- 15.or/1-14

Appendix 2. Risk of bias assessment tool

Potential source of bias	Assessment criteria
Random sequence generation Selection bias (biased allocation to interventions) due to inadequate generation of a randomised sequence	<i>Low risk of bias:</i> Random number table; computer random number generator; coin tossing; shuffling cards or envelopes; throwing dice; drawing of lots; minimisation (minimisation may be implemented without a random element, and this is considered to be equivalent to being random). <i>High risk of bias:</i> Sequence generated by odd or even date of birth; date (or day) of admission; sequence generated by hospital or clinic record number; allocation by judgement of the clinician; by preference of the participant; based on the results of a laboratory test or a series of tests; by availability of the intervention. <i>Unclear:</i> Insufficient information about the sequence generation process to permit judgement.
Allocation concealment Selection bias (biased allocation to interventions) due to inadequate concealment of allocations prior to assignment	<i>Low risk of bias:</i> Randomisation method described that would not allow investigator/participant to know or influence intervention group before eligible participant entered in the study (e.g. central allocation, including telephone, web-based, and pharmacy-controlled, randomisation; sequentially numbered drug containers of identical appearance; sequentially numbered, opaque, sealed envelopes). <i>High risk of bias:</i> Using an open random allocation schedule (e.g. a list of random numbers); assignment envelopes were used without appropriate safeguards (e.g. if envelopes were unsealed or non-opaque or not sequentially numbered); alternation or rotation; date of birth; case record number; any other explicitly unconcealed procedure. <i>Unclear:</i> Randomisation stated but no information on method used is available.
Blinding of participants and personnel Performance bias due to knowledge of the allocated interventions by participants	<i>Low risk of bias:</i> No blinding or incomplete blinding, but the review authors judge that the outcome is not likely to be influenced by lack of blinding; blinding of participants and key study personnel ensured, and unlikely that the blinding could have been broken.

(Continued)

and personnel during the study

High risk of bias: No blinding or incomplete blinding, and the outcome is likely to be influenced by lack of blinding; blinding of key study participants and personnel attempted, but likely that the blinding could have been broken, and the outcome is likely to be influenced by lack of blinding.

Unclear: Insufficient information to permit judgement

Blinding of outcome assessment

Detection bias due to knowledge of the allocated interventions by outcome assessors.

Low risk of bias: No blinding of outcome assessment, but the review authors judge that the outcome measurement is not likely to be influenced by lack of blinding; blinding of outcome assessment ensured, and unlikely that the blinding could have been broken.

High risk of bias: No blinding of outcome assessment, and the outcome measurement is likely to be influenced by lack of blinding; blinding of outcome assessment, but likely that the blinding could have been broken, and the outcome measurement is likely to be influenced by lack of blinding.

Unclear: Insufficient information to permit judgement

Incomplete outcome data

Attrition bias due to amount, nature or handling of incomplete outcome data.

Low risk of bias: No missing outcome data; reasons for missing outcome data unlikely to be related to true outcome (for survival data, censoring unlikely to be introducing bias); missing outcome data balanced in numbers across intervention groups, with similar reasons for missing data across groups; for dichotomous outcome data, the proportion of missing outcomes compared with observed event risk not enough to have a clinically relevant impact on the intervention effect estimate; for continuous outcome data, plausible effect size (difference in means or standardised difference in means) among missing outcomes not enough to have a clinically relevant impact on observed effect size; missing data have been imputed using appropriate methods.

High risk of bias: Reason for missing outcome data likely to be related to true outcome, with either imbalance in numbers or reasons for missing data across intervention groups; for dichotomous outcome data, the proportion of missing outcomes compared with observed event risk enough to induce clinically relevant bias in intervention effect estimate; for continuous outcome data, plausible effect size (difference in means or standardised difference in means) among missing outcomes enough to induce clinically relevant bias in observed effect size; 'as-treated' analysis done with substantial departure of the intervention received from that assigned at randomisation; potentially inappropriate application of simple imputation.

Unclear: Insufficient information to permit judgement

Selective reporting

Reporting bias due to selective outcome reporting

Low risk of bias: The study protocol is available and all of the study's pre-specified (primary and secondary) outcomes that are of interest in the review have been reported in the pre-specified way; the study protocol is not available but it is clear that the published reports include all expected outcomes, including those that were pre-specified (convincing text of this nature may be uncommon).

High risk of bias: Not all of the study's pre-specified primary outcomes have been reported; one or more primary outcomes is reported using measurements, analysis methods or subsets of the data (e.g. sub-scales) that were not pre-specified; one or more reported primary outcomes were not pre-specified (unless clear justification for their reporting is provided, such as an unexpected adverse effect); one or more outcomes of interest in the review are reported incompletely so that they cannot be entered in a meta-analysis; the study report fails to include results for a key outcome that would be expected to have been reported for such a study.

Unclear: Insufficient information to permit judgement

Other bias

Bias due to problems not covered elsewhere in the table

Low risk of bias: The study appears to be free of other sources of bias.

High risk of bias: Had a potential source of bias related to the specific study design used; stopped early due to some data-dependent process (including a formal-stopping rule); had extreme baseline imbalance; has been claimed to have been fraudulent; had some other problem.

(Continued)

Unclear: Insufficient information to assess whether an important risk of bias exists; insufficient rationale or evidence that an identified problem will introduce bias.

WHAT'S NEW

Date	Event	Description
2 February 2023	New citation required but conclusions have not changed	New studies added, no change to conclusions
2 February 2023	New search has been performed	Two new studies identified and included

HISTORY

Protocol first published: Issue 1, 2005

Review first published: Issue 3, 2009

Date	Event	Description
21 July 2015	New search has been performed	New studies identified
21 July 2015	New citation required and conclusions have changed	New study investigating treatment for severe kidney disease in adults included
18 March 2010	Amended	Contact details updated.
16 June 2008	Amended	Converted to new review format.

CONTRIBUTIONS OF AUTHORS

- Elisabeth Hodson: study selection, quality appraisal, data extraction, data analysis, writing original protocol, writing original review, updating review.
- Jonathan Craig: writing review, disagreement resolution
- Deirdre Hahn: study selection, quality appraisal, data extraction, data analysis, writing of manuscript for update of review

DECLARATIONS OF INTEREST

- Deirdre Hahn: no relevant interests were disclosed
- Elisabeth Hodson: no relevant interests were disclosed
- Jonathan Craig: no relevant interests were disclosed

SOURCES OF SUPPORT

Internal sources

- No sources of support provided

External sources

- No sources of support provided

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

Studies of therapies with herbal treatments and non-pharmacological interventions were excluded.

INDEX TERMS

Medical Subject Headings (MeSH)

Fosinopril; *IgA Vasculitis [complications] [drug therapy]; *Kidney Diseases; Leflunomide; Proteinuria; Tacrolimus; *Vasculitis

MeSH check words

Adult; Child; Humans